Theory of Mathematics

Volume IV: Complex Analysis and Riemann Surfaces

Contents

Part I

Holomorphic Functions

Chapter 1

Complex Differentiability

Complex differentiability is much more rigid than real differentiability. A single complex derivative must approximate increments from every direction while respecting multiplication by i, and this rigidity quickly leads to analyticity.

Learning goals

After this chapter, the reader should be able to:

- test complex differentiability at a point by comparing directional difference quotients or the real derivative;
- translate between the complex derivative and the corresponding real-linear map;
- verify holomorphicity of elementary functions and identify exceptional points where differentiability fails;
- apply product, quotient, and chain rules in complex-variable calculations;
- construct first-order complex linearizations with a controlled little-o remainder;
- produce examples showing why real differentiability alone does not imply complex differentiability;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

1.1 Complex derivative

A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions.

Example 1.1 (Derivative of a quadratic)**.** Let $f(z) = z^2$ and fix z_0. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{2z_0 h + h^2}{h} = 2z_0 + h.$$

The limit is $2z_0$ independently of the direction in which $h \to 0$. Hence f is complex differentiable everywhere and $f'(z_0) = 2z_0$. The calculation exhibits the decisive requirement: one complex limit must work for every approach direction.

1.2 Real-linear viewpoint

Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number.

1.3 Directional consistency

Approaching along real and imaginary directions gives necessary compatibility conditions.

1.4 Holomorphic functions

A function holomorphic on an open set is complex differentiable at every point there.

Example 1.2 (Conjugation fails the direction test)**.** For $f(z) = \overline{z}$ at the origin, real increments give

$$\frac{f(h) - f(0)}{h} = 1,$$

while purely imaginary increments $h = it$ give

$$\frac{f(it) - f(0)}{it} = \frac{-it}{it} = -1.$$

The directional limits disagree, so conjugation is not complex differentiable at 0. Translating the same calculation shows that it is nowhere complex differentiable.

1.5 Elementary examples

Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points.

1.6 Algebra rules

Sums, products, quotients, and compositions obey familiar derivative rules where defined.

1.7 Local linearization

$f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|)$ is the coordinate-free local approximation.

Example 1.3 (A function differentiable only at one point)**.** Consider $f(z) = |z|^2 = z\overline{z}$. At 0,

$$\frac{f(h) - f(0)}{h} = \frac{|h|^2}{h} = \overline{h} \longrightarrow 0,$$

so $f'(0) = 0$. At a nonzero point z_0, increments along the real and imaginary directions produce different linear terms because of the $\overline{h}$ contribution. Thus a function may be complex differentiable at an isolated point without being holomorphic on any neighborhood of that point.

1.8 Rigidity preview

Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles.

1.9 Core structural results

Theorem 1.4 (Complex-linear characterization). *A real-differentiable map $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 iff its real derivative commutes with multiplication by i.*

Proof. If Df is multiplication by $f'(z_0)$ it clearly commutes with i. Conversely any real-linear map commuting with i is determined by its value at 1 and equals complex multiplication by that value. □

Theorem 1.5 (Product rule). *If f, g are complex differentiable at z_0, then $(fg)' = f'g + fg'$.*

Proof. Use the increment identity $f(z)g(z) - f_0g_0 = (f - f_0)g + f_0(g - g_0)$ and divide by $z - z_0$, using continuity implied by differentiability. □

Theorem 1.6 (Chain rule). *If f is differentiable at z_0 and g at $f(z_0)$, then $(g \circ f)'(z_0) = g'(f(z_0))f'(z_0)$.*

Proof. Insert the complex-linear first-order expansions for f and g and combine their little-o remainders. □

1.10 Worked examples

Example 1.7 (Complex derivative diagnostic). Give a rigorous worked analysis of Complex derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.8 (Real-linear viewpoint diagnostic). Give a rigorous worked analysis of Real-linear viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.9 (Directional consistency diagnostic). Give a rigorous worked analysis of Directional consistency. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Approaching along real and imaginary directions gives necessary compatibility conditions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.10 (Holomorphic functions diagnostic). Give a rigorous worked analysis of Holomorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function holomorphic on an open set is complex differentiable at every point there. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

1.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 1.11 (Complex derivative diagnostic)**.** Give a rigorous worked analysis of Complex derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.12 (Real-linear viewpoint diagnostic)**.** Give a rigorous worked analysis of Real-linear viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.13 (Directional consistency diagnostic)**.** Give a rigorous worked analysis of Directional consistency. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Approaching along real and imaginary directions gives necessary compatibility conditions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.14 (Holomorphic functions diagnostic)**.** Give a rigorous worked analysis of Holomorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function holomorphic on an open set is complex differentiable at every point there. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.15 (Elementary examples diagnostic)**.** Give a rigorous worked analysis of Elementary examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.16 (Algebra rules diagnostic)**.** Give a rigorous worked analysis of Algebra rules. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Sums, products, quotients, and compositions obey familiar derivative rules where defined. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.17 (Local linearization diagnostic)**.** Give a rigorous worked analysis of Local linearization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|)$ is the coordinate-free local approximation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.18 (Rigidity preview diagnostic)**.** Give a rigorous worked analysis of Rigidity preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.19 (Complex-linear characterization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A real-differentiable map $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 iff its real derivative commutes with multiplication by i.

Solution. If Df is multiplication by $f'(z_0)$ it clearly commutes with i. Conversely any real-linear map commuting with i is determined by its value at 1 and equals complex multiplication by that value. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.20 (Product rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f, g are complex differentiable at z_0, then $(fg)' = f'g + fg'$.

Solution. Use the increment identity $f(z)g(z) - f_0g_0 = (f - f_0)g + f_0(g - g_0)$ and divide by $z - z_0$, using continuity implied by differentiability. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.21 (Chain rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is differentiable at z_0 and g at $f(z_0)$, then $(g \circ f)'(z_0) = g'(f(z_0))f'(z_0)$.

Solution. Insert the complex-linear first-order expansions for f and g and combine their little-o remainders. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Complex differentiability is much more rigid than real differentiability. A single complex derivative must approximate increments from every direction while respecting multiplication by i, and this rigidity quickly leads to analyticity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

1.12 Exercises with complete solutions

Exercise 1.23. State and apply the central principle behind Complex derivative. Give one concrete consequence.

Hint. Compare the difference quotient along the real and imaginary directions first; if the two limits disagree, complex differentiability fails immediately. □

Solution. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.24. State and apply the central principle behind Real-linear viewpoint. Give one concrete consequence.

Hint. Write the real derivative matrix and impose commutation with multiplication by i; this forces the Cauchy–Riemann form. □

Solution. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.25. State and apply the central principle behind Directional consistency. Give one concrete consequence.

Hint. Use two distinct approach directions to test whether a proposed derivative is direction-independent. □

Solution. Approaching along real and imaginary directions gives necessary compatibility conditions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.26. State and apply the central principle behind Holomorphic functions. Give one concrete consequence.

Hint. Holomorphicity is an open-set condition: check complex differentiability at every point of a neighborhood, not just at one point. □

Solution. A function holomorphic on an open set is complex differentiable at every point there. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.27. State and apply the central principle behind Elementary examples. Give one concrete consequence.

Hint. For conjugation or modulus, compute directional quotients at the suspected exceptional point before making a global claim. □

Solution. Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.28. State and apply the central principle behind Algebra rules. Give one concrete consequence.

Hint. Apply the algebra rule only after checking denominators stay nonzero where needed. □

Solution. Sums, products, quotients, and compositions obey familiar derivative rules where defined. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.29. State and apply the central principle behind Local linearization. Give one concrete consequence.

Hint. Subtract $f(z_0) + f'(z_0)h$ and divide the remainder by $|h|$; the quotient must tend to zero. $\square$

Solution. $f(z_0+h) = f(z_0)+f'(z_0)h+o(|h|)$ is the coordinate-free local approximation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 1.30. State and apply the central principle behind Rigidity preview. Give one concrete consequence.

Hint. Use local complex linearity to predict analyticity consequences, but do not invoke power-series machinery before it has been established. $\square$

Solution. Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

1.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 1.31 (Polynomial derivative)**.** Compute the complex derivative of $f(z) = 3z^4 - 2z + 7$.

Hint. Use the algebra rules for powers and sums. $\square$

Solution. The power rule gives $f'(z) = 12z^3 - 2$. $\square$

Exercise 1.32 (Reciprocal derivative)**.** Differentiate $f(z) = 1/(z-2i)$ on its domain.

Hint. Apply the reciprocal or quotient rule away from the pole. $\square$

Solution. For $z \neq 2i$, $f'(z) = -(z-2i)^{-2}$. $\square$

Exercise 1.33 (Chain rule)**.** Find the derivative of $f(z) = (z^2+1)^5$.

Hint. Differentiate the outer fifth power and then the inner quadratic. $\square$

Solution. The chain rule gives $f'(z) = 10z(z^2+1)^4$. $\square$

Exercise 1.34 (Directional diagnostic)**.** Test complex differentiability of $f(z) = \operatorname{Re} z$ at 0.

Hint. Compare real and imaginary increments. $\square$

Solution. Along real increments the quotient is 1, while along imaginary increments it is 0. Therefore the derivative does not exist. $\square$

Exercise 1.35 (Linearization)**.** Write the first order complex linearization of e^z at z_0.

Hint. Use the derivative e^{z_0}. ☐

Solution. The linearization is $e^{z_0} + e^{z_0}(z - z_0)$, with an error $o(|z - z_0|)$. ☐

Proofs

Exercise 1.36 (Differentiability implies continuity)**.** Prove that complex differentiability at z_0 implies continuity there.

Hint. Factor $f(z) - f(z_0)$ by $z - z_0$. ☐

Solution. Write $f(z) - f(z_0) = (z - z_0)Q(z)$, where $Q(z) \to f'(z_0)$. The factor Q is bounded near z_0, so the product tends to zero. ☐

Exercise 1.37 (Product rule)**.** Prove the complex product rule directly from difference quotients.

Hint. Add and subtract $f(z)g(z_0)$. ☐

Solution. Split the increment into $f(z)(g(z) - g(z_0)) + g(z_0)(f(z) - f(z_0))$, divide by $z - z_0$, and use continuity of f. ☐

Exercise 1.38 (Reciprocal rule)**.** If $f(z_0) \neq 0$, prove that $1/f$ is differentiable at z_0 and find its derivative.

Hint. Use a difference of reciprocals and continuity of f. ☐

Solution. The quotient becomes $-(f(z) - f(z_0))/((z - z_0)f(z)f(z_0))$. Passing to the limit gives $-(f'(z_0))/f(z_0)^2$. ☐

Exercise 1.39 (Complex linear real derivative)**.** Prove that a real linear map $T : \mathbb{C} \to \mathbb{C}$ is complex linear exactly when $T(i) = iT(1)$.

Hint. Every complex number is $x + iy$. ☐

Solution. If the condition holds, real linearity gives $T(x+iy) = xT(1)+yT(i) = (x+iy)T(1)$. The converse follows by evaluating a complex linear map at i. ☐

Counterexamples and hypothesis tests

Exercise 1.40 (Absolute value)**.** Is $f(z) = |z|$ complex differentiable at 0? Justify the answer.

Hint. Compare positive and negative real increments. ☐

Solution. For positive real h, the quotient is 1; for negative real h, it is -1. Hence no complex derivative exists. ☐

Exercise 1.41 (A single derivative is not holomorphicity)**.** Does complex differentiability of a function at one point imply holomorphicity near that point?

Hint. Use $|z|^2$. ☐

Solution. No. The function $|z|^2$ is complex differentiable at 0 with derivative zero, but it is not complex differentiable at any nonzero point. ☐

Exercise 1.42 (Quotient rule hypothesis)**.** What fails if the quotient rule is applied to f/g at a point where $g = 0$?

Hint. First ask whether the quotient is even defined near the point. □

Solution. The quotient need not be defined at the point and its reciprocal factor can be singular. The nonvanishing denominator hypothesis is essential. □

Applications and investigations

Exercise 1.43 (Error estimate from linearization)**.** Suppose $f'(z_0) = a$. Explain how the derivative predicts $f(z_0 + h)$ for small h.

Hint. Use the definition of the little o remainder. □

Solution. One has $f(z_0+h) = f(z_0)+ah+r(h)$ with $|r(h)|/|h| \to 0$. Thus multiplication by a gives the first order change in both magnitude and direction. □

Exercise 1.44 (Real linear classifier)**.** For $T(z) = az + b\overline{z}$, determine when T is complex linear.

Hint. Compare $T(iz)$ with $iT(z)$. □

Solution. The conjugate term changes sign under multiplication by i. Equality for every z forces $b = 0$; then $T(z) = az$ is complex linear. □

Challenge problems

Exercise 1.45 (Two direction agreement is not enough)**.** Explain why agreement of the real and imaginary directional quotients alone does not prove complex differentiability without additional regularity.

Hint. Directional tests provide necessary conditions, not uniform control over all directions. □

Solution. The derivative requires one limit as $h \to 0$ through arbitrary complex directions. A function can be engineered so the quotients along the coordinate axes agree while quotients along a curved or oblique path do not. □

Exercise 1.46 (Derivative of inversion as geometry)**.** Show that $f(z) = 1/z$ has derivative $-1/z^2$ and interpret the local map as a rotation-dilation.

Hint. Compute the difference quotient and then write the nonzero derivative in polar form. □

Solution. The quotient simplifies to $-1/(z(z + h))$, giving $-1/z^2$. Any nonzero complex derivative is multiplication by a complex number, hence locally a dilation by its modulus followed by a rotation by its argument. □

Chapter 2

Cauchy–Riemann Equations

The Cauchy–Riemann equations translate complex differentiability into real partial-derivative relations. They reveal the geometric link between holomorphic maps, conformality, harmonic functions, and complex linearity.

Learning goals

After this chapter, the reader should be able to:

- derive the Cauchy–Riemann equations from complex differentiability;
- check the Cauchy–Riemann equations for explicit functions and combine them with regularity hypotheses correctly;
- reconstruct harmonic conjugates from compatible partial-derivative data;
- distinguish necessary Cauchy–Riemann conditions from sufficient conditions for holomorphicity;
- express the derivative in terms of real partial derivatives once holomorphicity is established;
- construct examples where the Cauchy–Riemann equations hold at a point but complex differentiability still fails;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

2.1 Cartesian form

For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$.

Example 2.1 (Cauchy–Riemann check for a cubic)**.** Write $z = x + iy$. For $f(z) = z^3$,

$$u = x^3 - 3xy^2, \qquad v = 3x^2y - y^3.$$

Then $u_x = 3x^2 - 3y^2 = v_y$ and $u_y = -6xy = -v_x$. The partial derivatives are polynomials and therefore continuous, so the sufficiency theorem applies everywhere. Moreover $f'(z) = u_x + iv_x = 3z^2$, agreeing with direct complex differentiation.

2.2 Derivation from directions

Real and imaginary difference quotients must agree, yielding the equations.

2.3 Sufficiency with regularity

If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby.

2.4 Polar form

In polar coordinates, the equations adapt to radial and angular derivatives.

2.5 Harmonic components

Twice differentiable holomorphic functions have harmonic real and imaginary parts.

Example 2.2 (Recovering a harmonic conjugate)**.** Let $u(x, y) = x^2 - y^2$. A conjugate v must satisfy

$$v_y = u_x = 2x, \qquad v_x = -u_y = 2y.$$

Integrating the first equation gives $v = 2xy + g(x)$. Differentiating in x and comparing with the second equation yields $g'(x) = 0$. Hence $v = 2xy + C$, and $u + iv = z^2 + iC$.

2.6 Harmonic conjugates

Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition.

2.7 Jacobian geometry

The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix.

2.8 Failure examples

Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity.

Example 2.3 (Pointwise equations without differentiability)**.** Define $f(0) = 0$ and, for $z = x + iy \neq 0$,

$$f(z) = \frac{x^3}{x^2 + y^2} + i\frac{y^3}{x^2 + y^2}.$$

The coordinate partial derivatives at the origin exist and give $u_x(0) = v_y(0) = 1$ and $u_y(0) = -v_x(0) = 0$, so the Cauchy–Riemann equations hold at that point. Along $y = x$, however, $f(x + ix)/(x + ix) = 1/2$, whereas along the real axis the quotient is 1. Thus the equations at one point, without a suitable differentiability hypothesis, are not sufficient.

2.9 Core structural results

Theorem 2.4 (Cauchy–Riemann necessity). *If $f = u + iv$ is complex differentiable at z_0, then $u_x = v_y$ and $u_y = -v_x$ at z_0.*

Proof. Evaluate the difference quotient along real increments and along purely imaginary increments and equate the two complex limits. □

Theorem 2.5 (Sufficiency under C^1 hypotheses). *If u, v have continuous first partials near z_0 satisfying the Cauchy–Riemann equations, then $f = u + iv$ is complex differentiable there.*

Proof. Use the real differentiability expansion and the equations to identify the real derivative matrix with complex multiplication by $u_x + iv_x$. □

Theorem 2.6 (Harmonicity). *If $f = u + iv$ is holomorphic with continuous second partials, then $\Delta u = \Delta v = 0$.*

Proof. Differentiate the Cauchy–Riemann equations and use equality of mixed partials to obtain cancellation in each Laplacian. □

2.10 Worked examples

Example 2.7 (Cartesian form diagnostic). Give a rigorous worked analysis of Cartesian form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.8 (Derivation from directions diagnostic). Give a rigorous worked analysis of Derivation from directions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Real and imaginary difference quotients must agree, yielding the equations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.9 (Sufficiency with regularity diagnostic). Give a rigorous worked analysis of Sufficiency with regularity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.10 (Polar form diagnostic). Give a rigorous worked analysis of Polar form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. In polar coordinates, the equations adapt to radial and angular derivatives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

2.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 2.11 (Cartesian form diagnostic)**.** Give a rigorous worked analysis of Cartesian form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.12 (Derivation from directions diagnostic)**.** Give a rigorous worked analysis of Derivation from directions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Real and imaginary difference quotients must agree, yielding the equations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.13 (Sufficiency with regularity diagnostic)**.** Give a rigorous worked analysis of Sufficiency with regularity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.14 (Polar form diagnostic)**.** Give a rigorous worked analysis of Polar form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. In polar coordinates, the equations adapt to radial and angular derivatives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.15 (Harmonic components diagnostic)**.** Give a rigorous worked analysis of Harmonic components. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Twice differentiable holomorphic functions have harmonic real and imaginary parts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.16 (Harmonic conjugates diagnostic)**.** Give a rigorous worked analysis of Harmonic conjugates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.17 (Jacobian geometry diagnostic)**.** Give a rigorous worked analysis of Jacobian geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.18 (Failure examples diagnostic)**.** Give a rigorous worked analysis of Failure examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.19 (Cauchy–Riemann necessity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = u + iv$ is complex differentiable at z_0, then $u_x = v_y$ and $u_y = -v_x$ at z_0.

Solution. Evaluate the difference quotient along real increments and along purely imaginary increments and equate the two complex limits. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.20 (Sufficiency under C^1 hypotheses proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If u, v have continuous first partials near z_0 satisfying the Cauchy–Riemann equations, then $f = u + iv$ is complex differentiable there.

Solution. Use the real differentiability expansion and the equations to identify the real derivative matrix with complex multiplication by $u_x + iv_x$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.21 (Harmonicity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = u + iv$ is holomorphic with continuous second partials, then $\Delta u = \Delta v = 0$.

Solution. Differentiate the Cauchy–Riemann equations and use equality of mixed partials to obtain cancellation in each Laplacian. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Cauchy–Riemann equations translate complex differentiability into real partial-derivative relations. They reveal the geometric link between holomorphic maps, conformality, harmonic functions, and complex linearity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

2.12 Exercises with complete solutions

Exercise 2.23. State and apply the central principle behind Cartesian form. Give one concrete consequence.

Hint. Write $f = u + iv$ and compare derivatives in the x- and y-directions to obtain the Cauchy–Riemann equations. □

Solution. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.24. State and apply the central principle behind Derivation from directions. Give one concrete consequence.

Hint. Check both equations $u_x = v_y$ and $u_y = -v_x$ at the point, then inspect the regularity needed for sufficiency. □

Solution. Real and imaginary difference quotients must agree, yielding the equations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.25. State and apply the central principle behind Sufficiency with regularity. Give one concrete consequence.

Hint. Integrate one of the equations to recover a candidate harmonic conjugate, then use the other equation to determine the missing function of one variable. □

Solution. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.26. State and apply the central principle behind Polar form. Give one concrete consequence.

Hint. The Cauchy–Riemann equations at a single point are only necessary without additional differentiability assumptions; test the remainder term. □

Solution. In polar coordinates, the equations adapt to radial and angular derivatives. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.27. State and apply the central principle behind Harmonic components. Give one concrete consequence.

Hint. Once holomorphicity is known, compute $f' = u_x + iv_x = v_y - iu_y$ using the most convenient partial derivatives. □

Solution. Twice differentiable holomorphic functions have harmonic real and imaginary parts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.28. State and apply the central principle behind Harmonic conjugates. Give one concrete consequence.

Hint. Differentiate the Cauchy–Riemann equations once more to show both real and imaginary parts are harmonic when second derivatives exist. □

Solution. Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.29. State and apply the central principle behind Jacobian geometry. Give one concrete consequence.

Hint. For a harmonic conjugate, verify the compatibility condition by checking that the proposed differential is closed. □

Solution. The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.30. State and apply the central principle behind Failure examples. Give one concrete consequence.

Hint. To build a counterexample, make the function satisfy Cauchy–Riemann at the origin while its difference quotient depends on direction. □

Solution. Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

2.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 2.31 (Quadratic CR check)**.** Verify the Cauchy–Riemann equations for $f(z) = z^2$.

Hint. Expand $(x + iy)^2$. □

Solution. Here $u = x^2 - y^2$ and $v = 2xy$. Thus $u_x = 2x = v_y$ and $u_y = -2y = -v_x$. □

Exercise 2.32 (Derivative from partials)**.** For $f(z) = z^3$, compute f' from $u_x + iv_x$.

Hint. Use the real and imaginary parts of the cubic. □

Solution. The expression is $3x^2 - 3y^2 + i6xy = 3(x + iy)^2 = 3z^2$. □

Exercise 2.33 (Harmonic test)**.** Check that $u(x, y) = e^x \cos y$ is harmonic.

Hint. Compute the two second partial derivatives. □

Solution. One finds $u_{xx} = e^x \cos y$ and $u_{yy} = -e^x \cos y$, so $\Delta u = 0$. □

Exercise 2.34 (Find a conjugate)**.** Find a harmonic conjugate of $u = e^x \cos y$.

Hint. Use $v_y = u_x$ and then check $v_x = -u_y$. □

Solution. Integrating $v_y = e^x \cos y$ gives $v = e^x \sin y + C$, which satisfies the other equation. □

Exercise 2.35 (Jacobian determinant)**.** If $f'(z_0) = a + ib$, compute the determinant of the real Jacobian at z_0.

Hint. The matrix is $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. □

Solution. Its determinant is $a^2 + b^2 = |f'(z_0)|^2$. □

Proofs

Exercise 2.36 (Necessity of CR)**.** Derive the Cauchy–Riemann equations from the complex difference quotient.

Hint. Use real increments and then imaginary increments. □

Solution. The real increment limit is $u_x + iv_x$. The imaginary increment limit is $v_y - iu_y$. Equality yields $u_x = v_y$ and $u_y = -v_x$. □

Exercise 2.37 (Harmonic real part)**.** Assuming continuous second partials, prove that the real part of a holomorphic function is harmonic.

Hint. Differentiate the two CR equations once more. □

Solution. From $u_x = v_y$ obtain $u_{xx} = v_{yx}$, and from $u_y = -v_x$ obtain $u_{yy} = -v_{xy}$. Equality of mixed partials gives $u_{xx} + u_{yy} = 0$. □

Exercise 2.38 (Harmonic imaginary part)**.** Prove the analogous harmonicity statement for the imaginary part.

Hint. Differentiate in the opposite order. □

Solution. The differentiated CR equations give $v_{xx} = -u_{yx}$ and $v_{yy} = u_{xy}$, whose sum is zero. □

Exercise 2.39 (Local angle scaling)**.** Show that when $f'(z_0) \neq 0$, the real derivative preserves oriented angles.

Hint. Write the Jacobian as multiplication by the complex number $f'(z_0)$. □

Solution. Multiplication by a nonzero complex number is a positive dilation followed by a rotation. It therefore changes every tangent direction by the same angle and preserves oriented angle differences. □

Counterexamples and hypothesis tests

Exercise 2.40 (Conjugation)**.** Test the CR equations for $f(z) = \overline{z}$.

Hint. Use $u = x$ and $v = -y$. □

Solution. One has $u_x = 1$ and $v_y = -1$, so the first equation fails everywhere. □

Exercise 2.41 (Real valued holomorphic function)**.** Can a nonconstant real valued function be holomorphic on a connected open set?

Hint. Set the imaginary part equal to zero in the CR equations. □

Solution. No. The equations force both first partial derivatives of the real part to vanish, so the function is locally constant and hence constant on each connected component. □

Exercise 2.42 (Pointwise CR)**.** Why is satisfying the CR equations at one point not by itself sufficient for complex differentiability?

Hint. The equations concern only coordinate partial derivatives. □

Solution. Complex differentiability requires a two dimensional linear approximation. Without real differentiability or suitable continuity of partials, coordinate partial information may miss oblique approaches. □

Applications and investigations

Exercise 2.43 (Potential and stream function)**.** If u is a harmonic potential and v is its harmonic conjugate, what geometric relation holds between their gradients?

Hint. Use the CR equations. □

Solution. The equations give $\nabla v = (-u_y, u_x)$, a quarter turn of ∇u. Their level curves are therefore orthogonal where the gradients are nonzero. □

Exercise 2.44 (Local area scaling)**.** How does a holomorphic map with derivative $f'(z_0) \neq 0$ scale infinitesimal area?

Hint. Use the Jacobian determinant. □

Solution. The local area factor is $|f'(z_0)|^2$, the determinant of the real derivative matrix. □

Challenge problems

Exercise 2.45 (Polar CR equations)**.** Derive the polar Cauchy–Riemann equations for $f = u + iv$ away from the origin.

Hint. Express u_x, u_y, v_x, v_y through radial and angular derivatives. □

Solution. Substituting the polar chain rule into the Cartesian equations yields $u_r = v_\theta / r$ and $v_r = -u_\theta / r$. □

Exercise 2.46 (Conjugate obstruction)**.** Explain why a harmonic function on a punctured plane can fail to have a single valued harmonic conjugate.

Hint. Consider $u(z) = \log |z|$. □

Solution. Locally its conjugate is an argument function. Going once around the origin changes the argument by 2π, so no globally single valued conjugate exists on the punctured plane. □

Chapter 3

Power Series and Analytic Functions

Power series are the local algebra of holomorphic functions. Their radius of convergence controls a disk on which termwise differentiation and integration are valid, and analytic functions inherit remarkable rigidity.

Learning goals

After this chapter, the reader should be able to:

- compute radii of convergence for power series and determine behavior at boundary points separately;
- differentiate and integrate power series term by term inside their disks of convergence;
- derive Taylor coefficients of holomorphic functions from derivatives at the center;
- use local power-series expansions to prove analyticity and uniqueness statements;
- manipulate standard expansions to obtain new series representations;
- identify the nearest singularity controlling the radius of convergence in concrete examples;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

3.1 Complex power series

A series $\sum a_n(z-z_0)^n$ has a radius of convergence determined by coefficient growth.

Example 3.1 (Radius from coefficient growth)**.** For

$$\sum_{n=0}^{\infty} \frac{z^n}{3^n},$$

the ratio of successive absolute terms tends to $|z|/3$. Thus the series converges for $|z| < 3$ and diverges for $|z| > 3$, so its radius is $R = 3$. Inside the disk it sums to $1/(1-z/3)$, but the radius is determined before any closed form is used.

3.2 Absolute convergence

Inside the radius the series converges absolutely; outside it diverges.

3.3 Uniform convergence on compact subdisks

On every smaller closed disk, convergence is uniform and normally controlled.

3.4 Termwise differentiation

Differentiated power series has the same radius of convergence and represents the derivative.

Example 3.2 (Taylor expansion about a shifted center)**.** To expand $1/z$ about $z_0 = 2$, write

$$\frac{1}{z} = \frac{1}{2+(z-2)} = \frac{1}{2}\frac{1}{1+(z-2)/2} = \frac{1}{2}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z-2}{2}\right)^n.$$

The nearest singularity to the center is at 0, at distance 2, so the expansion is valid exactly for $|z-2| < 2$.

3.5 Termwise integration

Primitive series are obtained coefficientwise on disks.

3.6 Analytic functions

A function is analytic when locally represented by a convergent power series.

3.7 Taylor coefficients

For analytic functions, coefficients are derivatives divided by factorials.

Example 3.3 (Termwise differentiation keeps the radius)**.** Starting from $e^z = \sum_{n\geq 0} z^n/n!$, termwise differentiation gives

$$\sum_{n\geq 1}\frac{nz^{n-1}}{n!} = \sum_{m\geq 0}\frac{z^m}{m!} = e^z.$$

The factorial growth makes the radius infinite both before and after differentiation. This concrete calculation models the general theorem that differentiation preserves the radius of a power series.

3.8 Uniqueness of expansion

A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center.

3.9 Core structural results

Theorem 3.4 (Uniform convergence on compact subdisks)**.** *If a power series has radius R, it converges uniformly on $|z - z_0| \leq r < R$.*

Proof. Choose ρ with $r < \rho < R$. Convergence at radius ρ bounds $|a_n|\rho^n$, giving a geometric majorant for $|a_n|r^n$. ☐

Theorem 3.5 (Termwise differentiation)**.** *Inside the radius of convergence, a power series may be differentiated termwise and the derivative series has the same radius.*

Proof. Use uniform convergence on compact subdisks together with the standard derivative-series convergence criterion, or compare coefficients with the root test. ☐

Theorem 3.6 (Coefficient uniqueness)**.** *If $\sum a_n(z - z_0)^n = 0$ on a neighborhood of z_0, then every $a_n = 0$.*

Proof. Differentiate k times and evaluate at z_0 to get $k!a_k = 0$. ☐

3.10 Worked examples

Example 3.7 (Complex power series diagnostic)**.** Give a rigorous worked analysis of Complex power series. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A series $\sum a_n(z - z_0)^n$ has a radius of convergence determined by coefficient growth. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.8 (Absolute convergence diagnostic)**.** Give a rigorous worked analysis of Absolute convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Inside the radius the series converges absolutely; outside it diverges. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.9 (Uniform convergence on compact subdisks diagnostic)**.** Give a rigorous worked analysis of Uniform convergence on compact subdisks. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On every smaller closed disk, convergence is uniform and normally controlled. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.10 (Termwise differentiation diagnostic)**.** Give a rigorous worked analysis of Termwise differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Differentiated power series has the same radius of convergence and represents the derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

3.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 3.11 (Complex power series diagnostic)**.** Give a rigorous worked analysis of Complex power series. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A series $\sum a_n(z-z_0)^n$ has a radius of convergence determined by coefficient growth. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.12 (Absolute convergence diagnostic)**.** Give a rigorous worked analysis of Absolute convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Inside the radius the series converges absolutely; outside it diverges. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.13 (Uniform convergence on compact subdisks diagnostic)**.** Give a rigorous worked analysis of Uniform convergence on compact subdisks. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On every smaller closed disk, convergence is uniform and normally controlled. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.14 (Termwise differentiation diagnostic)**.** Give a rigorous worked analysis of Termwise differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiated power series has the same radius of convergence and represents the derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.15 (Termwise integration diagnostic)**.** Give a rigorous worked analysis of Termwise integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Primitive series are obtained coefficientwise on disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.16 (Analytic functions diagnostic)**.** Give a rigorous worked analysis of Analytic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function is analytic when locally represented by a convergent power series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.17 (Taylor coefficients diagnostic)**.** Give a rigorous worked analysis of Taylor coefficients. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For analytic functions, coefficients are derivatives divided by factorials. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 3.18 (Uniqueness of expansion diagnostic)**.** Give a rigorous worked analysis of Uniqueness of expansion. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.19 (Uniform convergence on compact subdisks proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If a power series has radius R, it converges uniformly on $|z - z_0| \leq r < R$.

Solution. Choose ρ with $r < \rho < R$. Convergence at radius ρ bounds $|a_n|\rho^n$, giving a geometric majorant for $|a_n|r^n$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.20 (Termwise differentiation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Inside the radius of convergence, a power series may be differentiated termwise and the derivative series has the same radius.

Solution. Use uniform convergence on compact subdisks together with the standard derivative-series convergence criterion, or compare coefficients with the root test. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.21 (Coefficient uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $\sum a_n(z - z_0)^n = 0$ on a neighborhood of z_0, then every $a_n = 0$.

Solution. Differentiate k times and evaluate at z_0 to get $k!a_k = 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Power series are the local algebra of holomorphic functions. Their radius of convergence controls a disk on which termwise differentiation and integration are valid, and analytic functions inherit remarkable rigidity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

3.12 Exercises with complete solutions

Exercise 3.23. State and apply the central principle behind Complex power series. Give one concrete consequence.

Hint. Use the root or ratio test on the coefficients to find the radius; analyze points with $|z - z_0| = R$ separately. ☐

Solution. A series $\sum a_n(z - z_0)^n$ has a radius of convergence determined by coefficient growth. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.24. State and apply the central principle behind Absolute convergence. Give one concrete consequence.

Hint. Inside a smaller closed disk, uniform convergence justifies termwise differentiation and integration. ☐

Solution. Inside the radius the series converges absolutely; outside it diverges. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.25. State and apply the central principle behind Uniform convergence on compact subdisks. Give one concrete consequence.

Hint. The Taylor coefficient is $a_n = f^{(n)}(z_0)/n!$; compute derivatives only after locating the expansion center. ☐

Solution. On every smaller closed disk, convergence is uniform and normally controlled. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.26. State and apply the central principle behind Termwise differentiation. Give one concrete consequence.

Hint. Use the local series to identify the first nonzero coefficient; it controls local zero order and uniqueness. ☐

Solution. Differentiated power series has the same radius of convergence and represents the derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.27. State and apply the central principle behind Termwise integration. Give one concrete consequence.

Hint. Rewrite the target into a known geometric, exponential, logarithmic, or binomial series before multiplying or composing expansions. ☐

Solution. Primitive series are obtained coefficientwise on disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.28. State and apply the central principle behind Analytic functions. Give one concrete consequence.

Hint. The nearest singularity in the complex plane bounds the Taylor disk even when the real-variable expression looks harmless. ☐

Solution. A function is analytic when locally represented by a convergent power series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.29. State and apply the central principle behind Taylor coefficients. Give one concrete consequence.

Hint. To prove equality of two analytic expressions, compare their Taylor coefficients on a common disk of convergence. ☐

Solution. For analytic functions, coefficients are derivatives divided by factorials. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 3.30. State and apply the central principle behind Uniqueness of expansion. Give one concrete consequence.

Hint. Do not substitute outside the convergence disk; first determine the annular or disk region where each manipulated series is valid. □

Solution. A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

3.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 3.31 (Geometric radius)**.** Find the radius of convergence of $\sum z^n/5^n$.

Hint. Apply the ratio test. □

Solution. The ratio tends to $|z|/5$, so $R = 5$. □

Exercise 3.32 (Factorial radius)**.** Find the radius of convergence of $\sum z^n/n!$.

Hint. Use the ratio test. □

Solution. The ratio of successive absolute terms is $|z|/(n+1) \to 0$ for every z, hence $R = \infty$. □

Exercise 3.33 (Differentiate a series)**.** Differentiate $\sum_{n\geq 0} z^n$ termwise inside its disk of convergence.

Hint. The geometric series has radius one. □

Solution. For $|z| < 1$, differentiation gives $\sum_{n\geq 1} nz^{n-1} = 1/(1-z)^2$. □

Exercise 3.34 (Taylor coefficients)**.** Find the first four Taylor coefficients of e^z at zero.

Hint. Use $a_n = f^{(n)}(0)/n!$. □

Solution. They are $1, 1, 1/2, 1/6$. □

Exercise 3.35 (Shifted geometric series)**.** Expand $1/(3-z)$ in powers of $z-1$ and state the radius.

Hint. Write $3 - z = 2 - (z - 1)$. □

Solution. One obtains $\frac{1}{2}\sum_{n\geq 0}((z-1)/2)^n$, valid for $|z-1| < 2$. □

Proofs

Exercise 3.36 (Absolute convergence inside the radius)**.** Prove that a power series converges absolutely at every point strictly inside its radius.

Hint. Choose a larger radius where the coefficient terms are bounded. ☐

Solution. If $|z - z_0| < r < R$, convergence at radius r bounds $|a_n|r^n$. Multiplying by the geometric factor $(|z - z_0|/r)^n$ gives a summable majorant. ☐

Exercise 3.37 (Uniform convergence on smaller disks)**.** Prove uniform convergence on $|z - z_0| \leq r < R$.

Hint. Use the same geometric majorant independently of z. ☐

Solution. Choose ρ with $r < \rho < R$. Boundedness of $|a_n|\rho^n$ yields the uniform majorant $M(r/\rho)^n$, so the Weierstrass test applies. ☐

Exercise 3.38 (Uniqueness of coefficients)**.** Show that a power series representation near z_0 has unique coefficients.

Hint. Differentiate repeatedly and evaluate at the center. ☐

Solution. Termwise differentiation gives $f^{(n)}(z_0) = n!a_n$. Hence $a_n = f^{(n)}(z_0)/n!$, fixing every coefficient. ☐

Exercise 3.39 (Derivative radius)**.** Explain why the derivative series has the same radius as the original series.

Hint. Compare $|na_n|^{1/n}$ with $|a_n|^{1/n}$. ☐

Solution. Since $n^{1/n} \to 1$, the limsup controlling the Cauchy–Hadamard radius is unchanged by multiplying coefficients by n. ☐

Counterexamples and hypothesis tests

Exercise 3.40 (Boundary is separate)**.** Does the radius of convergence determine convergence at every boundary point?

Hint. Compare the geometric series with an alternating harmonic type series. ☐

Solution. No. The radius separates the open convergence and exterior divergence regions, but individual points on the boundary require separate analysis. ☐

Exercise 3.41 (Nearest singularity)**.** Why can the Taylor series of $1/(1 - z)$ at zero not have radius larger than one?

Hint. The represented analytic function has a singularity at $z = 1$. ☐

Solution. A convergent power series is analytic throughout its disk. A radius larger than one would make the series analytic at 1, contradicting the pole there. ☐

Exercise 3.42 (Smooth is not analytic)**.** Does infinite real differentiability imply a complex power series representation?

Hint. Complex analyticity is much more rigid than real smoothness. ☐

Solution. No. Real smoothness alone imposes no Cauchy–Riemann structure. A function of x and y can be smooth as a real map but fail complex differentiability everywhere. ☐

Applications and investigations

Exercise 3.43 (Summing a weighted geometric series)**.** Use power series to evaluate $\sum_{n\geq 1} nr^{n-1}$ for $|r| < 1$.

Hint. Differentiate the geometric identity. ☐

Solution. Differentiating $\sum r^n = 1/(1-r)$ gives $\sum_{n\geq 1} nr^{n-1} = 1/(1-r)^2$. ☐

Exercise 3.44 (Local analytic model)**.** Explain how a Taylor polynomial approximates an analytic function near its center.

Hint. Separate the first N terms from the convergent tail. ☐

Solution. Inside a smaller disk the tail is uniformly controlled, so truncating after degree N gives a computable local polynomial model whose error tends to zero as N grows. ☐

Challenge problems

Exercise 3.45 (Cauchy–Hadamard)**.** State the coefficient formula for the radius of convergence and explain its meaning.

Hint. Use the root test on $|a_n(z-z_0)^n|$. ☐

Solution. The formula is $1/R = \limsup |a_n|^{1/n}$, with the usual conventions. It measures the exponential growth rate of the coefficients. ☐

Exercise 3.46 (Multiple centers)**.** The function $1/(1-z)$ is expanded about $z_0 = -1$. Find the radius without computing coefficients.

Hint. Measure the distance from the center to the nearest singularity. ☐

Solution. The only finite singularity is at 1, distance 2 from -1. Therefore the Taylor radius is 2. ☐

Chapter 4

Complex Integration

Complex integration integrates a complex-valued function along an oriented curve. Parameterization, estimates, primitives, and deformation ideas prepare the way for Cauchy's theorem.

Learning goals

After this chapter, the reader should be able to:

- parameterize piecewise smooth contours and compute complex line integrals directly;
- apply contour reversal, concatenation, and reparameterization rules correctly;
- estimate contour integrals using the ML inequality;
- identify primitives and use them to evaluate path-independent integrals;
- distinguish path independence from mere equality along one family of curves;
- choose contours adapted to the geometry of the integrand and domain;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

4.1 Parametrized curves

A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data.

Example 4.1 (Parametrizing a line segment)**.** Let $\gamma(t) = t(1 + i)$ for $0 \leq t \leq 1$. Then $\gamma'(t) = 1 + i$, and

$$\int_\gamma z\,dz = \int_0^1 t(1+i)(1+i)\,dt = \frac{(1+i)^2}{2} = i.$$

The calculation makes the definition operational: substitute the parametrization and multiply by its complex velocity.

4.2 Contour integral

Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$.

4.3 Reparameterization

Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign.

4.4 ML estimate

$|\int_\gamma f dz| \leq L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound.

Example 4.2 (A circle integral with orientation)**.** For the positively oriented circle $\gamma(t) = Re^{it}$, $0 \leq t \leq 2\pi$,

$$\int_\gamma \overline{z}\, dz = \int_0^{2\pi} Re^{-it}\, iRe^{it}\, dt = 2\pi i R^2.$$

Reversing the orientation changes the sign. The example also shows that closed contour integrals need not vanish for nonholomorphic integrands.

4.5 Primitives

If $F' = f$, then contour integrals of f depend only on endpoints.

4.6 Closed curves

Functions with primitives integrate to zero around every closed curve.

4.7 Polygonal and circular contours

Line segments, circles, and piecewise polygonal paths provide standard computational contours.

Example 4.3 (A primitive makes the path irrelevant)**.** For $f(z) = 3z^2$, a primitive is $F(z) = z^3$. Therefore every piecewise smooth path from $z = -1$ to $z = 2 + i$ has

$$\int_\gamma 3z^2\, dz = F(2 + i) - F(-1) = (2 + i)^3 + 1.$$

No parametrization of the chosen path is needed once a primitive is known.

4.8 Deformation preview

When holomorphicity holds in the intervening region, later theorems will make integrals invariant under suitable contour deformations.

4.9 Parameter singularities

Care is required when curves pass through poles, branch cuts, or points where the integrand is not defined.

4.10 Core structural results

Theorem 4.4 (Reparameterization invariance)**.** *An orientation-preserving C^1 change of parameter leaves the contour integral unchanged.*

Proof. Substitute the new parameter into the real integral and use the chain rule and ordinary change-of-variables formula. ☐

Theorem 4.5 (ML inequality)**.** *If f is continuous on a rectifiable curve γ, then $|\int_\gamma f dz| \leq L(\gamma) \max_\gamma |f|$.*

Proof. Take absolute values in the parametrized integral and bound $|f(\gamma(t))|$ by its maximum; the remaining integral of $|\gamma'|$ is the curve length. ☐

Theorem 4.6 (Fundamental theorem for contour integrals)**.** *If $F' = f$ on a domain containing γ, then $\int_\gamma f dz = F(\gamma(b)) - F(\gamma(a))$.*

Proof. Differentiate $F(\gamma(t))$ by the chain rule and integrate the resulting ordinary derivative over the parameter interval. ☐

Theorem 4.7 (Primitive criterion on connected domains)**.** *If every closed contour integral of continuous f vanishes in a connected open set, then f has a primitive.*

Proof. Fix a base point and define $F(z)$ by integrating f along any path from the base point to z; closed-integral vanishing gives path independence, and a short terminal segment proves $F' = f$. ☐

4.11 Worked examples

Example 4.8 (Parametrized curves diagnostic)**.** Give a rigorous worked analysis of Parametrized curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.9 (Contour integral diagnostic)**.** Give a rigorous worked analysis of Contour integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.10 (Reparameterization diagnostic)**.** Give a rigorous worked analysis of Reparameterization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.11 (ML estimate diagnostic)**.** Give a rigorous worked analysis of ML estimate. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. $|\int_\gamma f dz| \leq L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

4.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 4.12 (Parametrized curves diagnostic)**.** Give a rigorous worked analysis of Parametrized curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.13 (Contour integral diagnostic)**.** Give a rigorous worked analysis of Contour integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.14 (Reparameterization diagnostic)**.** Give a rigorous worked analysis of Reparameterization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.15 (ML estimate diagnostic)**.** Give a rigorous worked analysis of ML estimate. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $|\int_\gamma f dz| \le L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.16 (Primitives diagnostic)**.** Give a rigorous worked analysis of Primitives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If $F' = f$, then contour integrals of f depend only on endpoints. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.17 (Closed curves diagnostic)**.** Give a rigorous worked analysis of Closed curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functions with primitives integrate to zero around every closed curve. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.18 (Polygonal and circular contours diagnostic)**.** Give a rigorous worked analysis of Polygonal and circular contours. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Line segments, circles, and piecewise polygonal paths provide standard computational contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.19 (Reparameterization invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: An orientation-preserving C^1 change of parameter leaves the contour integral unchanged.

Solution. Substitute the new parameter into the real integral and use the chain rule and ordinary change-of-variables formula. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.20 (ML inequality proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is continuous on a rectifiable curve γ, then $|\int_\gamma f dz| \le L(\gamma)\max_\gamma |f|$.

Solution. Take absolute values in the parametrized integral and bound $|f(\gamma(t))|$ by its maximum; the remaining integral of $|\gamma'|$ is the curve length. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.21 (Fundamental theorem for contour integrals proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $F' = f$ on a domain containing γ, then $\int_\gamma f dz = F(\gamma(b)) - F(\gamma(a))$.

Solution. Differentiate $F(\gamma(t))$ by the chain rule and integrate the resulting ordinary derivative over the parameter interval. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.22 (Primitive criterion on connected domains proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If every closed contour integral of continuous f vanishes in a connected open set, then f has a primitive.

Solution. Fix a base point and define $F(z)$ by integrating f along any path from the base point to z; closed-integral vanishing gives path independence, and a short terminal segment proves $F' = f$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Complex integration integrates a complex-valued function along an oriented curve. Parameterization, estimates, primitives, and deformation ideas prepare the way for Cauchy's theorem. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

4.13 Exercises with complete solutions

Exercise 4.24. State and apply the central principle behind Parametrized curves. Give one concrete consequence.

Hint. Parameterize each smooth contour segment $z = \gamma(t)$ and convert $\int_\Gamma f(z)dz$ to $\int f(\gamma(t))\gamma'(t)dt$. ☐

Solution. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.25. State and apply the central principle behind Contour integral. Give one concrete consequence.

Hint. Reverse orientation by replacing t with the reversed parameter; this changes the sign of the contour integral. □

Solution. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.26. State and apply the central principle behind Reparameterization. Give one concrete consequence.

Hint. For concatenated paths, split the parameter interval and add the segment integrals. □

Solution. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.27. State and apply the central principle behind ML estimate. Give one concrete consequence.

Hint. Apply the ML bound $|\int_\Gamma f dz| \leq \max_\Gamma |f| \, L(\Gamma)$ after estimating both maximum and length. □

Solution. $|\int_\gamma f dz| \leq L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.28. State and apply the central principle behind Primitives. Give one concrete consequence.

Hint. If $F' = f$, evaluate by $F(\gamma(b)) - F(\gamma(a))$; this is the quickest path-independence test. □

Solution. If $F' = f$, then contour integrals of f depend only on endpoints. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.29. State and apply the central principle behind Closed curves. Give one concrete consequence.

Hint. To disprove path independence, compare integrals around two paths whose difference forms a closed loop enclosing a singularity. □

Solution. Functions with primitives integrate to zero around every closed curve. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.30. State and apply the central principle behind Polygonal and circular contours. Give one concrete consequence.

Hint. Choose contours so that the integrand or parameterization simplifies on each segment rather than forcing one formula globally. □

Solution. Line segments, circles, and piecewise polygonal paths provide standard computational contours. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 4.31. State and apply the central principle behind Deformation preview. Give one concrete consequence.

Hint. Before any contour deformation, list the singularities and make sure none are crossed. ☐

Solution. When holomorphicity holds in the intervening region, later theorems will make integrals invariant under suitable contour deformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

4.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 4.32 (Straight segment)**.** Evaluate $\int_\gamma z\,dz$ for $\gamma(t) = t$, $0 \le t \le 2$.

Hint. Substitute the parametrization. ☐

Solution. The integral is $\int_0^2 t\,dt = 2$. ☐

Exercise 4.33 (Quarter circle)**.** Parametrize the counterclockwise quarter circle of radius two from 2 to $2i$.

Hint. Use an exponential parametrization. ☐

Solution. A convenient choice is $\gamma(t) = 2e^{it}$ for $0 \le t \le \pi/2$. ☐

Exercise 4.34 (Reverse path)**.** If $\int_\gamma f(z)\,dz = 3 - i$, find the integral over the reversed path.

Hint. Reversal changes the sign of the velocity. ☐

Solution. The value is $-3 + i$. ☐

Exercise 4.35 (Primitive evaluation)**.** Evaluate $\int_\gamma 2z\,dz$ from 1 to i along any path.

Hint. Use the primitive z^2. ☐

Solution. The value is $i^2 - 1^2 = -2$. ☐

Exercise 4.36 (ML bound)**.** A contour has length 5 and $|f| \le 3$ on it. Bound $|\int_\gamma f\,dz|$.

Hint. Use the ML inequality. ☐

Solution. The integral has modulus at most 15. ☐

Proofs

Exercise 4.37 (Reparametrization)**.** Prove invariance of a contour integral under an orientation preserving smooth reparametrization.

Hint. Make a one variable substitution in the parameter integral. □

Solution. If $t = \phi(s)$ with $\phi' > 0$, then $(\gamma \circ \phi)' = \gamma'(\phi)\phi'$. The ordinary substitution formula gives the same integral. □

Exercise 4.38 (Concatenation)**.** Prove that the integral over a concatenated path is the sum of the two integrals.

Hint. Split the parameter interval at the joining time. □

Solution. After a standard reparametrization, the parameter integral splits into two ordinary integrals, one for each component path. □

Exercise 4.39 (ML inequality)**.** Prove $|\int_\gamma f\,dz| \leq ML$ when $|f| \leq M$ and the contour length is L.

Hint. Take absolute values inside the parameter integral. □

Solution. The triangle inequality gives at most $\int M|\gamma'(t)|dt = ML$. □

Exercise 4.40 (Primitive theorem)**.** If $F' = f$, prove $\int_\gamma f\,dz = F(\gamma(b)) - F(\gamma(a))$.

Hint. Differentiate $F(\gamma(t))$. □

Solution. The chain rule gives $(F \circ \gamma)' = f(\gamma(t))\gamma'(t)$. The real fundamental theorem of calculus then gives the endpoint difference. □

Counterexamples and hypothesis tests

Exercise 4.41 (Closed does not imply zero)**.** Give a continuous integrand whose integral around a closed circle is nonzero.

Hint. Use conjugation. □

Solution. For $f(z) = \overline{z}$ on $|z| = R$, direct parametrization gives $2\pi i R^2 \neq 0$. □

Exercise 4.42 (Endpoint data alone)**.** Can endpoint data determine $\int_\gamma f\,dz$ for an arbitrary continuous f?

Hint. Path independence requires additional structure such as a primitive. □

Solution. No. Without a primitive or a relevant holomorphic theorem, different paths with the same endpoints may give different integrals. □

Exercise 4.43 (Orientation forgotten)**.** What error results from reversing a contour but keeping the same integral value?

Hint. Track the sign of the derivative of the reversed parametrization. □

Solution. Reversal multiplies the integral by -1. Forgetting orientation therefore changes the answer by a sign. □

Applications and investigations

Exercise 4.44 (Arc contribution estimate)**.** A semicircular arc has radius R, and $|f(z)| \leq C/R^2$ there. Show its integral tends to zero as $R \to \infty$.

Hint. The arc length is πR. ☐

Solution. The ML bound is $\pi R \cdot C/R^2 = \pi C/R \to 0$. This estimate will be central in real integral contours. ☐

Exercise 4.45 (Path choice)**.** Why is a polygonal parametrization often useful for numerical contour integration?

Hint. Each segment has a constant derivative. ☐

Solution. The integral becomes a sum of ordinary one dimensional integrals on simple intervals, making both quadrature and error control transparent. ☐

Challenge problems

Exercise 4.46 (Unit circle logarithmic integrand)**.** Directly parametrize $\int_{|z|=1} dz/z$.

Hint. Use $z = e^{it}$. ☐

Solution. Then $dz = ie^{it}dt$, so the integral is $\int_0^{2\pi} i\,dt = 2\pi i$. ☐

Exercise 4.47 (Piecewise primitive)**.** Suppose a contour crosses a point where a chosen primitive formula changes branch. Why must the integral be split with care?

Hint. A primitive must be single valued and differentiable on a neighborhood of each relevant path piece. ☐

Solution. One may use valid local primitives on separate pieces, but endpoint cancellations only work after branch values are chosen consistently at the joins. ☐

Chapter 5

Cauchy's Theorem

Cauchy's theorem is the topological heart of elementary complex analysis: holomorphic functions have zero integral around contractible closed contours. This creates path independence, primitives, and deformation invariance.

Learning goals

After this chapter, the reader should be able to:

- verify the hypotheses of Cauchy's theorem on simply connected or triangulated domains;
- deform contours through holomorphic regions without crossing singularities;
- prove vanishing of closed contour integrals from holomorphicity and topology;
- identify when multiply connected geometry prevents a direct use of Cauchy's theorem;
- construct primitives from path integrals when closed integrals vanish;
- analyze counterexamples showing why a puncture or singularity changes the conclusion;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

5.1 Triangle theorem

The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error.

5.2 Polygonal domains

Triangulation extends the result to polygonal contours.

Example 5.1 (Triangle cancellation)**.** Suppose f is holomorphic on a neighborhood of a triangle and its interior. Cauchy's theorem gives

$$\int_{\partial T} f(z)\,dz = 0.$$

For a polynomial such as $f(z) = z^3 + 2z$, this can also be checked from the primitive $z^4/4 + z^2$. The theorem is stronger because it applies even when a primitive has not yet been constructed explicitly.

5.3 Simply connected domains

On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish.

5.4 Homotopy invariance

Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies.

5.5 Existence of primitives

Zero integrals on closed contours imply primitives.

Example 5.2 (Deforming a contour without crossing singularities)**.** Let $f(z) = 1/(z-3)$, and let γ_0 and γ_1 be two closed curves lying in the disk $|z| < 2$. Since the pole at 3 lies outside the disk and the curves can be deformed into one another there, Cauchy's theorem gives equal integrals. In fact each integral is zero because f has a primitive on the simply connected disk.

5.6 Goursat refinement

Continuity of the derivative is not required; complex differentiability alone suffices.

5.7 Multiply connected warning

Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain.

5.8 Topological content

Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities.

Example 5.3 (The puncture blocks Cauchy's theorem)**.** On $\mathbb{C} \setminus \{0\}$, the function $1/z$ is holomorphic, yet

$$\int_{|z|=1} \frac{dz}{z} = 2\pi i.$$

There is no contradiction: the unit circle does not bound a region contained in the domain. The missing point is a topological obstruction, and the example explains why domain hypotheses matter as much as pointwise holomorphicity.

5.9 Core structural results

Theorem 5.4 (Cauchy–Goursat theorem on triangles)**.** *If f is holomorphic on a neighborhood of a triangle and its interior, then the integral around the triangle is zero.*

Proof. Repeatedly subdivide into four similar triangles and choose one carrying at least a quarter of the integral magnitude. Diameters shrink to a point where differentiability gives a linear approximation whose constant and linear parts integrate to zero; the remainder estimate forces the original integral to vanish. □

Theorem 5.5 (Primitive on simply connected domains)**.** *Every holomorphic function on a simply connected domain has a primitive.*

Proof. Cauchy's theorem makes integrals around closed contours vanish, so path integration from a base point is well-defined and differentiates back to the integrand. □

Theorem 5.6 (Homotopy invariance)**.** *If two closed contours are homotopic through contours inside a domain where f is holomorphic, their integrals agree.*

Proof. Partition the homotopy strip into small cells whose boundary integrals vanish by local Cauchy theory; interior edges cancel, leaving the difference of the two boundary contour integrals. □

5.10 Worked examples

Example 5.7 (Triangle theorem diagnostic)**.** Give a rigorous worked analysis of Triangle theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.8 (Polygonal domains diagnostic)**.** Give a rigorous worked analysis of Polygonal domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Triangulation extends the result to polygonal contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.9 (Simply connected domains diagnostic)**.** Give a rigorous worked analysis of Simply connected domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.10 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

5.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 5.11 (Triangle theorem diagnostic)**.** Give a rigorous worked analysis of Triangle theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.12 (Polygonal domains diagnostic)**.** Give a rigorous worked analysis of Polygonal domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Triangulation extends the result to polygonal contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.13 (Simply connected domains diagnostic)**.** Give a rigorous worked analysis of Simply connected domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.14 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.15 (Existence of primitives diagnostic)**.** Give a rigorous worked analysis of Existence of primitives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Zero integrals on closed contours imply primitives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.16 (Goursat refinement diagnostic)**.** Give a rigorous worked analysis of Goursat refinement. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Continuity of the derivative is not required; complex differentiability alone suffices. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.17 (Multiply connected warning diagnostic)**.** Give a rigorous worked analysis of Multiply connected warning. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.18 (Topological content diagnostic)**.** Give a rigorous worked analysis of Topological content. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 5.19 (Cauchy–Goursat theorem on triangles proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic on a neighborhood of a triangle and its interior, then the integral around the triangle is zero.

Solution. Repeatedly subdivide into four similar triangles and choose one carrying at least a quarter of the integral magnitude. Diameters shrink to a point where differentiability gives a linear approximation whose constant and linear parts integrate to zero; the remainder estimate forces the original integral to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 5.20 (Primitive on simply connected domains proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every holomorphic function on a simply connected domain has a primitive.

Solution. Cauchy's theorem makes integrals around closed contours vanish, so path integration from a base point is well-defined and differentiates back to the integrand. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 5.21 (Homotopy invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two closed contours are homotopic through contours inside a domain where f is holomorphic, their integrals agree.

Solution. Partition the homotopy strip into small cells whose boundary integrals vanish by local Cauchy theory; interior edges cancel, leaving the difference of the two boundary contour integrals. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 5.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Cauchy's theorem is the topological heart of elementary complex analysis: holomorphic functions have zero integral around contractible closed contours. This creates path independence, primitives, and deformation invariance. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

5.12 Exercises with complete solutions

Exercise 5.23. State and apply the central principle behind Triangle theorem. Give one concrete consequence.

Hint. Check holomorphicity on an open set containing the contour and its interior before invoking Cauchy's theorem. □

Solution. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.24. State and apply the central principle behind Polygonal domains. Give one concrete consequence.

Hint. During contour deformation, verify the swept region is free of singularities; homotopy through a puncture is not allowed. ☐

Solution. Triangulation extends the result to polygonal contours. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.25. State and apply the central principle behind Simply connected domains. Give one concrete consequence.

Hint. Triangulate or decompose the region so interior edge integrals cancel pairwise. ☐

Solution. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.26. State and apply the central principle behind Homotopy invariance. Give one concrete consequence.

Hint. In a multiply connected region, inspect each hole; closed contour integrals need not vanish around omitted points. ☐

Solution. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.27. State and apply the central principle behind Existence of primitives. Give one concrete consequence.

Hint. Define a candidate primitive by a path integral from a base point and use vanishing of closed integrals to prove path independence. ☐

Solution. Zero integrals on closed contours imply primitives. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.28. State and apply the central principle behind Goursat refinement. Give one concrete consequence.

Hint. For a punctured-domain counterexample, integrate $1/z$ around a loop with nonzero winding number. ☐

Solution. Continuity of the derivative is not required; complex differentiability alone suffices. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.29. State and apply the central principle behind Multiply connected warning. Give one concrete consequence.

Hint. Use local primitives on small disks and patch them only when the overlap integrals are consistent. ☐

Solution. Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.30. State and apply the central principle behind Topological content. Give one concrete consequence.

Hint. Separate topology from regularity: holomorphicity alone does not guarantee every closed integral vanishes on a non-simply-connected domain. ☐

Solution. Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

5.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 5.31 (Polynomial loop)**.** Evaluate $\int_\gamma (z^4 + z)\,dz$ over any closed contour.

Hint. The integrand has an entire primitive. ☐

Solution. The value is zero. ☐

Exercise 5.32 (Disk holomorphic)**.** Let $f(z) = 1/(z-5)$. Evaluate its integral around $|z| = 2$.

Hint. The function is holomorphic on the disk bounded by the contour. ☐

Solution. Cauchy's theorem gives zero. ☐

Exercise 5.33 (Rectangle)**.** Evaluate the integral of e^z around the boundary of a rectangle.

Hint. The exponential is entire. ☐

Solution. The integral is zero. ☐

Exercise 5.34 (Two paths)**.** If f is holomorphic on a simply connected domain, compare integrals along two paths with the same endpoints.

Hint. Join one path to the reverse of the other. ☐

Solution. The resulting closed contour has integral zero, so the two path integrals agree. ☐

Exercise 5.35 (Primitive consequence)**.** What is $\int_\gamma \cos z\,dz$ from 0 to π?

Hint. Use the primitive $\sin z$. ☐

Solution. The value is $\sin\pi - \sin 0 = 0$. ☐

Proofs

Exercise 5.36 (Path independence from closed loops)**.** Prove that vanishing of every closed contour integral implies path independence.

Hint. Concatenate one path with the reverse of the other. ☐

Solution. The closed integral is the first path integral minus the second. If it vanishes, the two are equal. ☐

Exercise 5.37 (Primitive from path independence)**.** On a connected domain with path independent integrals, construct a primitive of f.

Hint. Fix a base point and integrate to z. ☐

Solution. Define $F(z) = \int_{z_*}^{z} f(w)dw$. Path independence makes this well defined. A short final segment and continuity of f show $F'(z) = f(z)$. ☐

Exercise 5.38 (Simply connected implication)**.** Explain why Cauchy's theorem yields primitives on a simply connected domain.

Hint. Closed curves can be contracted without leaving the domain. ☐

Solution. Cauchy's theorem makes every closed contour integral vanish; path independence follows, and the base point integral construction produces a primitive. ☐

Exercise 5.39 (Contour deformation)**.** Prove that two homologous boundary curves have equal integrals when f is holomorphic in the region between them.

Hint. Orient the boundary of the region between the curves. ☐

Solution. The boundary consists of one curve and the reverse of the other. Cauchy's theorem makes the total integral zero, hence the two original integrals are equal. ☐

Counterexamples and hypothesis tests

Exercise 5.40 (Punctured disk)**.** Why can Cauchy's theorem not be used to conclude $\int_{|z|=1} dz/z = 0$?

Hint. Inspect the interior of the contour. ☐

Solution. The interior contains the missing point 0, where the integrand is not holomorphic. The required filled region is not contained in the domain. ☐

Exercise 5.41 (Pole on the boundary)**.** Can the standard Cauchy theorem be applied when the integrand has a pole on the contour?

Hint. Holomorphicity is required on a neighborhood of the contour and its interior. ☐

Solution. No. The contour integral may even be undefined in the ordinary sense, so the hypothesis fails before the theorem is invoked. ☐

Exercise 5.42 (Disconnected domain)**.** Does simple connectedness make sense without connectedness?

Hint. Check the definition of a simply connected domain. ☐

Solution. A domain is connected and open by convention. Statements about simple connectedness presuppose that connected setting. ☐

Applications and investigations

Exercise 5.43 (Contour simplification)**.** Explain how Cauchy's theorem can simplify a difficult path to an easier one.

Hint. Deform the path through a region free of singularities. ☐

Solution. The integral is unchanged under such a deformation. One can therefore replace a complicated contour by a geometrically convenient representative of the same deformation class. ☐

Exercise 5.44 (Detecting holes)**.** How can a nonzero integral of a holomorphic function around a closed contour reveal domain topology?

Hint. Compare with the simply connected conclusion. ☐

Solution. A nonzero closed integral shows that no primitive exists on the whole domain and that the contour cannot be contracted through a region where the integrand remains holomorphic. ☐

Challenge problems

Exercise 5.45 (Morera converse)**.** State the idea behind the converse principle that vanishing triangle integrals can imply holomorphicity.

Hint. Use triangle integrals to construct local primitives. ☐

Solution. If a continuous function has zero integral around every triangle, one constructs a path independent local integral and hence a local primitive. Differentiating that primitive recovers the function, proving holomorphicity. ☐

Exercise 5.46 (Annulus comparison)**.** Two circles $|z| = 1$ and $|z| = 2$ lie in an annulus where f is holomorphic. Show their positively oriented integrals are equal.

Hint. Apply Cauchy's theorem to the region between them with boundary orientation. ☐

Solution. The boundary integral is the outer circle integral minus the inner circle integral. It vanishes, so the two are equal. ☐

Chapter 6

Cauchy's Integral Formula

Cauchy's integral formula reconstructs a holomorphic function from boundary values. Its differentiated forms yield analyticity, derivative estimates, Liouville's theorem, and the mean-value property.

Learning goals

After this chapter, the reader should be able to:

- apply Cauchy's integral formula to evaluate values and derivatives of holomorphic functions;
- derive Cauchy estimates from contour bounds;
- use Cauchy's formula to prove infinite differentiability and analyticity;
- compute contour integrals containing powers of $(z-a)^{-1}$ by matching the correct derivative formula;
- choose a circle contained in the domain to optimize an estimate;
- use Cauchy estimates to prove Liouville-type or derivative-vanishing consequences;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

6.1 Integral formula

Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$.

Example 6.1 (Cauchy formula in one line)**.** For $|z| = 2$, evaluate

$$\int_{|z|=2} \frac{e^z}{z-1}\, dz.$$

The point 1 lies inside the circle and e^z is holomorphic on a neighborhood of the closed disk. Cauchy's integral formula gives the value $2\pi i e$. No parametrization is required.

6.2 Circle form

On circles the formula becomes an explicit average weighted by the Cauchy kernel.

6.3 Derivative formulas

Differentiating under the integral gives formulas for every derivative.

6.4 Cauchy estimates

Derivative formulas bound derivatives by boundary maxima and powers of radius.

Example 6.2 (Derivative extraction)**.** For the positively oriented circle $|z| = 3$,

$$\int_{|z|=3} \frac{\sin z}{(z-1)^3}\, dz = \frac{2\pi i}{2!}(\sin z)''\big|_{z=1} = -\pi i \sin 1.$$

The denominator power determines which derivative is extracted.

6.5 Analyticity

Holomorphic functions are automatically represented by convergent Taylor series.

6.6 Mean-value property

Holomorphic functions and their harmonic components satisfy circle-average identities.

6.7 Liouville theorem

A bounded entire function must be constant.

Example 6.3 (Cauchy estimate)**.** Suppose $|f(z)| \leq 10$ on $|z - z_0| = 2$ and f is holomorphic on the closed disk. The derivative formula yields

$$|f^{(3)}(z_0)| \leq \frac{3!\,10}{2^3} = \frac{15}{2}.$$

This turns boundary control into quantitative bounds on every derivative at the center.

6.8 Fundamental theorem of algebra

Liouville's theorem yields a short proof that every nonconstant complex polynomial has a root.

6.9 Core structural results

Theorem 6.4 (Cauchy integral formula)**.** *If f is holomorphic on and inside a positively oriented simple closed contour Γ and z lies inside, then $f(z) = \frac{1}{2\pi i}\int_\Gamma \frac{f(\zeta)}{\zeta - z} d\zeta$.*

Proof. Subtract $f(z)$ from the numerator so the apparent singularity becomes removable, apply Cauchy's theorem to the regular part, and compute the integral of $(\zeta - z)^{-1}$ around the contour as $2\pi i$. $\square$

Theorem 6.5 (Cauchy derivative formula)**.** $f^{(n)}(z) = \frac{n!}{2\pi i} \int_\Gamma \frac{f(\zeta)}{(\zeta-z)^{n+1}} d\zeta.$

Proof. Differentiate the Cauchy formula with respect to z; uniform separation of z from the contour justifies differentiation under the integral. ☐

Theorem 6.6 (Cauchy estimate)**.** *If* $|f| \le M$ *on* $|\zeta - z_0| = R$*, then* $|f^{(n)}(z_0)| \le n!M/R^n$*.*

Proof. Apply the derivative formula on the circle and use the ML inequality with length $2\pi R$ and denominator magnitude R^{n+1}. ☐

Theorem 6.7 (Liouville theorem)**.** *Every bounded entire function is constant.*

Proof. Apply the Cauchy estimate for the first derivative on circles of radius R and let $R \to \infty$, forcing $f' = 0$. ☐

6.10 Worked examples

Example 6.8 (Integral formula diagnostic)**.** Give a rigorous worked analysis of Integral formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.9 (Circle form diagnostic)**.** Give a rigorous worked analysis of Circle form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.10 (Derivative formulas diagnostic)**.** Give a rigorous worked analysis of Derivative formulas. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Differentiating under the integral gives formulas for every derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.11 (Cauchy estimates diagnostic)**.** Give a rigorous worked analysis of Cauchy estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Derivative formulas bound derivatives by boundary maxima and powers of radius. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

6.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 6.12 (Integral formula diagnostic)**.** Give a rigorous worked analysis of Integral formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 6.13 (Circle form diagnostic)**.** Give a rigorous worked analysis of Circle form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.14 (Derivative formulas diagnostic)**.** Give a rigorous worked analysis of Derivative formulas. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiating under the integral gives formulas for every derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.15 (Cauchy estimates diagnostic)**.** Give a rigorous worked analysis of Cauchy estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Derivative formulas bound derivatives by boundary maxima and powers of radius. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.16 (Analyticity diagnostic)**.** Give a rigorous worked analysis of Analyticity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic functions are automatically represented by convergent Taylor series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.17 (Mean-value property diagnostic)**.** Give a rigorous worked analysis of Mean-value property. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic functions and their harmonic components satisfy circle-average identities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.18 (Liouville theorem diagnostic)**.** Give a rigorous worked analysis of Liouville theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A bounded entire function must be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.19 (Cauchy integral formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic on and inside a positively oriented simple closed contour Γ and z lies inside, then $f(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - z} d\zeta$.

Solution. Subtract $f(z)$ from the numerator so the apparent singularity becomes removable, apply Cauchy's theorem to the regular part, and compute the integral of $(\zeta - z)^{-1}$ around the contour as $2\pi i$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 6.20 (Cauchy derivative formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $f^{(n)}(z) = \frac{n!}{2\pi i}\int_\Gamma \frac{f(\zeta)}{(\zeta-z)^{n+1}}d\zeta$.

Solution. Differentiate the Cauchy formula with respect to z; uniform separation of z from the contour justifies differentiation under the integral. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 6.21 (Cauchy estimate proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $|f| \le M$ on $|\zeta - z_0| = R$, then $|f^{(n)}(z_0)| \le n!M/R^n$.

Solution. Apply the derivative formula on the circle and use the ML inequality with length $2\pi R$ and denominator magnitude R^{n+1}. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 6.22 (Liouville theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every bounded entire function is constant.

Solution. Apply the Cauchy estimate for the first derivative on circles of radius R and let $R \to \infty$, forcing $f' = 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 6.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Cauchy's integral formula reconstructs a holomorphic function from boundary values. Its differentiated forms yield analyticity, derivative estimates, Liouville's theorem, and the mean-value property. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

6.12 Exercises with complete solutions

Exercise 6.24. State and apply the central principle behind Integral formula. Give one concrete consequence.

Hint. Match the integrand to $f(z)/(z - a)$ and check that a lies inside the positively oriented contour. ☐

Solution. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 6.25. State and apply the central principle behind Circle form. Give one concrete consequence.

Hint. For higher derivatives, rewrite the denominator as $(z-a)^{n+1}$ and use $2\pi i f^{(n)}(a)/n!$. ☐

Solution. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 6.26. State and apply the central principle behind Derivative formulas. Give one concrete consequence.

Hint. Choose a circle centered at a contained in the domain, then bound the Cauchy integral by its maximum modulus. □

Solution. Differentiating under the integral gives formulas for every derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.27. State and apply the central principle behind Cauchy estimates. Give one concrete consequence.

Hint. Cauchy's estimate gives $|f^{(n)}(a)| \leq n!M/R^n$; select R to make the bound useful. □

Solution. Derivative formulas bound derivatives by boundary maxima and powers of radius. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.28. State and apply the central principle behind Analyticity. Give one concrete consequence.

Hint. To prove analyticity, expand $1/(z-w)$ geometrically for $|w-a| < |z-a|$ inside the Cauchy formula. □

Solution. Holomorphic functions are automatically represented by convergent Taylor series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.29. State and apply the central principle behind Mean-value property. Give one concrete consequence.

Hint. For Liouville-type reasoning, let the radius in a Cauchy estimate tend to infinity while the global bound remains fixed. □

Solution. Holomorphic functions and their harmonic components satisfy circle-average identities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.30. State and apply the central principle behind Liouville theorem. Give one concrete consequence.

Hint. If the contour does not wind once around a, include the winding number rather than using the simple formula blindly. □

Solution. A bounded entire function must be constant. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.31. State and apply the central principle behind Fundamental theorem of algebra. Give one concrete consequence.

Hint. Audit whether singularities of the auxiliary function lie inside the chosen contour before differentiating under the formula. □

Solution. Liouville's theorem yields a short proof that every nonconstant complex polynomial has a root. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

6.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 6.32 (Basic Cauchy value)**.** Evaluate $\int_{|z|=2}(z^2+1)/(z-i)\,dz$.

Hint. The point i lies inside the circle. ☐

Solution. The value is $2\pi i(i^2+1)=0$. ☐

Exercise 6.33 (Outside point)**.** Evaluate $\int_{|z|=1} e^z/(z-2)\,dz$.

Hint. The integrand is holomorphic on the closed unit disk. ☐

Solution. The value is zero by Cauchy's theorem. ☐

Exercise 6.34 (First derivative)**.** Evaluate $\int_{|z|=2} e^z/(z-1)^2\,dz$.

Hint. Use the first derivative form of Cauchy's formula. ☐

Solution. The result is $2\pi ie$. ☐

Exercise 6.35 (Second derivative)**.** Evaluate $\int_{|z|=2} z^5/(z-1)^3\,dz$.

Hint. Extract the second derivative and divide by $2!$. ☐

Solution. Since $(z^5)''|_1=20$, the integral is $2\pi i\cdot 20/2=20\pi i$. ☐

Exercise 6.36 (Derivative bound)**.** If $|f|\le 4$ on $|z|=3$, bound $|f''(0)|$.

Hint. Use the Cauchy estimate $n!M/R^n$. ☐

Solution. The bound is $2!\cdot 4/3^2=8/9$. ☐

Proofs

Exercise 6.37 (Higher derivative formula)**.** Derive the higher derivative Cauchy formula from the basic formula.

Hint. Differentiate the kernel with respect to the interior point. ☐

Solution. Uniform control on a smaller disk justifies differentiation under the contour integral. Repeating gives $f^{(n)}(a)=n!/(2\pi i)\int f(z)/(z-a)^{n+1}dz$. ☐

Exercise 6.38 (Cauchy estimates)**.** Prove the standard Cauchy derivative estimate.

Hint. Take moduli in the higher derivative formula on a circle of radius R. ☐

Solution. The kernel contributes $R^{-(n+1)}$, the contour length contributes $2\pi R$, and the prefactor gives $|f^{(n)}(a)|\le n!M/R^n$. ☐

Exercise 6.39 (Liouville theorem)**.** Use Cauchy estimates to prove that every bounded entire function is constant.

Hint. Let the radius tend to infinity in the first derivative estimate. ☐

Solution. If $|f| \leq M$ globally, then $|f'(a)| \leq M/R$ for every R. Letting $R \to \infty$ gives $f'(a) = 0$ for every a, so f is constant. ☐

Exercise 6.40 (Mean value identity)**.** Derive the mean value formula for a holomorphic function on a circle.

Hint. Parametrize the Cauchy formula on $z = a + Re^{it}$. ☐

Solution. The factors Re^{it} cancel, leaving $f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + Re^{it})dt$. ☐

Counterexamples and hypothesis tests

Exercise 6.41 (Singularity inside)**.** Can Cauchy's formula be applied with $f(z) = 1/z$ and center $a = 1$ on a circle enclosing zero?

Hint. The numerator function in the formula must be holomorphic throughout the interior. ☐

Solution. Not directly if the chosen circle encloses the pole at zero. One must isolate singularities or use later residue methods. ☐

Exercise 6.42 (Point on contour)**.** What fails if the evaluation point a lies on the contour in Cauchy's formula?

Hint. The kernel has a singularity on the path. ☐

Solution. The ordinary contour integral is not covered by the formula and may be undefined. The point must lie strictly inside the contour. ☐

Exercise 6.43 (Wrong orientation)**.** How does clockwise orientation change the Cauchy formula value?

Hint. Reverse the contour orientation. ☐

Solution. The integral changes sign, so the usual $2\pi i f(a)$ becomes $-2\pi i f(a)$. ☐

Applications and investigations

Exercise 6.44 (Coefficient recovery)**.** Explain how Cauchy's formula recovers Taylor coefficients from boundary data.

Hint. Combine the higher derivative formula with $a_n = f^{(n)}(a)/n!$. ☐

Solution. One gets $a_n = (2\pi i)^{-1} \int f(z)/(z-a)^{n+1}dz$. Thus each local coefficient is encoded by the function on any surrounding contour inside the analytic domain. ☐

Exercise 6.45 (Maximum growth constraint)**.** Why do Cauchy estimates make very slow growth restrictive for entire functions?

Hint. Choose a large circle and compare growth with the power of its radius in the denominator. ☐

Solution. If the maximum modulus grows more slowly than R^n, the estimate can force the n-th derivative to vanish. Repeating such bounds yields polynomial or constant rigidity results. ☐

Challenge problems

Exercise 6.46 (Fundamental theorem of algebra)**.** Use Liouville's theorem to outline a proof that a nonconstant complex polynomial has a zero.

Hint. Assume the polynomial has no zeros and examine its reciprocal. □

Solution. If p has no zeros, $1/p$ is entire. It tends to zero at infinity and is bounded on a large disk, so it is bounded everywhere. Liouville makes it constant, contradicting nonconstancy of p. □

Exercise 6.47 (All derivatives from one contour)**.** Explain why knowing a holomorphic function on one circle determines every derivative at its center.

Hint. Use the family of higher derivative formulas. □

Solution. For every n, the same boundary values enter $f^{(n)}(a) = n!/(2\pi i) \int f(z)/(z-a)^{n+1} dz$. Thus one contour determines the full Taylor germ at the center. □

Part II

Singularities and Residues

Chapter 7

Zeros and the Identity Theorem

Zeros of holomorphic functions are isolated unless the function vanishes identically. This local factorization yields the identity theorem, uniqueness from accumulation sets, and multiplicity as the correct notion of zero order.

Learning goals

After this chapter, the reader should be able to:

- factor zeros of holomorphic functions by their multiplicity;
- apply the identity theorem after verifying an accumulation point lies inside the domain;
- distinguish isolated zeros from identically vanishing functions;
- use local power series to compute zero multiplicity;
- prove uniqueness of analytic continuation on connected domains;
- construct examples showing why boundary accumulation of zeros does not trigger the identity theorem;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

7.1 Zeros and multiplicity

If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero.

7.2 Local factorization

Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0.

Example 7.1 (Multiplicity by factorization)**.** For $f(z) = z^3(z-2)^2e^z$, the zero at 0 has multiplicity three because $e^z(z-2)^2$ is holomorphic and nonzero at 0. Likewise the zero at 2 has multiplicity two. Writing a function as $(z-a)^m g(z)$ with $g(a) \neq 0$ isolates the exact order of vanishing.

7.3 Isolated zeros

A nonzero holomorphic function has isolated zeros.

7.4 Identity theorem

Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain.

7.5 Uniqueness of analytic continuation

A holomorphic continuation, when it exists on a connected overlap, is unique.

Example 7.2 (An accumulation point forces identity)**.** Let f be holomorphic on the unit disk and suppose $f(1/n) = 0$ for every positive integer n. The zeros accumulate at 0, which lies inside the domain. The identity theorem therefore gives $f \equiv 0$ on the entire connected disk. Merely having infinitely many zeros would not suffice if their accumulation occurred only at the boundary.

7.6 Zero sets

Nontrivial holomorphic zero sets in one complex variable are discrete.

7.7 Multiplicity under products

Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller.

7.8 Applications

Functional identities, symmetry extensions, and polynomial root counts all use the identity principle.

Example 7.3 (Boundary accumulation is different)**.** The function $\sin(1/(1-z))$ is holomorphic on the unit disk and has infinitely many zeros accumulating at the boundary point $z = 1$. It is not identically zero. This example isolates the interior accumulation hypothesis in the identity theorem.

7.9 Core structural results

Theorem 7.4 (Local zero factorization)**.** *If f is holomorphic near z_0, not identically zero, and $f(z_0) = 0$, then $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$ for a unique integer $m \geq 1$.*

Proof. Take the first nonzero Taylor coefficient a_m and factor $(z - z_0)^m$ from the Taylor series; the remaining power series defines g and has nonzero value a_m at the center. □

Theorem 7.5 (Isolated zeros)**.** *Zeros of a nonzero holomorphic function are isolated.*

Proof. The local factorization has nonvanishing factor g near z_0, so the only nearby zero is the explicit factor z_0. □

Theorem 7.6 (Identity theorem)**.** *If two holomorphic functions on a connected domain agree on a set with an accumulation point in the domain, then they agree everywhere.*

Proof. Their difference is holomorphic and has a nonisolated zero. The isolated-zero theorem forces the difference to vanish identically on the connected domain. □

7.10 Worked examples

Example 7.7 (Zeros and multiplicity diagnostic)**.** Give a rigorous worked analysis of Zeros and multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.8 (Local factorization diagnostic)**.** Give a rigorous worked analysis of Local factorization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.9 (Isolated zeros diagnostic)**.** Give a rigorous worked analysis of Isolated zeros. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A nonzero holomorphic function has isolated zeros. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.10 (Identity theorem diagnostic)**.** Give a rigorous worked analysis of Identity theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

7.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 7.11 (Zeros and multiplicity diagnostic)**.** Give a rigorous worked analysis of Zeros and multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.12 (Local factorization diagnostic)**.** Give a rigorous worked analysis of Local factorization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.13 (Isolated zeros diagnostic)**.** Give a rigorous worked analysis of Isolated zeros. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A nonzero holomorphic function has isolated zeros. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.14 (Identity theorem diagnostic)**.** Give a rigorous worked analysis of Identity theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.15 (Uniqueness of analytic continuation diagnostic)**.** Give a rigorous worked analysis of Uniqueness of analytic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic continuation, when it exists on a connected overlap, is unique. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.16 (Zero sets diagnostic)**.** Give a rigorous worked analysis of Zero sets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Nontrivial holomorphic zero sets in one complex variable are discrete. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.17 (Multiplicity under products diagnostic)**.** Give a rigorous worked analysis of Multiplicity under products. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.18 (Applications diagnostic)**.** Give a rigorous worked analysis of Applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functional identities, symmetry extensions, and polynomial root counts all use the identity principle. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.19 (Local zero factorization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic near z_0, not identically zero, and $f(z_0) = 0$, then $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$ for a unique integer $m \geq 1$.

Solution. Take the first nonzero Taylor coefficient a_m and factor $(z - z_0)^m$ from the Taylor series; the remaining power series defines g and has nonzero value a_m at the center. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 7.20 (Isolated zeros proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Zeros of a nonzero holomorphic function are isolated.

Solution. The local factorization has nonvanishing factor g near z_0, so the only nearby zero is the explicit factor z_0. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 7.21 (Identity theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two holomorphic functions on a connected domain agree on a set with an accumulation point in the domain, then they agree everywhere.

Solution. Their difference is holomorphic and has a nonisolated zero. The isolated-zero theorem forces the difference to vanish identically on the connected domain. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 7.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Zeros of holomorphic functions are isolated unless the function vanishes identically. This local factorization yields the identity theorem, uniqueness from accumulation sets, and multiplicity as the correct notion of zero order. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

7.12 Exercises with complete solutions

Exercise 7.23. State and apply the central principle behind Zeros and multiplicity. Give one concrete consequence.

Hint. Expand near the zero and factor out the first nonzero Taylor term $(z-a)^m$; the exponent m is the multiplicity. ☐

Solution. If $f(z_0)=0$, the smallest nonzero Taylor coefficient determines the order of the zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 7.24. State and apply the central principle behind Local factorization. Give one concrete consequence.

Hint. If zeros accumulate at an interior point, apply the local power series there; every coefficient must vanish. ☐

Solution. Near a zero of order m, write $f(z)=(z-z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 7.25. State and apply the central principle behind Isolated zeros. Give one concrete consequence.

Hint. The identity theorem requires the accumulation point to lie in the connected domain, not merely on its boundary. ☐

Solution. A nonzero holomorphic function has isolated zeros. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 7.26. State and apply the central principle behind Identity theorem. Give one concrete consequence.

Hint. To compare two holomorphic functions, apply the zero theorem to their difference. □

Solution. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.27. State and apply the central principle behind Uniqueness of analytic continuation. Give one concrete consequence.

Hint. A nonzero holomorphic function has isolated zeros; choose a disk containing no other zeros after factoring the local power. □

Solution. A holomorphic continuation, when it exists on a connected overlap, is unique. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.28. State and apply the central principle behind Zero sets. Give one concrete consequence.

Hint. Analytic continuation is unique because two continuations agree on an overlap with an interior accumulation set. □

Solution. Nontrivial holomorphic zero sets in one complex variable are discrete. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.29. State and apply the central principle behind Multiplicity under products. Give one concrete consequence.

Hint. Boundary accumulation gives no identity-theorem conclusion; use an example such as zeros approaching a boundary point. □

Solution. Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.30. State and apply the central principle behind Applications. Give one concrete consequence.

Hint. Track connected components: equality on one component does not automatically propagate to a different component. □

Solution. Functional identities, symmetry extensions, and polynomial root counts all use the identity principle. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

7.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 7.31 (Order of a polynomial zero)**.** Find the multiplicity of $z = 1$ for $f(z) = (z-1)^4(z+2)$.

Hint. Check whether the remaining factor vanishes at one. □

Solution. The remaining factor equals three, so the multiplicity is four. □

Exercise 7.32 (Derivative criterion)**.** If $f(a) = f'(a) = 0$ and $f''(a) \neq 0$, what is the multiplicity of the zero?

Hint. Use the Taylor expansion. □

Solution. The first nonzero Taylor term has degree two, so the multiplicity is two. □

Exercise 7.33 (Zeros of sine)**.** List the zeros of $\sin z$ and their multiplicities.

Hint. Use the ordinary sine zeros and evaluate the derivative there. □

Solution. The zeros are $n\pi$, $n \in \mathbb{Z}$. Since $\cos(n\pi) \neq 0$, each is simple. □

Exercise 7.34 (Product multiplicity)**.** If f has order three at a and g has order two there, find the order of fg.

Hint. Factor both functions. □

Solution. The product contains $(z-a)^5$ times a nonvanishing factor, so the order is five. □

Exercise 7.35 (Quotient order)**.** If f and g have zero orders five and two at a, what is the zero order of f/g after cancellation?

Hint. Subtract the orders. □

Solution. Provided the quotient is interpreted after local cancellation, it has a zero of order three. □

Proofs

Exercise 7.36 (Zeros are isolated)**.** Prove that a nonzero holomorphic function has isolated zeros.

Hint. Use the first nonzero Taylor coefficient at a zero. □

Solution. Factor $f(z) = (z-a)^m g(z)$ with $g(a) \neq 0$. Continuity keeps g nonzero on a small disk, so a is the only zero there. □

Exercise 7.37 (Identity theorem)**.** Prove that if zeros of a holomorphic function accumulate at an interior point, the function is identically zero on the connected domain.

Hint. First show all Taylor coefficients vanish at the accumulation point. □

Solution. If a first nonzero Taylor coefficient existed, the zero would be isolated. Thus the function vanishes on a disk. The set where it locally vanishes is both open and closed in the connected domain, so it is the whole domain. ☐

Exercise 7.38 (Agreement on a set)**.** Show that two holomorphic functions agreeing on a set with an interior accumulation point agree everywhere on a connected domain.

Hint. Apply the identity theorem to their difference. ☐

Solution. The difference is holomorphic and has zeros with the same accumulation point, hence vanishes identically. ☐

Exercise 7.39 (Order from derivatives)**.** Prove that a zero has multiplicity m exactly when the first $m-1$ derivatives vanish and the m-th does not.

Hint. Compare with the Taylor expansion. ☐

Solution. The least index with nonzero Taylor coefficient is exactly the least derivative order with nonzero value, and the coefficient is $f^{(m)}(a)/m!$. ☐

Counterexamples and hypothesis tests

Exercise 7.40 (Infinitely many zeros)**.** Is an infinite zero set enough to conclude that a holomorphic function is zero?

Hint. Ask where the zeros accumulate. ☐

Solution. No. The zeros of $\sin z$ are infinite but have no finite accumulation point. An interior accumulation point is essential. ☐

Exercise 7.41 (Disconnected domain)**.** If a holomorphic function vanishes on one component of a disconnected open set, must it vanish on every component?

Hint. The identity theorem propagates only through connectedness. ☐

Solution. No. The function can be defined differently on separate components while remaining holomorphic on each. ☐

Exercise 7.42 (Boundary accumulation)**.** Why do zeros tending to a boundary point not trigger the identity theorem?

Hint. The accumulation point must belong to the domain. ☐

Solution. Local Taylor analysis is unavailable at a boundary point not in the domain, so the theorem's key hypothesis fails. ☐

Applications and investigations

Exercise 7.43 (Uniqueness of analytic interpolation)**.** Two holomorphic models agree at a sequence of sample points converging to an interior point. What follows?

Hint. Apply the identity theorem to the difference. ☐

Solution. They agree throughout the connected domain. Thus sufficiently accumulating exact analytic data determine the model uniquely. ☐

Exercise 7.44 (Root stability locally)**.** Why does factoring a zero help study nearby perturbations?

Hint. Separate the vanishing factor from the nonzero analytic factor. ☐

Solution. The factorization identifies multiplicity and isolates a disk where no other zeros occur. Later contour arguments can then track how many perturbed zeros remain in that disk. ☐

Challenge problems

Exercise 7.45 (Even order and local square roots)**.** Suppose every zero of a holomorphic function in a simply connected neighborhood has even order. Explain locally why a holomorphic square root is plausible.

Hint. Factor each zero with an even exponent and take a root of the nonvanishing factor. ☐

Solution. Near a zero, write $f = (z-a)^{2m} g$ with $g(a) \neq 0$. A local logarithm of g gives a local square root, so $(z-a)^m \exp(\frac{1}{2}\log g)$ squares to f. ☐

Exercise 7.46 (Multiplicity under composition)**.** If f has a zero of order m at a and g has a zero of order n at $f(a) = 0$, determine the order of $g \circ f$ at a.

Hint. Factor both functions at the relevant points. ☐

Solution. Write $f(z) = (z-a)^m u(z)$ and $g(w) = w^n v(w)$ with nonvanishing factors. Then $g(f(z)) = (z-a)^{mn} u(z)^n v(f(z))$, so the order is mn. ☐

Chapter 8

Laurent Series

Laurent series extend power-series expansions to annuli and encode singular behavior through negative powers. The principal part distinguishes removable singularities, poles, and essential singularities.

Learning goals

After this chapter, the reader should be able to:

- construct Laurent expansions on specified annuli and identify the correct convergence region;
- compute principal and analytic parts of Laurent series;
- derive Laurent coefficients from contour integrals;
- choose among geometric-series expansions according to the relative sizes of $|z-a|$;
- use Laurent series to classify singular behavior and compute residues;
- distinguish expansions of the same function on different annuli;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

8.1 Laurent expansions

A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$.

Example 8.1 (One function, two Laurent annuli)**.** For $f(z) = 1/(z(z-1))$, partial fractions give $-1/z + 1/(z-1)$. On $0 < |z| < 1$,

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{n\geq 0} z^n,$$

so $f(z) = -z^{-1} - \sum_{n\geq 0} z^n$. For $|z| > 1$, instead

$$\frac{1}{z-1} = \frac{1}{z}\frac{1}{1-1/z} = \sum_{n\geq 1} z^{-n},$$

and cancellation of the first term gives $f(z) = \sum_{n\geq 2} z^{-n}$. The annulus is part of the answer.

8.2 Annuli of convergence

Different singularities determine concentric annuli on which different Laurent expansions may be valid.

8.3 Coefficient formula

Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$.

8.4 Principal part

Negative-power terms constitute the principal part and measure the singular component.

Example 8.2 (Essential singularity encoded by infinitely many negative powers)**.** Substituting $1/z$ into the exponential series gives

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots .$$

The Laurent series has infinitely many negative powers, so the singularity at zero is essential. The coefficient of z^{-1} is 1, anticipating the residue.

8.5 Positive part

Nonnegative powers form an ordinary Taylor series.

8.6 Geometric expansions

Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus.

8.7 Uniqueness

The Laurent coefficients are uniquely determined by contour integrals.

Example 8.3 (Expansion about a shifted point)**.** Expand $1/(z+1)$ about $z_0 = 1$. With $w = z - 1$,

$$\frac{1}{z+1} = \frac{1}{2+w} = \frac{1}{2} \sum_{n \geq 0} (-w/2)^n,$$

valid for $|w| < 2$. For $|w| > 2$, rewrite instead as $w^{-1}(1 + 2/w)^{-1}$, producing a Laurent series in negative powers. The same singularity at $z = -1$ sets the annular boundary.

8.8 Residue preview

The coefficient a_{-1} is the residue and controls contour integrals around the singularity.

8.9 Annular deformation

Coefficient integrals are independent of the chosen circle within the annulus.

8.10 Core structural results

Theorem 8.4 (Laurent theorem). *A function holomorphic on an annulus $r < |z - z_0| < R$ has a unique Laurent series converging normally on compact subannuli.*

Proof. Apply Cauchy's formula to a region bounded by two circles, expand the outer and inner Cauchy kernels geometrically in the appropriate ratios, and justify termwise integration by uniform convergence on compact subannuli. ☐

Theorem 8.5 (Laurent coefficient formula). *For any circle $|\zeta - z_0| = \rho$ inside the annulus, $a_n = \frac{1}{2\pi i} \int \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.*

Proof. Insert the normally convergent Laurent series into the integral and use orthogonality of powers around the circle; only the nth coefficient survives. ☐

Theorem 8.6 (Uniqueness). *A Laurent expansion on a fixed annulus is unique.*

Proof. The coefficient integral recovers every coefficient directly from the function, so two expansions must have identical coefficients. ☐

8.11 Worked examples

Example 8.7 (Laurent expansions diagnostic). Give a rigorous worked analysis of Laurent expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.8 (Annuli of convergence diagnostic). Give a rigorous worked analysis of Annuli of convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.9 (Coefficient formula diagnostic). Give a rigorous worked analysis of Coefficient formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.10 (Principal part diagnostic). Give a rigorous worked analysis of Principal part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Negative-power terms constitute the principal part and measure the singular component. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

8.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 8.11 (Laurent expansions diagnostic). Give a rigorous worked analysis of Laurent expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.12 (Annuli of convergence diagnostic)**.** Give a rigorous worked analysis of Annuli of convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.13 (Coefficient formula diagnostic)**.** Give a rigorous worked analysis of Coefficient formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.14 (Principal part diagnostic)**.** Give a rigorous worked analysis of Principal part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Negative-power terms constitute the principal part and measure the singular component. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.15 (Positive part diagnostic)**.** Give a rigorous worked analysis of Positive part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Nonnegative powers form an ordinary Taylor series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.16 (Geometric expansions diagnostic)**.** Give a rigorous worked analysis of Geometric expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.17 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Laurent coefficients are uniquely determined by contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 8.18 (Residue preview diagnostic)**.** Give a rigorous worked analysis of Residue preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The coefficient a_{-1} is the residue and controls contour integrals around the singularity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.19 (Laurent theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A function holomorphic on an annulus $r < |z - z_0| < R$ has a unique Laurent series converging normally on compact subannuli.

Solution. Apply Cauchy's formula to a region bounded by two circles, expand the outer and inner Cauchy kernels geometrically in the appropriate ratios, and justify termwise integration by uniform convergence on compact subannuli. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.20 (Laurent coefficient formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For any circle $|\zeta - z_0| = \rho$ inside the annulus, $a_n = \frac{1}{2\pi i} \int \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.

Solution. Insert the normally convergent Laurent series into the integral and use orthogonality of powers around the circle; only the nth coefficient survives. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.21 (Uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A Laurent expansion on a fixed annulus is unique.

Solution. The coefficient integral recovers every coefficient directly from the function, so two expansions must have identical coefficients. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Laurent series extend power-series expansions to annuli and encode singular behavior through negative powers. The principal part distinguishes removable singularities, poles, and essential singularities. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

8.13 Exercises with complete solutions

Exercise 8.23. State and apply the central principle behind Laurent expansions. Give one concrete consequence.

Hint. Locate the singularities relative to the center first; they determine the annuli on which different Laurent expansions are valid. □

Solution. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.24. State and apply the central principle behind Annuli of convergence. Give one concrete consequence.

Hint. For $|z - a| < R$, expand denominator factors in positive powers; for $|z - a| > R$, factor out powers of $z - a$ and expand inversely. □

Solution. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.25. State and apply the central principle behind Coefficient formula. Give one concrete consequence.

Hint. Separate negative powers from nonnegative powers to identify principal and analytic parts. □

Solution. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.26. State and apply the central principle behind Principal part. Give one concrete consequence.

Hint. Use $a_n = (2\pi i)^{-1} \int f(z)(z - a)^{-n-1} dz$ on any circle lying inside the annulus. □

Solution. Negative-power terms constitute the principal part and measure the singular component. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.27. State and apply the central principle behind Positive part. Give one concrete consequence.

Hint. The coefficient of $(z - a)^{-1}$ is the residue; read it directly once the Laurent series is correct. □

Solution. Nonnegative powers form an ordinary Taylor series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.28. State and apply the central principle behind Geometric expansions. Give one concrete consequence.

Hint. Do not merge expansions valid on different annuli even if their algebraic expressions look similar. □

Solution. Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.29. State and apply the central principle behind Uniqueness. Give one concrete consequence.

Hint. For rational functions, partial fractions often expose the correct geometric-series expansions immediately. □

Solution. The Laurent coefficients are uniquely determined by contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.30. State and apply the central principle behind Residue preview. Give one concrete consequence.

Hint. Check convergence by comparing the geometric ratio magnitude with one on the intended annulus. □

Solution. The coefficient a_{-1} is the residue and controls contour integrals around the singularity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

8.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 8.31 (Inner expansion). Find the Laurent series of $1/(z(1-z))$ for $0 < |z| < 1$.

Hint. Use the geometric series for $1/(1-z)$. ☐

Solution. The series is $z^{-1} + 1 + z + z^2 + \cdots$. ☐

Exercise 8.32 (Outer expansion). Expand $1/(z-2)$ for $|z| > 2$.

Hint. Factor out z. ☐

Solution. One obtains $z^{-1}(1-2/z)^{-1} = \sum_{n\geq 0} 2^n z^{-n-1}$. ☐

Exercise 8.33 (Coefficient extraction). What is the coefficient of z^{-1} in $e^{1/z}$?

Hint. Read it from the exponential series. ☐

Solution. The coefficient is 1. ☐

Exercise 8.34 (Principal part). Find the principal part of $3/z^4 - 2/z + 7 + z$ at zero.

Hint. Keep only negative powers. ☐

Solution. The principal part is $3z^{-4} - 2z^{-1}$. ☐

Exercise 8.35 (Annulus boundaries). For a Laurent expansion about zero of $1/((z-1)(z-3))$, name the natural radial regions.

Hint. Use distances from the center to the singularities. ☐

Solution. The regions are $|z| < 1$, $1 < |z| < 3$, and $|z| > 3$, excluding the singular circles themselves. ☐

Proofs

Exercise 8.36 (Laurent coefficient formula)**.** Derive the contour formula for a Laurent coefficient a_n on an annulus.

Hint. Multiply by $(z - z_0)^{-n-1}$ and integrate around a circle in the annulus. □

Solution. Termwise integration kills every power except the one with exponent -1, yielding $a_n = (2\pi i)^{-1} \int f(z)/(z - z_0)^{n+1} dz$. □

Exercise 8.37 (Uniqueness)**.** Prove uniqueness of the Laurent expansion on an annulus.

Hint. Use the coefficient formula. □

Solution. Every coefficient is determined by the contour integral formula, so two Laurent expansions of the same function must have identical coefficients. □

Exercise 8.38 (Independence of radius)**.** Show that Laurent coefficients do not depend on which circle inside the annulus is used.

Hint. Apply Cauchy's theorem to the region between two circles. □

Solution. The relevant coefficient integrand is holomorphic throughout the intervening annulus, so its integrals over the two positively oriented circles are equal. □

Exercise 8.39 (Taylor as a special Laurent series)**.** Explain why a Laurent series with no negative powers is a Taylor series.

Hint. The nonnegative coefficients define an ordinary power series. □

Solution. With all negative coefficients zero, the representation extends analytically across the center and is precisely the Taylor expansion there. □

Counterexamples and hypothesis tests

Exercise 8.40 (Wrong geometric expansion)**.** Why is $1/(1 - z) = \sum z^n$ invalid for $|z| > 1$?

Hint. Check the geometric ratio. □

Solution. The terms z^n do not even tend to zero when $|z| > 1$. One must expand in powers of $1/z$ instead. □

Exercise 8.41 (Annulus omitted)**.** Why is a Laurent formula incomplete without its annulus of validity?

Hint. Different geometric rewritings can produce different series. □

Solution. The same function can have distinct Laurent expansions in different annuli centered at the same point. The convergence region determines which series represents the function. □

Exercise 8.42 (Finite principal part)**.** Does a finite principal part necessarily indicate an essential singularity?

Hint. Compare the singularity classification by negative terms. □

Solution. No. A finite nonempty principal part corresponds to a pole; infinitely many negative terms characterize an essential singularity. □

Applications and investigations

Exercise 8.43 (Residue preview). Why is the $(z - z_0)^{-1}$ Laurent coefficient especially important for contour integration?

Hint. Integrate each Laurent monomial around a circle. ☐

Solution. Every monomial integrates to zero except exponent -1, whose integral is $2\pi i$. Thus that coefficient alone controls the local contour contribution. ☐

Exercise 8.44 (Local singularity diagnosis). How does a computed Laurent series distinguish removable, pole, and essential behavior?

Hint. Inspect the principal part. ☐

Solution. No negative powers means removable; finitely many negative powers means a pole; infinitely many negative powers means an essential singularity. ☐

Challenge problems

Exercise 8.45 (Three radial regions). Find the form of the Laurent expansion of $1/(z(z-1)(z-2))$ on each of $0 < |z| < 1$, $1 < |z| < 2$, and $|z| > 2$.

Hint. Use partial fractions and choose a geometric expansion for each factor appropriate to the region. ☐

Solution. Partial fractions reduce the problem to terms at $0, 1, 2$. In each annulus, expand factors with singularities outside in nonnegative powers of z and factors inside in negative powers of z; the resulting series are different but unique on their regions. ☐

Exercise 8.46 (Center one). Describe the natural annuli for the Laurent expansion of $1/(z(z-2))$ about $z_0 = 1$.

Hint. Measure the two singularity distances from the new center. ☐

Solution. Both singularities are at distance one from the center. Thus there are only two radial regions: $|z-1| < 1$ and $|z-1| > 1$, with the singular circle separating them. ☐

Chapter 9

Isolated Singularities

An isolated singularity is classified by its Laurent principal part. Removable singularities have no negative powers, poles have finitely many, and essential singularities have infinitely many.

Learning goals

After this chapter, the reader should be able to:

- classify isolated singularities as removable, poles, or essential using Laurent coefficients or growth criteria;
- determine pole order from local factorizations;
- apply removable-singularity theorems from boundedness or vanishing conditions;
- use Casorati–Weierstrass or related behavior to recognize essential singularities;
- construct local normal forms near poles and removable points;
- distinguish singularity classification from global behavior of the function;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

9.1 Removable singularities

A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture.

Example 9.1 (Removing a fake singularity)**.** The quotient $\sin z/z$ is undefined at zero in its displayed form, but

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots .$$

The right side is analytic at zero and has value 1 there. Defining $f(0) = 1$ removes the singularity.

9.2 Poles

A pole of order m is characterized by $(z - z_0)^m f(z)$ extending holomorphically and nonzero at z_0.

9.3 Essential singularities

An essential singularity has infinitely many negative Laurent coefficients.

9.4 Classification theorem

Every isolated singularity is exactly one of removable, pole, or essential.

Example 9.2 (Detecting a pole and its order)**.** For $f(z) = e^z/z^3$, the numerator is nonzero at zero. Hence $z^3 f(z) = e^z$ extends holomorphically with nonzero value at zero. Therefore zero is a pole of order three. The leading Laurent term is z^{-3}.

9.5 Riemann removable theorem

Boundedness near the puncture is enough for removability.

9.6 Casorati–Weierstrass

Near an essential singularity the image is dense in the complex plane.

9.7 Behavior of reciprocals

Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order.

Example 9.3 (An essential singularity)**.** The Laurent series

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n! z^n}$$

contains infinitely many negative powers. Thus zero is essential. No multiplication by a finite power of z can make the function bounded or holomorphic at the origin.

9.8 Singularity at infinity

Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains.

9.9 Core structural results

Theorem 9.4 (Classification by Laurent series)**.** *An isolated singularity is removable iff all negative Laurent coefficients vanish, a pole iff only finitely many negative coefficients are nonzero, and essential otherwise.*

Proof. The three cases correspond exactly to whether the Laurent series extends as a Taylor series, has a finite principal part with highest negative order, or has an infinite principal part. □

Theorem 9.5 (Riemann removable singularity theorem)**.** *If f is holomorphic and bounded on a punctured disk around z_0, then f extends holomorphically to z_0.*

Proof. Use Cauchy's formula on a punctured annulus and estimate negative Laurent coefficients; boundedness forces every negative coefficient to vanish. □

Theorem 9.6 (Pole-zero reciprocity)**.** *f has a zero of order m at z_0 iff $1/f$ has a pole of order m there.*

Proof. Factor $f = (z - z_0)^m g$ with $g(z_0) \neq 0$ and invert to obtain $(1/f) = (z - z_0)^{-m}(1/g)$ with holomorphic nonvanishing factor. □

9.10 Worked examples

Example 9.7 (Removable singularities diagnostic)**.** Give a rigorous worked analysis of Removable singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.8 (Poles diagnostic)**.** Give a rigorous worked analysis of Poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A pole of order m is characterized by $(z - z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.9 (Essential singularities diagnostic)**.** Give a rigorous worked analysis of Essential singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. An essential singularity has infinitely many negative Laurent coefficients. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.10 (Classification theorem diagnostic)**.** Give a rigorous worked analysis of Classification theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Every isolated singularity is exactly one of removable, pole, or essential. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

9.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 9.11 (Removable singularities diagnostic)**.** Give a rigorous worked analysis of Removable singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.12 (Poles diagnostic)**.** Give a rigorous worked analysis of Poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A pole of order m is characterized by $(z-z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.13 (Essential singularities diagnostic)**.** Give a rigorous worked analysis of Essential singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. An essential singularity has infinitely many negative Laurent coefficients. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.14 (Classification theorem diagnostic)**.** Give a rigorous worked analysis of Classification theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every isolated singularity is exactly one of removable, pole, or essential. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.15 (Riemann removable theorem diagnostic)**.** Give a rigorous worked analysis of Riemann removable theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Boundedness near the puncture is enough for removability. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.16 (Casorati–Weierstrass diagnostic)**.** Give a rigorous worked analysis of Casorati–Weierstrass. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near an essential singularity the image is dense in the complex plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.17 (Behavior of reciprocals diagnostic)**.** Give a rigorous worked analysis of Behavior of reciprocals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.18 (Singularity at infinity diagnostic)**.** Give a rigorous worked analysis of Singularity at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.19 (Classification by Laurent series proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: An isolated singularity is removable iff all negative Laurent coefficients vanish, a pole iff only finitely many negative coefficients are nonzero, and essential otherwise.

Solution. The three cases correspond exactly to whether the Laurent series extends as a Taylor series, has a finite principal part with highest negative order, or has an infinite principal part. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 9.20 (Riemann removable singularity theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic and bounded on a punctured disk around z_0, then f extends holomorphically to z_0.

Solution. Use Cauchy's formula on a punctured annulus and estimate negative Laurent coefficients; boundedness forces every negative coefficient to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 9.21 (Pole-zero reciprocity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: f has a zero of order m at z_0 iff $1/f$ has a pole of order m there.

Solution. Factor $f = (z - z_0)^m g$ with $g(z_0) \neq 0$ and invert to obtain $(1/f) = (z - z_0)^{-m}(1/g)$ with holomorphic nonvanishing factor. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 9.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. An isolated singularity is classified by its Laurent principal part. Removable singularities have no negative powers, poles have finitely many, and essential singularities have infinitely many. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

9.12 Exercises with complete solutions

Exercise 9.23. State and apply the central principle behind Removable singularities. Give one concrete consequence.

Hint. If all negative Laurent coefficients vanish, the singularity is removable; if finitely many occur, it is a pole; infinitely many signal essential behavior. ☐

Solution. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.24. State and apply the central principle behind Poles. Give one concrete consequence.

Hint. For a pole, factor $f(z) = (z - a)^{-m} g(z)$ with $g(a) \neq 0$ and identify the smallest valid m. ☐

Solution. A pole of order m is characterized by $(z - z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.25. State and apply the central principle behind Essential singularities. Give one concrete consequence.

Hint. A bounded punctured-neighborhood function has a removable singularity; extend it by the Cauchy-limit value. ☐

Solution. An essential singularity has infinitely many negative Laurent coefficients. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.26. State and apply the central principle behind Classification theorem. Give one concrete consequence.

Hint. Use $(z-a)^m f(z) \to c \neq 0$ to certify a pole of order m. ☐

Solution. Every isolated singularity is exactly one of removable, pole, or essential. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.27. State and apply the central principle behind Riemann removable theorem. Give one concrete consequence.

Hint. For essential singularities, inspect the Laurent principal part or invoke Casorati–Weierstrass only after ruling out removable and pole cases. ☐

Solution. Boundedness near the puncture is enough for removability. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.28. State and apply the central principle behind Casorati–Weierstrass. Give one concrete consequence.

Hint. Classification is local: ignore remote poles and zeros when analyzing the singularity at a. ☐

Solution. Near an essential singularity the image is dense in the complex plane. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.29. State and apply the central principle behind Behavior of reciprocals. Give one concrete consequence.

Hint. Zeros of the reciprocal correspond to poles of the original function; use this to switch perspectives when convenient. ☐

Solution. Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 9.30. State and apply the central principle behind Singularity at infinity. Give one concrete consequence.

Hint. Check whether cancellation removes an apparent singularity before classifying the unreduced formula. ☐

Solution. Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

9.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 9.31 (Removable quotient)**.** Classify the singularity of $(e^z - 1)/z$ at zero.

Hint. Expand the numerator or take a limit. ☐

Solution. The quotient tends to one and has a Taylor expansion beginning $1 + z/2 + \cdots$, so the singularity is removable. ☐

Exercise 9.32 (Pole order)**.** Classify zero for $(1 + z)/z^5$.

Hint. The numerator is nonzero at zero. ☐

Solution. Zero is a pole of order five. ☐

Exercise 9.33 (Essential exponential)**.** Classify zero for e^{1/z^2}.

Hint. Expand the exponential. ☐

Solution. The Laurent series has infinitely many negative powers, so zero is essential. ☐

Exercise 9.34 (Zero versus pole)**.** Classify zero for $z^3/(e^z - 1)^2$.

Hint. Use $e^z - 1 = z + O(z^2)$. ☐

Solution. The denominator has order two and the numerator order three, leaving a zero of order one after cancellation; the apparent singularity is removable. ☐

Exercise 9.35 (Limit criterion)**.** If $\lim_{z\to a}(z - a)^2 f(z) = 7 \neq 0$, what singularity does f have at a?

Hint. Use the pole characterization. ☐

Solution. The point is a pole of order two. ☐

Proofs

Exercise 9.36 (Bounded removable theorem)**.** Prove that a bounded holomorphic function on a punctured disk has a removable singularity at the center.

Hint. Apply Cauchy's formula on circles avoiding the point or examine Laurent coefficients. ☐

Solution. For negative Laurent coefficients, the coefficient estimate contains a positive power of the circle radius. Boundedness forces each such coefficient to zero as the radius shrinks, leaving a Taylor series. ☐

Exercise 9.37 (Pole characterization)**.** Prove that a is a pole of order m exactly when $(z-a)^m f(z)$ extends holomorphically and nonvanishingly at a.

Hint. Use the finite principal part factorization. ☐

Solution. A pole of order m has Laurent form $(z-a)^{-m}g(z)$ with $g(a) \neq 0$. Multiplying removes exactly that pole; the converse follows by dividing the nonvanishing extension by $(z-a)^m$. ☐

Exercise 9.38 (Removable from finite limit)**.** Show that a finite limit of $f(z)$ as $z \to a$ makes the isolated singularity removable.

Hint. A finite limit gives local boundedness. ☐

Solution. Near a, the function is bounded, so the removable singularity theorem applies. Defining the missing value to be the limit yields continuity and holomorphicity. ☐

Exercise 9.39 (Essential criterion)**.** Show that an isolated singularity is essential exactly when its Laurent principal part has infinitely many nonzero terms.

Hint. Use the exhaustive Laurent classification. ☐

Solution. No negative terms gives a removable point; finitely many gives a pole. The only remaining case is infinitely many negative terms, which therefore characterizes an essential singularity. ☐

Counterexamples and hypothesis tests

Exercise 9.40 (Unbounded means pole)**.** Does unboundedness near an isolated singularity always imply a pole?

Hint. Recall essential singularities. ☐

Solution. No. Essential singularities are also unbounded in every punctured neighborhood. One needs finite principal part or equivalent pole behavior. ☐

Exercise 9.41 (Vanishing product)**.** If $(z-a)f(z) \to 0$, must f have a removable singularity?

Hint. Apply the removable theorem first to $g(z) = (z-a)f(z)$. ☐

Solution. Yes. The function g extends holomorphically with $g(a) = 0$, so $g(z) = (z-a)h(z)$ for a holomorphic h. Hence $f = h$ on the punctured disk and extends holomorphically across a. ☐

Exercise 9.42 (Classification requires isolation)**.** Can the removable, pole, essential trichotomy be applied at an accumulation point of singularities?

Hint. Check the word isolated. ☐

Solution. No. The Laurent annulus around the point may not exist, so the isolated singularity classification does not apply. ☐

Applications and investigations

Exercise 9.43 (Regularizing a formula)**.** Why is removing a removable singularity useful in analysis or computation?

Hint. The extended function is genuinely holomorphic at the point. □

Solution. The extension eliminates artificial division by zero, restores stable local evaluation, and permits direct use of Taylor series and derivative formulas across the point. □

Exercise 9.44 (Pole order and blowup)**.** How does pole order describe the leading local growth?

Hint. Use the factorization $f = (z-a)^{-m}g$ with $g(a) \neq 0$. □

Solution. The magnitude behaves like a nonzero constant times $|z-a|^{-m}$ to leading order, so the integer m quantifies the algebraic blowup rate. □

Challenge problems

Exercise 9.45 (Reciprocal classification)**.** If f has a pole of order m at a, classify $1/f$ after extension.

Hint. Invert the local factorization. □

Solution. Writing $f = (z-a)^{-m}g$ with $g(a) \neq 0$ gives $1/f = (z-a)^m/g$, which extends holomorphically with a zero of order m. □

Exercise 9.46 (Behavior at infinity)**.** Explain how to classify the point at infinity for a function by studying $g(w) = f(1/w)$ at $w = 0$.

Hint. The map $w = 1/z$ turns large z into small w. □

Solution. The singularity type of g at zero defines the type of f at infinity. For example, a polynomial of degree m gives a pole of order m at infinity. □

Chapter 10

Residues and the Residue Theorem

Residues compress the local singular data relevant to contour integration into one Laurent coefficient. The residue theorem converts global contour integrals into finite sums of local contributions.

Learning goals

After this chapter, the reader should be able to:

- compute residues at simple and higher-order poles using the most efficient available formula;
- apply the residue theorem after identifying all enclosed singularities and contour orientations;
- compute residues at infinity and use global residue balance;
- use logarithmic derivatives to count zeros and poles with multiplicity;
- choose between Laurent expansion, limit, derivative, and partial-fraction methods for residue computation;
- audit winding numbers and contour geometry before summing residue contributions;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

10.1 Residue definition

The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$.

Example 10.1 (Residue at a simple pole)**.** For $f(z) = e^z/(z-1)$,

$$\operatorname{Res}_{z=1} f = \lim_{z \to 1} (z-1) f(z) = e.$$

Therefore a positively oriented contour enclosing 1 and no other singularities has integral $2\pi i e$.

10.2 Simple poles

For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$.

10.3 Higher-order poles

Derivative formulas compute residues at poles of finite order.

10.4 Residue theorem

A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form.

Example 10.2 (A second order pole)**.** Consider $f(z) = 1/(z^2(z-1))$ at $z = 0$. Since the pole has order two,

$$\operatorname{Res}_{z=0} f = \frac{d}{dz}\frac{1}{z-1}\bigg|_{z=0} = -1.$$

The derivative formula avoids computing the full Laurent series.

10.5 Local-to-global principle

Only the -1 Laurent terms contribute to closed contour integrals.

10.6 Residue at infinity

For meromorphic functions on the sphere, the residue at infinity balances the finite residues.

10.7 Logarithmic derivative

Residues of f'/f record zero and pole multiplicities.

Example 10.3 (Summing local contributions)**.** For

$$f(z) = \frac{1}{(z-1)(z+1)},$$

the residues at 1 and -1 are $1/2$ and $-1/2$. A contour enclosing both has zero integral, while a contour enclosing only 1 has integral πi. The residue theorem converts a global contour integral into a finite sum of local coefficients.

10.8 Computational strategy

Factor denominators, classify enclosed poles, choose the simplest residue formula, and audit contour orientation.

10.9 Parameter dependence

Residue calculations often reduce integral identities to algebraic sums depending analytically on parameters.

10.10 Core structural results

Theorem 10.4 (Residue theorem)**.** *If f is holomorphic inside and on a positively oriented contour except for finitely many isolated singularities a_k, then $\int_\Gamma f\,dz = 2\pi i \sum_k \operatorname{Res}(f, a_k)$ in the simple winding-one setting.*

Proof. Remove small disks around the singularities, apply Cauchy's theorem to the remaining multiply connected region, and let the small circles shrink; each circle integral tends to $2\pi i$ times the Laurent -1 coefficient. □

Theorem 10.5 (Simple-pole formula)**.** *If $f = g/h$ with g, h holomorphic, $h(a) = 0$, and $h'(a) \neq 0$, then $\operatorname{Res}(f, a) = g(a)/h'(a)$.*

Proof. Factor $h(z) = (z-a)k(z)$ with $k(a) = h'(a)$; then $(z-a)f(z) = g(z)/k(z)$ and evaluate at a. □

Theorem 10.6 (Higher-pole residue formula)**.** *If f has a pole of order m at a, then $\operatorname{Res}(f, a) = \frac{1}{(m-1)!}\left[\frac{d^{m-1}}{dz^{m-1}}((z-a)^m f(z))\right]_{z=a}$.*

Proof. The bracketed factor is holomorphic; its Taylor coefficient of degree $m-1$ is exactly the Laurent coefficient multiplying $(z-a)^{-1}$. □

Theorem 10.7 (Global residue balance)**.** *For a meromorphic function on the Riemann sphere, the sum of all residues including infinity is zero.*

Proof. Integrate around a sufficiently large circle and compare the exterior coordinate $w = 1/z$ with the finite residues inside; the residue at infinity is defined to cancel the large-circle contribution. □

10.11 Worked examples

Example 10.8 (Residue definition diagnostic)**.** Give a rigorous worked analysis of Residue definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The residue at an isolated singularity is the Laurent coefficient of $(z-z_0)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.9 (Simple poles diagnostic)**.** Give a rigorous worked analysis of Simple poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.10 (Higher-order poles diagnostic)**.** Give a rigorous worked analysis of Higher-order poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Derivative formulas compute residues at poles of finite order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.11 (Residue theorem diagnostic)**.** Give a rigorous worked analysis of Residue theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

10.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 10.12 (Residue definition diagnostic)**.** Give a rigorous worked analysis of Residue definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.13 (Simple poles diagnostic)**.** Give a rigorous worked analysis of Simple poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.14 (Higher-order poles diagnostic)**.** Give a rigorous worked analysis of Higher-order poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Derivative formulas compute residues at poles of finite order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.15 (Residue theorem diagnostic)**.** Give a rigorous worked analysis of Residue theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.16 (Local-to-global principle diagnostic)**.** Give a rigorous worked analysis of Local-to-global principle. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Only the -1 Laurent terms contribute to closed contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.17 (Residue at infinity diagnostic)**.** Give a rigorous worked analysis of Residue at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For meromorphic functions on the sphere, the residue at infinity balances the finite residues. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.18 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Residues of f'/f record zero and pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 10.19 (Residue theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic inside and on a positively oriented contour except for finitely many isolated singularities a_k, then $\int_\Gamma f dz = 2\pi i \sum_k \operatorname{Res}(f, a_k)$ in the simple winding-one setting.

Solution. Remove small disks around the singularities, apply Cauchy's theorem to the remaining multiply connected region, and let the small circles shrink; each circle integral tends to $2\pi i$ times the Laurent -1 coefficient. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 10.20 (Simple-pole formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = g/h$ with g, h holomorphic, $h(a) = 0$, and $h'(a) \neq 0$, then $\operatorname{Res}(f, a) = g(a)/h'(a)$.

Solution. Factor $h(z) = (z - a)k(z)$ with $k(a) = h'(a)$; then $(z - a)f(z) = g(z)/k(z)$ and evaluate at a. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 10.21 (Higher-pole residue formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f has a pole of order m at a, then $\operatorname{Res}(f, a) = \frac{1}{(m-1)!}\left[\frac{d^{m-1}}{dz^{m-1}}((z - a)^m f(z))\right]_{z=a}$.

Solution. The bracketed factor is holomorphic; its Taylor coefficient of degree $m-1$ is exactly the Laurent coefficient multiplying $(z - a)^{-1}$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 10.22 (Global residue balance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a meromorphic function on the Riemann sphere, the sum of all residues including infinity is zero.

Solution. Integrate around a sufficiently large circle and compare the exterior coordinate $w = 1/z$ with the finite residues inside; the residue at infinity is defined to cancel the large-circle contribution. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 10.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Residues compress the local singular data relevant to contour integration into one Laurent coefficient. The residue theorem converts global contour integrals into finite sums of local contributions. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

10.13 Exercises with complete solutions

Exercise 10.24. State and apply the central principle behind Residue definition. Give one concrete consequence.

Hint. At a simple pole of g/h, verify $h(a) = 0$ and $h'(a) \neq 0$, then use $g(a)/h'(a)$. ☐

Solution. The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.25. State and apply the central principle behind Simple poles. Give one concrete consequence.

Hint. For a pole of order m, multiply by $(z - a)^m$, differentiate $m - 1$ times, and evaluate at a. ☐

Solution. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.26. State and apply the central principle behind Higher-order poles. Give one concrete consequence.

Hint. List every singularity inside the contour before computing any residue; missing one pole invalidates the sum. ☐

Solution. Derivative formulas compute residues at poles of finite order. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.27. State and apply the central principle behind Residue theorem. Give one concrete consequence.

Hint. Include orientation and winding numbers in the residue theorem rather than assuming every contour is positively oriented once. ☐

Solution. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.28. State and apply the central principle behind Local-to-global principle. Give one concrete consequence.

Hint. For a large collection of finite poles, compute the residue at infinity and use total residue zero when it is shorter. ☐

Solution. Only the -1 Laurent terms contribute to closed contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.29. State and apply the central principle behind Residue at infinity. Give one concrete consequence.

Hint. Residues of f'/f equal zero multiplicities minus pole multiplicities; check the contour avoids zeros and poles. ☐

Solution. For meromorphic functions on the sphere, the residue at infinity balances the finite residues. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.30. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Choose Laurent, limit, derivative, or partial-fraction methods by local pole structure, not by habit. ☐

Solution. Residues of f'/f record zero and pole multiplicities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.31. State and apply the central principle behind Computational strategy. Give one concrete consequence.

Hint. After summing residues, perform a quick scale or symmetry check on the final contour integral. ☐

Solution. Factor denominators, classify enclosed poles, choose the simplest residue formula, and audit contour orientation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

10.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 10.32 (Simple rational residue)**.** Find $\operatorname{Res}_{z=2} 1/(z(z-2))$.

Hint. Multiply by $z-2$ and take the limit. ☐

Solution. The residue is $1/2$. ☐

Exercise 10.33 (Exponential residue)**.** Find the residue of $e^z/(z-i)$ at i.

Hint. It is a simple pole. ☐

Solution. The residue is e^i. ☐

Exercise 10.34 (Second order residue)**.** Find $\operatorname{Res}_{z=0} e^z/z^2$.

Hint. Read the coefficient of z^{-1} or differentiate the numerator. ☐

Solution. Since $e^z = 1 + z + \cdots$, the residue is 1. ☐

Exercise 10.35 (Contour sum)**.** A contour encloses simple poles with residues 2 and $-3i$. Find the integral.

Hint. Multiply the residue sum by $2\pi i$. ☐

Solution. The integral is $2\pi i(2-3i) = 6\pi + 4\pi i$. ☐

Exercise 10.36 (Residue from Laurent series)**.** What is the residue at zero of $4z^{-3} - 7z^{-1} + 2 + z$?

Hint. Select the coefficient of z^{-1}. ☐

Solution. The residue is -7. ☐

Proofs

Exercise 10.37 (Residue theorem from Laurent series)**.** Prove the residue theorem for finitely many isolated singularities inside a contour.

Hint. Excise small circles around the singularities and apply Cauchy's theorem to the remaining region. □

Solution. The outer integral equals the sum of the small circle integrals. Each small circle integral is $2\pi i$ times the local z^{-1} Laurent coefficient, yielding the residue sum. □

Exercise 10.38 (Simple pole formula)**.** Prove $\mathrm{Res}_a(g/h) = g(a)/h'(a)$ when $h(a) = 0$, $h'(a) \neq 0$, and $g(a) \neq 0$.

Hint. Factor the simple zero of h. □

Solution. Write $h(z) = (z - a)k(z)$ with $k(a) = h'(a)$. Then $(z - a)g/h = g/k$, whose limit at a is $g(a)/h'(a)$. □

Exercise 10.39 (Higher pole formula)**.** Derive the residue formula for a pole of order m.

Hint. Multiply by $(z - a)^m$ and take the Taylor coefficient of order $m - 1$. □

Solution. If $g = (z-a)^m f$ is holomorphic, the $(z - a)^{-1}$ term of f comes from the $(m - 1)$-st Taylor coefficient of g, giving $g^{(m-1)}(a)/(m - 1)!$. □

Exercise 10.40 (Residue of a derivative)**.** Show that the residue of f' at an isolated singularity is zero.

Hint. Differentiate the Laurent series termwise. □

Solution. A derivative term has exponent $n - 1$ with coefficient na_n. Exponent -1 would require $n = 0$, whose coefficient is zero. Hence there is no $(z - a)^{-1}$ term. □

Counterexamples and hypothesis tests

Exercise 10.41 (Pole on contour)**.** Why is the ordinary residue theorem not directly applicable when a pole lies on the contour?

Hint. The integrand is not defined continuously along the path. □

Solution. The contour integral is not an ordinary integral under the theorem's hypotheses. Indentation or principal value methods require separate arguments. □

Exercise 10.42 (Orientation)**.** What happens to the residue theorem for clockwise orientation?

Hint. Reverse the contour. □

Solution. The integral becomes $-2\pi i$ times the enclosed residue sum. □

Exercise 10.43 (Outside residues)**.** Should poles outside a contour be included in the residue sum?

Hint. Only singularities in the interior contribute. □

Solution. No. Exterior poles do not affect the contour integral as long as the integrand is holomorphic on a neighborhood of the contour itself. □

Applications and investigations

Exercise 10.44 (Logarithmic derivative)**.** If f has a zero of order m at a, find the residue of f'/f there.

Hint. Factor $f = (z-a)^m g$. □

Solution. Then $f'/f = m/(z-a) + g'/g$, and the second term is holomorphic at a. The residue is m. □

Exercise 10.45 (Residue at infinity)**.** For a rational function decaying as $1/z^2$ or faster, what relation holds among all finite residues?

Hint. Integrate on a sufficiently large circle and let its radius grow. □

Solution. The large circle integral tends to zero, so the sum of all finite residues is zero. Equivalently the residue at infinity vanishes. □

Challenge problems

Exercise 10.46 (Counting zeros and poles)**.** Explain why residues of f'/f naturally count zeros positively and poles negatively.

Hint. Use local factorizations at zeros and poles. □

Solution. A zero of order m contributes residue m; a pole of order n contributes $-n$. Integrating f'/f therefore counts the divisor inside the contour. □

Exercise 10.47 (Residue at infinity formula)**.** Show formally that $\operatorname{Res}_\infty f = -\sum \operatorname{Res}_{a_k} f$ for a meromorphic function on the sphere.

Hint. Take a large positively oriented circle enclosing all finite poles. □

Solution. The finite residue theorem gives the large circle integral as $2\pi i$ times the finite sum. Under $w = 1/z$, the same contour is a negatively oriented small circle around infinity, producing the stated negative relation. □

Chapter 11

Evaluation of Real Integrals

Real integrals can often be evaluated by embedding the integrand into a meromorphic complex function and choosing contours whose auxiliary pieces vanish or are explicitly computable.

Learning goals

After this chapter, the reader should be able to:

- select contour geometries that convert real integrals into residue calculations;
- estimate large-arc and small-arc contributions and justify their vanishing limits;
- handle even rational integrands with semicircular contours;
- use indentation or principal-value contours when poles lie on the real axis;
- track orientation and branch choices when extracting the desired real integral;
- distinguish ordinary improper integrals from Cauchy principal values;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

11.1 Contour selection

Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries.

Example 11.1 (The basic rational integral)**.** For $a > 0$, integrate $1/(z^2 + a^2)$ over the upper semicircle. The only enclosed pole is ia, with residue $1/(2ia)$. The arc contribution tends to zero because the integrand is $O(R^{-2})$. Hence

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2} = 2\pi i \frac{1}{2ia} = \frac{\pi}{a}.$$

11.2 Even rational integrals

Symmetry can reduce whole-line contour integrals to half-line real integrals.

11.3 Decay on large arcs

Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish.

11.4 Jordan-type estimates

Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign.

Example 11.2 (Fourier damping by contour choice)**.** For $b > 0$, consider $e^{ibz}/(z^2 + a^2)$ in the upper half-plane. Since $|e^{ib(x+iy)}| = e^{-by}$, the exponential damps the large arc. The pole at ia contributes

$$\int_{-\infty}^{\infty} \frac{e^{ibx}}{x^2 + a^2}\, dx = \frac{\pi}{a} e^{-ab}.$$

Taking real parts gives the cosine integral.

11.5 Principal values

Poles on the contour require indentation and lead naturally to Cauchy principal values.

11.6 Trigonometric integrals

The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals.

11.7 Parameter differentiation

Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas.

Example 11.3 (A quartic denominator)**.** For $1/(z^4 + 1)$, the upper half-plane poles are $e^{i\pi/4}$ and $e^{3i\pi/4}$. Their residues are $1/(4z_k^3)$. Summing them and multiplying by $2\pi i$, while the large arc vanishes, gives

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} = \frac{\pi}{\sqrt{2}}.$$

Evenness then yields $\int_0^\infty dx/(x^4 + 1) = \pi/(2\sqrt{2})$.

11.8 Branch-aware previews

Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later.

11.9 Verification

Residue methods should be checked against convergence, orientation, enclosed singularities, and real-valued symmetry.

11.10 Core structural results

Theorem 11.4 (Upper-half-plane residue method)**.** *Suppose a rational function has no real poles and decays fast enough that the upper semicircle contribution vanishes. Then its real-line integral equals $2\pi i$ times the residues of poles in the upper half-plane.*

Proof. Apply the residue theorem to the closed contour consisting of the real segment and upper semicircle, then send the radius to infinity using the decay estimate. □

Theorem 11.5 (Unit-circle substitution)**.** *For a 2π-periodic rational expression in $\sin t$ and $\cos t$, the substitution $z = e^{it}$ converts the integral into a contour integral on $|z| = 1$.*

Proof. Use $dt = dz/(iz)$, $\cos t = (z + z^{-1})/2$, and $\sin t = (z - z^{-1})/(2i)$; simplify to a rational function of z and apply residues inside the unit circle. □

Theorem 11.6 (Principal-value indentation rule)**.** *A simple pole on a smooth contour contributes half of the usual residue term under a symmetric indentation, with sign determined by the indentation orientation.*

Proof. Replace a small neighborhood of the pole by a semicircular detour, approximate the integrand by its residue divided by $z - a$, and compute the limiting half-circle integral. □

11.11 Worked examples

Example 11.7 (Contour selection diagnostic)**.** Give a rigorous worked analysis of Contour selection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.8 (Even rational integrals diagnostic)**.** Give a rigorous worked analysis of Even rational integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Symmetry can reduce whole-line contour integrals to half-line real integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.9 (Decay on large arcs diagnostic)**.** Give a rigorous worked analysis of Decay on large arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.10 (Jordan-type estimates diagnostic)**.** Give a rigorous worked analysis of Jordan-type estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

11.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 11.11 (Contour selection diagnostic)**.** Give a rigorous worked analysis of Contour selection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.12 (Even rational integrals diagnostic)**.** Give a rigorous worked analysis of Even rational integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Symmetry can reduce whole-line contour integrals to half-line real integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.13 (Decay on large arcs diagnostic)**.** Give a rigorous worked analysis of Decay on large arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.14 (Jordan-type estimates diagnostic)**.** Give a rigorous worked analysis of Jordan-type estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.15 (Principal values diagnostic)**.** Give a rigorous worked analysis of Principal values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles on the contour require indentation and lead naturally to Cauchy principal values. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.16 (Trigonometric integrals diagnostic)**.** Give a rigorous worked analysis of Trigonometric integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.17 (Parameter differentiation diagnostic)**.** Give a rigorous worked analysis of Parameter differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.18 (Branch-aware previews diagnostic)**.** Give a rigorous worked analysis of Branch-aware previews. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 11.19 (Upper-half-plane residue method proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Suppose a rational function has no real poles and decays fast enough that the upper semicircle contribution vanishes. Then its real-line integral equals $2\pi i$ times the residues of poles in the upper half-plane.

Solution. Apply the residue theorem to the closed contour consisting of the real segment and upper semicircle, then send the radius to infinity using the decay estimate. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 11.20 (Unit-circle substitution proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a 2π-periodic rational expression in $\sin t$ and $\cos t$, the substitution $z = e^{it}$ converts the integral into a contour integral on $|z| = 1$.

Solution. Use $dt = dz/(iz)$, $\cos t = (z + z^{-1})/2$, and $\sin t = (z - z^{-1})/(2i)$; simplify to a rational function of z and apply residues inside the unit circle. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 11.21 (Principal-value indentation rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A simple pole on a smooth contour contributes half of the usual residue term under a symmetric indentation, with sign determined by the indentation orientation.

Solution. Replace a small neighborhood of the pole by a semicircular detour, approximate the integrand by its residue divided by $z - a$, and compute the limiting half-circle integral. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 11.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Real integrals can often be evaluated by embedding the integrand into a meromorphic complex function and choosing contours whose auxiliary pieces vanish or are explicitly computable. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

11.13 Exercises with complete solutions

Exercise 11.23. State and apply the central principle behind Contour selection. Give one concrete consequence.

Hint. For an even rational integrand, use an upper semicircle and relate the full real-axis integral to twice the half-line integral if symmetry permits. ☐

Solution. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.24. State and apply the central principle behind Even rational integrals. Give one concrete consequence.

Hint. Estimate the arc by ML: bound the integrand on $|z| = R$ and multiply by the arc length before sending $R \to \infty$. ☐

Solution. Symmetry can reduce whole-line contour integrals to half-line real integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.25. State and apply the central principle behind Decay on large arcs. Give one concrete consequence.

Hint. If a pole lies on the real axis, indent around it and decide whether the problem asks for an ordinary integral or principal value. ☐

Solution. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.26. State and apply the central principle behind Jordan-type estimates. Give one concrete consequence.

Hint. Choose upper or lower half-plane according to the exponential factor so the arc contribution decays. ☐

Solution. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.27. State and apply the central principle behind Principal values. Give one concrete consequence.

Hint. Track the orientation of small indentation arcs; their residue contribution often carries a half-residue factor. ☐

Solution. Poles on the contour require indentation and lead naturally to Cauchy principal values. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.28. State and apply the central principle behind Trigonometric integrals. Give one concrete consequence.

Hint. For trigonometric integrals, substitute $z = e^{it}$, $dt = dz/(iz)$, and convert sine/cosine to rational functions of z. ☐

Solution. The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.29. State and apply the central principle behind Parameter differentiation. Give one concrete consequence.

Hint. Before closing the contour, check decay degree: the large arc may not vanish for the chosen integrand. ☐

Solution. Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 11.30. State and apply the central principle behind Branch-aware previews. Give one concrete consequence.

Hint. Extract the desired real quantity only after simplifying the complex contour identity and taking real or imaginary parts as needed. ☐

Solution. Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

11.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 11.31 (Cauchy kernel integral)**.** Evaluate $\int_{-\infty}^{\infty} dx/(x^2+4)$.

Hint. Use the formula with $a = 2$. ☐

Solution. The value is $\pi/2$. ☐

Exercise 11.32 (Cosine transform)**.** For $b > 0$, evaluate $\int_{-\infty}^{\infty} \cos(bx)/(x^2+1)\,dx$.

Hint. Use the upper half-plane exponential contour. ☐

Solution. The value is πe^{-b}. ☐

Exercise 11.33 (Half line)**.** Evaluate $\int_0^{\infty} dx/(x^2+1)$.

Hint. Use evenness. ☐

Solution. It is half of π, so the value is $\pi/2$. ☐

Exercise 11.34 (Arc decay)**.** If a rational integrand is $O(R^{-3})$ on a semicircle of radius R, what order bound holds for the arc integral?

Hint. Multiply the supremum by arc length. ☐

Solution. The arc length is $O(R)$, so the integral is $O(R^{-2})$ and tends to zero. ☐

Exercise 11.35 (Quartic half line)**.** Use the full line value to compute $\int_0^{\infty} dx/(x^4+1)$.

Hint. The integrand is even. ☐

Solution. The value is $\pi/(2\sqrt{2})$. ☐

Proofs

Exercise 11.36 (Semicircle arc lemma for rational decay)**.** Prove that if $|f(z)| \leq C/R^{1+\epsilon}$ on the upper semicircle of radius R, then the arc integral tends to zero.

Hint. Use the ML inequality and arc length πR. □

Solution. The bound is $\pi RC/R^{1+\epsilon} = \pi C/R^{\epsilon} \to 0$. □

Exercise 11.37 (Cosine from exponential integral)**.** Justify extracting the cosine integral as the real part of an exponential contour integral.

Hint. Use absolute convergence or a suitable limiting argument. □

Solution. Since $e^{ibx} = \cos(bx) + i\sin(bx)$, linearity separates real and imaginary parts. For the rational denominator used here, both limits are controlled by the contour estimates. □

Exercise 11.38 (Even reduction)**.** Prove that an integrable even function satisfies $\int_{-\infty}^{\infty} f = 2\int_0^{\infty} f$.

Hint. Substitute $x = -t$ on the negative half line. □

Solution. The negative half line integral becomes $\int_0^{\infty} f(-t)dt = \int_0^{\infty} f(t)dt$, and adding the positive half gives twice that value. □

Exercise 11.39 (Choosing the half-plane)**.** For $b < 0$, explain why the lower half-plane is the natural choice for e^{ibz}.

Hint. Inspect $|e^{ib(x+iy)}|$. □

Solution. The modulus is e^{-by}. When $b < 0$, this decays for $y < 0$, so the lower semicircle provides damping rather than growth. □

Counterexamples and hypothesis tests

Exercise 11.40 (Insufficient decay)**.** Why does an $O(1/R)$ bound on a semicircle not by itself force the arc integral to vanish?

Hint. Multiply by the arc length. □

Solution. The ML bound is only $O(1)$, so it need not tend to zero. Stronger decay or oscillatory cancellation is required. □

Exercise 11.41 (Real pole)**.** What obstructs applying the ordinary upper semicircle argument when the integrand has a pole on the real axis?

Hint. The contour passes through the singularity. □

Solution. The real integral and contour integral are not ordinary integrals under the standard theorem. One needs indentation and often a principal value interpretation. □

Exercise 11.42 (Wrong exponential half-plane)**.** What happens if $b > 0$ but the contour for e^{ibz} is closed in the lower half-plane?

Hint. Check the exponential modulus. □

Solution. There $y < 0$, so e^{-by} grows exponentially. The large arc estimate generally fails. □

Applications and investigations

Exercise 11.43 (Cauchy density normalization)**.** Use the rational integral to verify that $p(x) = a/(\pi(x^2 + a^2))$ integrates to one for $a > 0$.

Hint. Factor the constant outside the known integral. □

Solution. The integral is $a/\pi \cdot \pi/a = 1$, confirming normalization of the Cauchy density. □

Exercise 11.44 (Transform pair)**.** What qualitative information does $\int e^{ibx}/(x^2 + a^2)dx = (\pi/a)e^{-a|b|}$ reveal?

Hint. Compare spatial pole distance with transform decay. □

Solution. The imaginary pole distance a becomes the exponential decay rate in frequency. Singularities away from the real axis control transform decay. □

Challenge problems

Exercise 11.45 (Squared denominator)**.** Outline how to evaluate $\int_{-\infty}^{\infty} dx/(x^2 + a^2)^2$.

Hint. There is a second order pole at ia; use the higher pole residue formula. □

Solution. The upper half-plane residue calculation yields $\pi/(2a^3)$. The arc still vanishes because the integrand decays as R^{-4}. □

Exercise 11.46 (Indented contour preview)**.** Explain the sign of the small upper indentation around a simple real pole when computing a principal value.

Hint. The small arc is traversed clockwise as the main contour detours above the pole. □

Solution. A clockwise half circle contributes asymptotically $-i\pi$ times the residue. Moving that term to the other side of the contour identity produces the familiar principal value contribution. □

Part III

Global Complex Analysis

Chapter 12

Winding Numbers and the Argument Principle

Winding numbers measure how closed curves wrap around points, while the argument principle converts winding into an exact count of zeros and poles. This chapter is the bridge from contour integration to global counting theorems.

Learning goals

After this chapter, the reader should be able to:

- compute winding numbers of explicit closed curves around specified points;
- apply the argument principle to count zeros minus poles inside a contour;
- derive zero/pole counts from the logarithmic derivative f'/f;
- track multiplicities correctly in argument-principle calculations;
- relate change of argument along a contour to topological winding;
- identify failures when the contour passes through zeros or poles;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

12.1 Index of a closed curve

For a closed curve Γ avoiding a, the index $\mathrm{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a.

Example 12.1 (Winding number of a translated circle)**.** Let $\gamma(t) = 2 + 3e^{it}$, $0 \le t \le 2\pi$. Around the point $a = 2$,

$$\mathrm{Ind}(\gamma, 2) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-2} = 1.$$

Around $a = 6$, the point lies outside the circle and the index is zero. The index records how the curve winds around a chosen point, not merely the shape of the curve itself.

12.2 Homotopy invariance

The winding number is unchanged under deformations of the curve that avoid the reference point.

12.3 Local constancy

As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement.

12.4 Logarithmic derivative

For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities.

Example 12.2 (Argument principle for a polynomial)**.** Take $f(z) = z^3 - 1$ on $|z| = 2$. There are no zeros or poles on the contour, and all three roots lie inside. Therefore

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{f'(z)}{f(z)} \, dz = 3.$$

Equivalently, the image curve $f(2e^{it})$ winds three times around the origin.

12.5 Argument principle

The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity.

12.6 Change of argument

The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$.

12.7 Counting polynomial roots

Contours can count roots in regions without locating them explicitly.

Example 12.3 (Zeros minus poles)**.** For $f(z) = (z-1)^2/(z+1)$ on $|z| = 2$, the contour encloses a double zero at 1 and a simple pole at -1. The argument principle gives

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{f'}{f} \, dz = 2 - 1 = 1.$$

Multiplicity and pole order enter with opposite signs.

12.8 General residue formulation

The winding-number form of the residue theorem allows multiply wound contours and non-simple geometry.

12.9 Boundary hypotheses

The argument principle requires the contour to avoid zeros and poles of the meromorphic function.

12.10 Topological synthesis

Winding, residues, multiplicity, and image curves are different faces of the same integer-valued invariant.

12.11 Core structural results

Theorem 12.4 (Integer-valued index). *For a closed piecewise C^1 curve Γ and $a \notin \Gamma$, $\operatorname{Ind}(\Gamma, a)$ is an integer.*

Proof. Locally choose a branch of the logarithm along successive short pieces of the curve. The integral of $(z-a)^{-1}$ is the net change of logarithm, and after one circuit the logarithm can change only by an integral multiple of $2\pi i$. ☐

Theorem 12.5 (Argument principle). *If f is meromorphic inside and on Γ, with no zeros or poles on Γ, then $\frac{1}{2\pi i}\int_\Gamma f'(z)/f(z)\,dz = N - P$, counting zeros and poles with multiplicity in the simple winding-one case.*

Proof. The logarithmic derivative has a simple pole at every zero with residue equal to its multiplicity and at every pole with residue minus its order. Apply the residue theorem. ☐

Theorem 12.6 (Homotopy invariance of index). *If two closed curves are homotopic in $\mathbb{C} \setminus \{a\}$, then they have the same winding number about a.*

Proof. Apply homotopy invariance of contour integrals to the holomorphic function $(z-a)^{-1}$ on the punctured plane. ☐

Theorem 12.7 (Local constancy of index). *The map $a \mapsto \operatorname{Ind}(\Gamma, a)$ is constant on every connected component of $\mathbb{C} \setminus \Gamma$.*

Proof. The integral depends continuously on a off the curve, while its values are integers. A continuous integer-valued function is locally constant, hence constant on connected components. ☐

12.12 Worked examples

Example 12.8 (Index of a closed curve diagnostic). Give a rigorous worked analysis of Index of a closed curve. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a closed curve Γ avoiding a, the index $\operatorname{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.9 (Homotopy invariance diagnostic). Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The winding number is unchanged under deformations of the curve that avoid the reference point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.10 (Local constancy diagnostic). Give a rigorous worked analysis of Local constancy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.11 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

12.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 12.12 (Index of a closed curve diagnostic)**.** Give a rigorous worked analysis of Index of a closed curve. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a closed curve Γ avoiding a, the index $\mathrm{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.13 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The winding number is unchanged under deformations of the curve that avoid the reference point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.14 (Local constancy diagnostic)**.** Give a rigorous worked analysis of Local constancy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.15 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.16 (Argument principle diagnostic)**.** Give a rigorous worked analysis of Argument principle. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.17 (Change of argument diagnostic)**.** Give a rigorous worked analysis of Change of argument. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.18 (Counting polynomial roots diagnostic)**.** Give a rigorous worked analysis of Counting polynomial roots. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Contours can count roots in regions without locating them explicitly. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.19 (Integer-valued index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a closed piecewise C^1 curve Γ and $a \notin \Gamma$, $\mathrm{Ind}(\Gamma, a)$ is an integer.

Solution. Locally choose a branch of the logarithm along successive short pieces of the curve. The integral of $(z-a)^{-1}$ is the net change of logarithm, and after one circuit the logarithm can change only by an integral multiple of $2\pi i$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 12.20 (Argument principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is meromorphic inside and on Γ, with no zeros or poles on Γ, then $\frac{1}{2\pi i}\int_\Gamma f'(z)/f(z)\,dz = N - P$, counting zeros and poles with multiplicity in the simple winding-one case.

Solution. The logarithmic derivative has a simple pole at every zero with residue equal to its multiplicity and at every pole with residue minus its order. Apply the residue theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 12.21 (Homotopy invariance of index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two closed curves are homotopic in $\mathbb{C} \setminus \{a\}$, then they have the same winding number about a.

Solution. Apply homotopy invariance of contour integrals to the holomorphic function $(z-a)^{-1}$ on the punctured plane. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 12.22 (Local constancy of index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The map $a \mapsto \mathrm{Ind}(\Gamma, a)$ is constant on every connected component of $\mathbb{C} \setminus \Gamma$.

Solution. The integral depends continuously on a off the curve, while its values are integers. A continuous integer-valued function is locally constant, hence constant on connected components. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 12.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Winding numbers measure how closed curves wrap around points, while the argument principle converts winding into an exact count of zeros and poles. This chapter is the bridge from contour integration to global counting theorems. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

12.14 Exercises with complete solutions

Exercise 12.24. State and apply the central principle behind Index of a closed curve. Give one concrete consequence.

Hint. Compute winding number as $(2\pi i)^{-1} \int_\Gamma dz/(z-a)$ or from net change of argument. ☐

Solution. For a closed curve Γ avoiding a, the index $\operatorname{Ind}(\Gamma, a) = \frac{1}{2\pi i} \int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.25. State and apply the central principle behind Homotopy invariance. Give one concrete consequence.

Hint. Deform the curve without crossing a; winding number stays unchanged under such homotopies. ☐

Solution. The winding number is unchanged under deformations of the curve that avoid the reference point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.26. State and apply the central principle behind Local constancy. Give one concrete consequence.

Hint. For the argument principle, apply the residue theorem to f'/f and interpret residues as multiplicities. ☐

Solution. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.27. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Count zeros and poles with multiplicity; a double zero contributes two, not one. ☐

Solution. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.28. State and apply the central principle behind Argument principle. Give one concrete consequence.

Hint. Make sure f has no zeros or poles on the contour before applying the principle. ☐

Solution. The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.29. State and apply the central principle behind Change of argument. Give one concrete consequence.

Hint. Track the change of $\arg f(\Gamma(t))$; its total increment is $2\pi(N-P)$. ☐

Solution. The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.30. State and apply the central principle behind Counting polynomial roots. Give one concrete consequence.

Hint. Use winding numbers around individual poles when the contour is not a simple Jordan curve. ☐

Solution. Contours can count roots in regions without locating them explicitly. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.31. State and apply the central principle behind General residue formulation. Give one concrete consequence.

Hint. For polynomial root counts, choose a contour on which the leading behavior makes f'/f or Rouché comparison manageable. ☐

Solution. The winding-number form of the residue theorem allows multiply wound contours and non-simple geometry. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

12.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 12.32 (Unit circle index)**.** Compute the winding number of e^{it}, $0 \le t \le 2\pi$, about zero.

Hint. Use the logarithmic integral definition. ☐

Solution. The index is one. ☐

Exercise 12.33 (Double traversal)**.** Compute the winding number of e^{2it} about zero for $0 \le t \le 2\pi$.

Hint. Track the change of argument. ☐

Solution. The argument increases by 4π, so the index is two. ☐

Exercise 12.34 (Clockwise circle)**.** Find the index of e^{-it} about zero.

Hint. Orientation matters. ☐

Solution. The index is minus one. ☐

Exercise 12.35 (Zero count)**.** Use the argument principle to count zeros of $z^4 + 1$ in $|z| = 2$.

Hint. All four roots have modulus one. ☐

Solution. There are four zeros, counted with multiplicity, and no poles. □

Exercise 12.36 (Zero minus pole)**.** For $f(z) = z^3/(z-1)^2$, compute the argument-principle count on $|z| = 2$.

Hint. Count orders inside. □

Solution. The result is $3 - 2 = 1$. □

Proofs

Exercise 12.37 (Index is integer)**.** Explain why the winding number of a closed curve about a point not on the curve is an integer.

Hint. Use a continuous change of argument along the curve. □

Solution. Choose a continuous argument along the parameter interval. Closedness forces the final argument to differ from the initial one by an integer multiple of 2π, and that integer is the index. □

Exercise 12.38 (Homotopy invariance)**.** Prove that winding number is unchanged under a deformation avoiding the reference point.

Hint. The index varies continuously with the deformation parameter but takes integer values. □

Solution. The integral defining the index depends continuously on the homotopy. A continuous integer valued function is locally constant, hence constant on the connected parameter interval. □

Exercise 12.39 (Argument principle)**.** Prove the argument principle from residues of f'/f.

Hint. Find the local residue at a zero or pole. □

Solution. A zero of order m gives residue m, while a pole of order n gives residue $-n$. The residue theorem sums these local integers to give zeros minus poles. □

Exercise 12.40 (Change of argument form)**.** Relate $\int f'/f$ to the net change of argument of f along a contour.

Hint. Parametrize the contour and differentiate a local logarithm. □

Solution. Along intervals where a logarithm is chosen, $d \log f = f'/f\, dz$. Its real part returns to the initial value on a closed contour, while the imaginary part changes by the total argument increment. □

Counterexamples and hypothesis tests

Exercise 12.41 (Point on curve)**.** Can the winding number about a be defined by the usual integral if a lies on the curve?

Hint. The kernel has a pole on the path. □

Solution. No. The integral is singular and the standard index is undefined. □

Exercise 12.42 (Zero on contour)**.** Can the argument principle be applied if f has a zero on the contour?

Hint. Then f'/f is singular on the path. □

Solution. No. Zeros and poles must be absent from the contour. □

Exercise 12.43 (Ignoring multiplicity)**.** What error arises if a double zero is counted only once in the argument principle?

Hint. The local residue equals the order. □

Solution. The count is off by one because a double zero contributes two, not one. □

Applications and investigations

Exercise 12.44 (Root counting without solving)**.** Why is the argument principle useful for a high degree polynomial?

Hint. It converts root counting into a contour integral or argument change. □

Solution. One can determine the number of roots in a region without finding their individual values, provided the boundary contains no roots. □

Exercise 12.45 (Nyquist style interpretation)**.** Interpret the argument principle geometrically in terms of the image of the boundary under f.

Hint. Track how $f(\gamma)$ winds around zero. □

Solution. The net winding of the image about zero equals the number of enclosed zeros minus poles, counted with multiplicity. □

Challenge problems

Exercise 12.46 (Meromorphic divisor count)**.** Show that a meromorphic function on a bounded region with no boundary zeros or poles has total divisor degree equal to the winding of its boundary values about zero.

Hint. Apply the argument principle. □

Solution. The divisor degree is the sum of zero orders minus pole orders. The argument principle identifies this integer with $(2\pi i)^{-1} \int f'/f$, which is the winding number of the image curve. □

Exercise 12.47 (Nested contours)**.** Two nested contours contain no zeros or poles in the annulus between them. Compare their argument-principle counts.

Hint. Apply the residue theorem to f'/f on the annulus. □

Solution. The two counts are equal because the difference of the contour integrals is the integral over the annular boundary, which vanishes when no zeros or poles lie between them. □

Chapter 13

Rouché's Theorem

Rouché's theorem compares two holomorphic functions on a boundary and concludes that they have the same number of interior zeros. It is one of the most efficient tools for root counting and perturbative existence.

Learning goals

After this chapter, the reader should be able to:

- verify Rouché inequalities on a contour and choose the dominant term strategically;
- use Rouché's theorem to count zeros in disks, annuli, or half-planes after suitable changes of variable;
- compare candidate functions on the boundary rather than throughout the interior;
- track multiplicities in the zero count;
- combine Rouché with the argument principle when direct estimates are awkward;
- construct examples where a non-strict boundary inequality is insufficient for the theorem;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

13.1 Boundary domination

The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary.

Example 13.1 (One zero in the unit disk)**.** On $|z| = 1$, compare $f(z) = z^5 + 3z + 1$ with $g(z) = 3z$. We have $|f - g| = |z^5 + 1| \leq 2 < 3 = |g|$. Rouché's theorem therefore says that f and $3z$ have the same number of zeros in the unit disk: exactly one, counted with multiplicity.

13.2 Homotopy viewpoint

The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change.

13.3 Zero counting

The argument principle turns that invariant winding number into equality of zero counts.

13.4 Polynomial applications

Dominant monomials on circles can count roots inside disks.

Example 13.2 (All five zeros in a larger disk)**.** On $|z| = 2$, compare the same polynomial with $g(z) = z^5$. Then

$$|3z + 1| \leq 7 < 32 = |z^5|.$$

Thus $z^5 + 3z + 1$ has five zeros in $|z| < 2$. Combined with the unit disk calculation, four zeros lie in the annulus $1 < |z| < 2$.

13.5 Perturbation stability

Small holomorphic perturbations preserve the number of zeros inside a protected contour.

13.6 Localization

Choosing several contours can isolate different root clusters.

13.7 Comparison choices

The art is to choose the part of the function that dominates on the chosen boundary.

Example 13.3 (Perturbing a known zero count)**.** Suppose p has no zeros on a contour C, and $|h| < |p|$ on C. Then $p + h$ has exactly the same number of zeros inside. This is not merely qualitative continuity: the strict boundary inequality supplies a verifiable certificate that the zero count is unchanged.

13.8 Multiplicity

Rouché counts zeros with multiplicity, exactly as the argument principle does.

13.9 Core structural results

Theorem 13.4 (Rouché theorem)**.** *If f, g are holomorphic inside and on a simple closed contour Γ and $|g| < |f|$ on Γ, then f and $f + g$ have the same number of zeros inside, counted with multiplicity.*

Proof. For $0 \leq t \leq 1$, $f + tg$ is nonzero on Γ because $|tg| < |f|$. Hence the winding number of $(f + tg)(\Gamma)$ about zero is constant in t. Apply the argument principle at $t = 0$ and $t = 1$. □

Theorem 13.5 (Polynomial root localization)**.** *If on $|z| = R$ one term of a polynomial strictly dominates the sum of all others, then the polynomial has the same number of zeros in $|z| < R$ as that dominant term.*

Proof. Apply Rouché with the dominant term as f and the sum of the remaining terms as g. □

Theorem 13.6 (Perturbative persistence)**.** *If f has no zeros on Γ and $\|g\|_\Gamma < \min_\Gamma |f|$, then $f + g$ has the same interior zero count as f.*

Proof. The displayed inequality is precisely the strict Rouché hypothesis. □

13.10 Worked examples

Example 13.7 (Boundary domination diagnostic)**.** Give a rigorous worked analysis of Boundary domination. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.8 (Homotopy viewpoint diagnostic)**.** Give a rigorous worked analysis of Homotopy viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.9 (Zero counting diagnostic)**.** Give a rigorous worked analysis of Zero counting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The argument principle turns that invariant winding number into equality of zero counts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.10 (Polynomial applications diagnostic)**.** Give a rigorous worked analysis of Polynomial applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Dominant monomials on circles can count roots inside disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

13.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 13.11 (Boundary domination diagnostic)**.** Give a rigorous worked analysis of Boundary domination. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.12 (Homotopy viewpoint diagnostic)**.** Give a rigorous worked analysis of Homotopy viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.13 (Zero counting diagnostic)**.** Give a rigorous worked analysis of Zero counting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument principle turns that invariant winding number into equality of zero counts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.14 (Polynomial applications diagnostic)**.** Give a rigorous worked analysis of Polynomial applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Dominant monomials on circles can count roots inside disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.15 (Perturbation stability diagnostic)**.** Give a rigorous worked analysis of Perturbation stability. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Small holomorphic perturbations preserve the number of zeros inside a protected contour. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.16 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Choosing several contours can isolate different root clusters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.17 (Comparison choices diagnostic)**.** Give a rigorous worked analysis of Comparison choices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The art is to choose the part of the function that dominates on the chosen boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.18 (Multiplicity diagnostic)**.** Give a rigorous worked analysis of Multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Rouché counts zeros with multiplicity, exactly as the argument principle does. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.19 (Rouché theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f, g are holomorphic inside and on a simple closed contour Γ and $|g| < |f|$ on Γ, then f and $f + g$ have the same number of zeros inside, counted with multiplicity.

Solution. For $0 \le t \le 1$, $f + tg$ is nonzero on Γ because $|tg| < |f|$. Hence the winding number of $(f + tg)(\Gamma)$ about zero is constant in t. Apply the argument principle at $t = 0$ and $t = 1$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.20 (Polynomial root localization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If on $|z| = R$ one term of a polynomial strictly dominates the sum of all others, then the polynomial has the same number of zeros in $|z| < R$ as that dominant term.

Solution. Apply Rouché with the dominant term as f and the sum of the remaining terms as g. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.21 (Perturbative persistence proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f has no zeros on Γ and $\|g\|_\Gamma < \min_\Gamma |f|$, then $f + g$ has the same interior zero count as f.

Solution. The displayed inequality is precisely the strict Rouché hypothesis. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Rouché's theorem compares two holomorphic functions on a boundary and concludes that they have the same number of interior zeros. It is one of the most efficient tools for root counting and perturbative existence. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

13.12 Exercises with complete solutions

Exercise 13.23. State and apply the central principle behind Boundary domination. Give one concrete consequence.

Hint. Choose a decomposition $f = g + h$ so that $|h| < |g|$ on the contour; the inequality need not hold inside. □

Solution. The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.24. State and apply the central principle behind Homotopy viewpoint. Give one concrete consequence.

Hint. Estimate each term uniformly on the boundary using the contour radius rather than pointwise interior bounds. □

Solution. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.25. State and apply the central principle behind Zero counting. Give one concrete consequence.

Hint. Once the strict inequality is proved, transfer the zero count with multiplicity from g to f. ☐

Solution. The argument principle turns that invariant winding number into equality of zero counts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 13.26. State and apply the central principle behind Polynomial applications. Give one concrete consequence.

Hint. If neither term dominates on one contour, change the radius or regroup terms before abandoning Rouché. ☐

Solution. Dominant monomials on circles can count roots inside disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 13.27. State and apply the central principle behind Perturbation stability. Give one concrete consequence.

Hint. Use the theorem on an annulus by comparing zero counts on outer and inner boundaries separately if needed. ☐

Solution. Small holomorphic perturbations preserve the number of zeros inside a protected contour. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 13.28. State and apply the central principle behind Localization. Give one concrete consequence.

Hint. Non-strict $|h| \leq |g|$ is not the standard hypothesis; identify where equality could allow zeros on the boundary. ☐

Solution. Choosing several contours can isolate different root clusters. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 13.29. State and apply the central principle behind Comparison choices. Give one concrete consequence.

Hint. Combine Rouché with a simple model polynomial whose zeros are already known. ☐

Solution. The art is to choose the part of the function that dominates on the chosen boundary. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 13.30. State and apply the central principle behind Multiplicity. Give one concrete consequence.

Hint. After counting, check whether any zeros could lie exactly on the contour; the theorem excludes that situation. ☐

Solution. Rouché counts zeros with multiplicity, exactly as the argument principle does. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

13.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 13.31 (Dominant monomial)**.** How many zeros does $z^7 + 2$ have in $|z| < 2$?

Hint. Compare with z^7 on $|z| = 2$. □

Solution. Since $2 < 2^7$, Rouché gives seven zeros. □

Exercise 13.32 (Small disk)**.** How many zeros does $z^4 + 5z + 1$ have in $|z| < 1$?

Hint. Compare with $5z$. □

Solution. On the unit circle, $|z^4 + 1| \leq 2 < 5$, so there is one zero. □

Exercise 13.33 (Constant dominance)**.** Show that $2 + z + z^2$ has no zeros in $|z| < 1/2$.

Hint. Compare with the constant two. □

Solution. On $|z| = 1/2$, $|z + z^2| \leq 3/4 < 2$. Hence the polynomial has the same zero count as the nonzero constant, namely zero. □

Exercise 13.34 (Annular subtraction)**.** A polynomial has two zeros in $|z| < 1$ and six in $|z| < 3$. How many lie in $1 < |z| < 3$?

Hint. Subtract the counts. □

Solution. There are four zeros in the annulus. □

Exercise 13.35 (Multiplicity)**.** If Rouché compares a function to $(z - a)^4$, how many zeros are certified inside the contour?

Hint. Zeros are counted with multiplicity. □

Solution. Exactly four zeros, counted with multiplicity. □

Proofs

Exercise 13.36 (Rouché from argument principle)**.** Prove Rouché's theorem using the family $f_t = f + tg$ when $|g| < |f|$ on the contour.

Hint. Show no f_t vanishes on the boundary. □

Solution. The inequality gives $|f_t| \geq |f| - t|g| > 0$. The boundary winding number of f_t about zero is integer valued and continuous in t, hence constant. The argument principle gives equal zero counts at $t = 0$ and $t = 1$. □

Exercise 13.37 (Symmetric version)**.** Show that if $|f - g| < |f|$ on a contour, then f and g have the same number of zeros inside.

Hint. Apply Rouché to f and perturbation $g - f$. ☐

Solution. The hypothesis is precisely $|g - f| < |f|$, so $f + (g - f) = g$ has the same zero count as f. ☐

Exercise 13.38 (Local stability)**.** Prove that a sufficiently small uniform perturbation preserves the number of zeros in a disk whose boundary contains no zeros.

Hint. The minimum of $|f|$ on the compact boundary is positive. ☐

Solution. Let $m = \min_C |f| > 0$. Any perturbation h with $\max_C |h| < m$ satisfies Rouché, so $f + h$ has the same number of zeros. ☐

Exercise 13.39 (FTA by Rouché)**.** Use Rouché to prove that a degree n polynomial has n zeros counted with multiplicity.

Hint. On a sufficiently large circle, the leading term dominates all lower terms. ☐

Solution. For large R, $|a_n z^n| > \sum_{k<n} |a_k| R^k$ on $|z| = R$. Thus the polynomial has the same zero count as $a_n z^n$, namely n. ☐

Counterexamples and hypothesis tests

Exercise 13.40 (Non-strict inequality)**.** Why is $|g| \leq |f|$ not the standard sufficient hypothesis in Rouché's theorem?

Hint. Equality can allow the homotopy to pass through zero on the boundary. ☐

Solution. The strict inequality guarantees boundary nonvanishing for the entire deformation. Without it, additional analysis is needed and the conclusion can fail. ☐

Exercise 13.41 (Zero on boundary)**.** Why must the comparison be arranged so the relevant functions have no boundary zeros?

Hint. The zero count can jump when a zero crosses the contour. ☐

Solution. A boundary zero makes the argument-principle winding undefined and destroys the homotopy certificate used in the proof. ☐

Exercise 13.42 (Poor comparison)**.** If neither term dominates on the chosen circle, does Rouché imply that the zero counts differ?

Hint. Failure of a sufficient condition is not a negative conclusion. ☐

Solution. No. It only means that this comparison and contour do not certify the count. A different comparison or contour may work. ☐

Applications and investigations

Exercise 13.43 (Root localization)**.** How can two Rouché counts localize polynomial roots to an annulus?

Hint. Count inside an inner and an outer circle. □

Solution. Subtracting the certified counts gives exactly how many roots lie between the circles, without solving the polynomial. □

Exercise 13.44 (Perturbed characteristic equation)**.** A characteristic polynomial p has a known zero count in a region. How can Rouché certify that modeling error h does not change it?

Hint. Bound $|h|$ by the minimum boundary magnitude of p. □

Solution. If $\max_C |h| < \min_C |p|$, then $p + h$ and p have identical zero counts inside the boundary. □

Challenge problems

Exercise 13.45 (Two-scale count)**.** For $z^6 + 4z^2 + 1$, find useful circles on which different terms dominate and explain how to infer an annular root count.

Hint. Try a small circle where the constant dominates and a larger circle where z^6 dominates. □

Solution. Choose radii satisfying $4R^2 + R^6 < 1$ for the inner disk and $4R^2 + 1 < R^6$ for the outer disk. Rouché then gives zero inner roots and six outer roots, so all six lie in the intervening annulus. □

Exercise 13.46 (Comparison with a factor)**.** Suppose $f(z) = p(z)+h(z)$ and p has a multiple zero. Does Rouché preserve the multiplicity count under a small boundary perturbation?

Hint. Rouché counts zeros with multiplicity rather than tracking individual labels. □

Solution. Yes. It preserves the total number counted with multiplicity inside the contour, even though a multiple zero may split into several nearby simple zeros. □

Chapter 14

Branches of the Logarithm and Roots

Branches of logarithms and roots exist exactly when the topology of the domain allows one to choose arguments continuously. The obstruction is winding around the origin.

Learning goals

After this chapter, the reader should be able to:

- construct branches of the logarithm on simply connected domains avoiding the origin;
- define branches of powers and roots from a chosen logarithm;
- track argument ranges and branch cuts explicitly;
- determine how analytic continuation around the origin changes logarithm and root branches;
- distinguish local branches from globally single-valued functions;
- choose branch cuts adapted to contour-integral and mapping problems;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

14.1 Multivalued logarithm

Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$.

Example 14.1 (Principal logarithm)**.** On $\mathbb{C} \setminus (-\infty, 0]$, define

$$\operatorname{Log} z = \log |z| + i \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z < \pi.$$

This is holomorphic and satisfies $(\operatorname{Log} z)' = 1/z$. The removed ray prevents a loop around zero from forcing the argument to jump by 2π.

14.2 Branches

A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$.

14.3 Argument functions

Writing $L(z) = \log|z| + i\operatorname{Arg}(z)$ links branches of logarithm to continuous arguments.

14.4 Slit domains

Removing a ray from the plane yields standard simply connected domains supporting logarithm branches.

Example 14.2 (A square-root branch)**.** On the same slit plane, define

$$\sqrt{z} = \exp\Bigl(\tfrac{1}{2}\operatorname{Log} z\Bigr).$$

Then $(\sqrt{z})^2 = z$, and the chosen branch has positive real value on the positive real axis. The second branch is its negative.

14.5 Nth roots

A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation.

14.6 Powers

For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch.

14.7 Monodromy

Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity.

Example 14.3 (Why the punctured plane has no global logarithm)**.** If a holomorphic logarithm L existed on $\mathbb{C} \setminus \{0\}$, then $L' = 1/z$. But

$$\int_{|z|=1} \frac{dz}{z} = 2\pi i \neq 0,$$

whereas the integral of a derivative around a closed curve is zero. Thus no single valued holomorphic logarithm exists on the punctured plane.

14.8 Integral criterion

A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes.

14.9 Topological obstruction

Nonzero winding around zero prevents a globally single-valued logarithm.

14.10 Core structural results

Theorem 14.4 (Logarithm branch criterion)**.** *On a domain $D \subset \mathbb{C} \setminus \{0\}$, a holomorphic logarithm exists if $\int_\gamma dz/z = 0$ for every closed contour γ in D.*

Proof. The vanishing integral condition gives a primitive L of $1/z$. Then $e^{-L(z)}z$ has derivative zero, so after adding a constant to L one obtains $e^{L(z)} = z$. □

Theorem 14.5 (Simply connected logarithm theorem)**.** *Every simply connected domain avoiding zero admits a holomorphic logarithm.*

Proof. The function $1/z$ is holomorphic on the domain, and Cauchy's theorem makes all its closed contour integrals vanish. Apply the logarithm branch criterion. □

Theorem 14.6 (Root from logarithm)**.** *If L is a logarithm branch on D, then $R(z) = e^{L(z)/n}$ is a holomorphic branch of the nth root.*

Proof. Directly, $R(z)^n = e^{L(z)} = z$, and holomorphicity follows from composition. □

14.11 Worked examples

Example 14.7 (Multivalued logarithm diagnostic)**.** Give a rigorous worked analysis of Multivalued logarithm. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.8 (Branches diagnostic)**.** Give a rigorous worked analysis of Branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.9 (Argument functions diagnostic)**.** Give a rigorous worked analysis of Argument functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Writing $L(z) = \log|z| + i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.10 (Slit domains diagnostic)**.** Give a rigorous worked analysis of Slit domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

14.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 14.11 (Multivalued logarithm diagnostic)**.** Give a rigorous worked analysis of Multivalued logarithm. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.12 (Branches diagnostic)**.** Give a rigorous worked analysis of Branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.13 (Argument functions diagnostic)**.** Give a rigorous worked analysis of Argument functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Writing $L(z) = \log|z| + i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.14 (Slit domains diagnostic)**.** Give a rigorous worked analysis of Slit domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.15 (Nth roots diagnostic)**.** Give a rigorous worked analysis of Nth roots. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.16 (Powers diagnostic)**.** Give a rigorous worked analysis of Powers. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.17 (Monodromy diagnostic)**.** Give a rigorous worked analysis of Monodromy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.18 (Integral criterion diagnostic)**.** Give a rigorous worked analysis of Integral criterion. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 14.19 (Logarithm branch criterion proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: On a domain $D \subset \mathbb{C} \setminus \{0\}$, a holomorphic logarithm exists if $\int_\gamma dz/z = 0$ for every closed contour γ in D.

Solution. The vanishing integral condition gives a primitive L of $1/z$. Then $e^{-L(z)}z$ has derivative zero, so after adding a constant to L one obtains $e^{L(z)} = z$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.20 (Simply connected logarithm theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every simply connected domain avoiding zero admits a holomorphic logarithm.

Solution. The function $1/z$ is holomorphic on the domain, and Cauchy's theorem makes all its closed contour integrals vanish. Apply the logarithm branch criterion. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.21 (Root from logarithm proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If L is a logarithm branch on D, then $R(z) = e^{L(z)/n}$ is a holomorphic branch of the nth root.

Solution. Directly, $R(z)^n = e^{L(z)} = z$, and holomorphicity follows from composition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Branches of logarithms and roots exist exactly when the topology of the domain allows one to choose arguments continuously. The obstruction is winding around the origin. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

14.13 Exercises with complete solutions

Exercise 14.23. State and apply the central principle behind Multivalued logarithm. Give one concrete consequence.

Hint. Choose a continuous argument range on the domain first; then define $\log z = \ln|z| + i \arg z$. ☐

Solution. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.24. State and apply the central principle behind Branches. Give one concrete consequence.

Hint. A simply connected domain avoiding zero admits a logarithm branch; verify both topology and omission of zero. ☐

Solution. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.25. State and apply the central principle behind Argument functions. Give one concrete consequence.

Hint. Define $z^\alpha = e^{\alpha \log z}$ only after fixing the logarithm branch. ☐

Solution. Writing $L(z) = \log |z| + i \operatorname{Arg}(z)$ links branches of logarithm to continuous arguments. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.26. State and apply the central principle behind Slit domains. Give one concrete consequence.

Hint. For an n-th root, changing the logarithm by $2\pi i$ multiplies the value by an n-th root of unity. ☐

Solution. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.27. State and apply the central principle behind Nth roots. Give one concrete consequence.

Hint. Analytic continuation once around zero adds $2\pi i$ to logarithm; track this monodromy explicitly. ☐

Solution. A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.28. State and apply the central principle behind Powers. Give one concrete consequence.

Hint. Place branch cuts so the contour does not cross them and the argument remains continuous along each side. ☐

Solution. For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.29. State and apply the central principle behind Monodromy. Give one concrete consequence.

Hint. Different branch choices differ by integer multiples of $2\pi i$; use this to compare formulas. ☐

Solution. Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.30. State and apply the central principle behind Integral criterion. Give one concrete consequence.

Hint. Do not treat a principal branch identity as globally valid across its cut or around the origin. □

Solution. A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

14.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 14.31 (Principal value)**.** Compute the principal logarithm of $-i$.

Hint. Use principal argument $-\pi/2$. □

Solution. The value is $-i\pi/2$. □

Exercise 14.32 (Principal square root)**.** Compute the principal square root of -1.

Hint. Use principal logarithm with argument π. □

Solution. The principal square root is i. □

Exercise 14.33 (Branch interval)**.** Give an argument interval defining a logarithm on the plane cut along the positive real axis.

Hint. Use any interval of length 2π that excludes the cut direction. □

Solution. For example, $0 < \arg z < 2\pi$. □

Exercise 14.34 (Complex power)**.** On a chosen logarithm branch, define z^α.

Hint. Exponentiate $\alpha \operatorname{Log} z$. □

Solution. Set $z^\alpha = \exp(\alpha \operatorname{Log} z)$ on the branch domain. □

Exercise 14.35 (Root branches)**.** How many holomorphic branches of the n-th root differ by constant factors on a connected branch domain?

Hint. Multiply one branch by the n-th roots of unity. □

Solution. There are n choices, obtained from one branch by factors $e^{2\pi ik/n}$. □

Proofs

Exercise 14.36 (Derivative of a logarithm branch)**.** Prove that any holomorphic branch L with $e^L = z$ satisfies $L' = 1/z$.

Hint. Differentiate the identity $e^{L(z)} = z$. ☐

Solution. The chain rule gives $e^{L(z)}L'(z) = 1$. Since $e^{L(z)} = z$, one gets $L'(z) = 1/z$. ☐

Exercise 14.37 (Branches differ by constants)**.** Show that two logarithm branches on a connected domain differ by an integer multiple of $2\pi i$.

Hint. Exponentiate their difference. ☐

Solution. If L_1, L_2 are branches, $e^{L_1 - L_2} = 1$. The difference is continuous and takes values in the discrete set $2\pi i\mathbb{Z}$, hence is constant on the connected domain. ☐

Exercise 14.38 (Root from logarithm)**.** Prove that $\exp(L/n)$ is an n-th root branch when L is a logarithm branch.

Hint. Raise the expression to the n-th power. ☐

Solution. Its n-th power is $e^L = z$, and holomorphicity follows by composition. ☐

Exercise 14.39 (No logarithm on a winding domain)**.** Show that if a domain contains a closed curve with nonzero winding about zero, it cannot admit a holomorphic logarithm of z.

Hint. A logarithm would have derivative $1/z$. ☐

Solution. The integral of its derivative around the curve would be zero, but $\int dz/z = 2\pi i$ times the nonzero winding number, a contradiction. ☐

Counterexamples and hypothesis tests

Exercise 14.40 (Branch crossing)**.** Why is the principal logarithm discontinuous if one tries to include both sides of its cut with the same argument convention?

Hint. Approach a negative real point from above and below. ☐

Solution. The principal arguments approach π and $-\pi$, differing by 2π, so no continuous single valued extension crosses the cut. ☐

Exercise 14.41 (Punctured annulus)**.** Does every annulus around zero admit a single valued logarithm?

Hint. An annulus contains loops winding once around zero. ☐

Solution. No. The integral of $1/z$ on such a loop is nonzero, obstructing a logarithm. ☐

Exercise 14.42 (Branch independence of powers)**.** Is z^α independent of logarithm branch for arbitrary complex α?

Hint. Changing a logarithm by $2\pi ik$ changes the exponential by a factor. ☐

Solution. In general no. The factor is $e^{2\pi ik\alpha}$, which equals one for every integer k only in special cases such as integer α. ☐

Applications and investigations

Exercise 14.43 (Fractional power domain)**.** Why is a slit plane a natural domain for $z^{1/3}$?

Hint. It supports a single valued logarithm. ☐

Solution. Once a logarithm is fixed, $z^{1/3} = e^{L/3}$ is holomorphic there, avoiding the monodromy caused by circling zero. ☐

Exercise 14.44 (Primitive of reciprocal)**.** On what kind of region can $1/z$ have a holomorphic primitive?

Hint. A primitive is a logarithm branch. ☐

Solution. Exactly on regions where a logarithm branch exists; in particular, simply connected regions avoiding zero support such a primitive. ☐

Challenge problems

Exercise 14.45 (Branch on a sector)**.** Construct a logarithm on the sector $\alpha < \arg z < \beta$ when $\beta - \alpha < 2\pi$.

Hint. Choose the argument continuously in that interval. ☐

Solution. Define $L(z) = \log|z| + i\arg z$ with $\arg z \in (\alpha, \beta)$. The sector cannot wrap fully around zero, so the argument is single valued. ☐

Exercise 14.46 (Monodromy of the square root)**.** What happens to a locally chosen square root after analytic continuation once around zero?

Hint. The logarithm changes by $2\pi i$. ☐

Solution. Half the logarithm changes by πi, so the exponential is multiplied by -1. A second circuit returns to the original value. ☐

Chapter 15

Analytic Continuation

Analytic continuation extends local holomorphic data across overlapping domains. The identity theorem guarantees uniqueness, while singularities and monodromy determine how far continuation can proceed.

Learning goals

After this chapter, the reader should be able to:

- continue analytic germs along paths by overlapping local expansions;
- use the identity theorem to prove uniqueness of continuation on overlaps;
- identify natural barriers to analytic continuation from singularities or monodromy;
- compare continuations along different paths in simply connected and non-simply-connected domains;
- construct examples of multivalued continuation such as logarithms or roots;
- state precisely when analytic continuation yields a single-valued global function;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

15.1 Germs

A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods.

Example 15.1 (Continuing the geometric series)**.** The series $\sum_{n\geq 0} z^n$ defines $1/(1-z)$ only for $|z| < 1$. The rational formula, however, is holomorphic on $\mathbb{C} \setminus \{1\}$ and agrees with the series on the disk. It therefore supplies an analytic continuation far beyond the original circle of convergence, except across the genuine singularity at 1.

15.2 Direct continuation

Two local representatives continue one another when they agree on a connected overlap.

15.3 Chains of continuation

A function can be propagated through a sequence of overlapping disks.

15.4 Uniqueness

The identity theorem prevents two different continuations from branching on the same connected overlap.

Example 15.2 (Continuation of a square root changes sheet)**.** Begin with a local square root near $z = 1$. Continue it along a loop encircling zero once. Locally write the root as $e^{L/2}$. After the loop, the continued logarithm has gained $2\pi i$, so the root has gained a factor $e^{\pi i} = -1$. The germ returns to the starting point with the opposite value.

15.5 Continuation along paths

Local elements can be transported step by step along a curve when singularities are avoided.

15.6 Natural boundaries

Some power series encounter singularities densely along their circle of convergence and cannot continue through it.

15.7 Singular obstruction

Poles, branch points, and essential singularities can stop single-valued continuation.

Example 15.3 (Uniqueness on overlaps)**.** Suppose two analytic continuations of the same germ are defined on overlapping connected regions. On a neighborhood where both agree with the original germ, their difference vanishes. The identity theorem then forces them to agree throughout the connected overlap. This is the local uniqueness mechanism behind analytic continuation.

15.8 Monodromy preview

Continuation around loops may return a different branch unless topology forces uniqueness globally.

15.9 Maximal continuation

The collection of all continuations naturally suggests a Riemann surface rather than a single planar domain.

15.10 Core structural results

Theorem 15.4 (Uniqueness of analytic continuation)**.** *If two holomorphic continuations agree on one nonempty open subset of a connected overlap, then they agree on the entire overlap.*

Proof. Their difference is holomorphic and vanishes on an open set, hence by the identity theorem vanishes everywhere on the connected overlap. ☐

Theorem 15.5 (Continuation along a fixed chain is unique)**.** *Given a chain of connected overlapping domains and an initial germ, at most one compatible holomorphic element can occupy each successive domain.*

Proof. Apply uniqueness of analytic continuation at each overlap inductively. ☐

Theorem 15.6 (Pathwise local uniqueness)**.** *Two continuations of the same initial germ along sufficiently close subdivisions of the same path agree wherever both are defined.*

Proof. Refine the subdivisions to common overlapping neighborhoods and repeatedly apply the identity theorem. ☐

15.11 Worked examples

Example 15.7 (Germs diagnostic)**.** Give a rigorous worked analysis of Germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.8 (Direct continuation diagnostic)**.** Give a rigorous worked analysis of Direct continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Two local representatives continue one another when they agree on a connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.9 (Chains of continuation diagnostic)**.** Give a rigorous worked analysis of Chains of continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function can be propagated through a sequence of overlapping disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.10 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The identity theorem prevents two different continuations from branching on the same connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

15.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 15.11 (Germs diagnostic)**.** Give a rigorous worked analysis of Germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 15.12 (Direct continuation diagnostic)**.** Give a rigorous worked analysis of Direct continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Two local representatives continue one another when they agree on a connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.13 (Chains of continuation diagnostic)**.** Give a rigorous worked analysis of Chains of continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function can be propagated through a sequence of overlapping disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.14 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The identity theorem prevents two different continuations from branching on the same connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.15 (Continuation along paths diagnostic)**.** Give a rigorous worked analysis of Continuation along paths. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local elements can be transported step by step along a curve when singularities are avoided. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.16 (Natural boundaries diagnostic)**.** Give a rigorous worked analysis of Natural boundaries. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Some power series encounter singularities densely along their circle of convergence and cannot continue through it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.17 (Singular obstruction diagnostic)**.** Give a rigorous worked analysis of Singular obstruction. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles, branch points, and essential singularities can stop single-valued continuation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.18 (Monodromy preview diagnostic)**.** Give a rigorous worked analysis of Monodromy preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Continuation around loops may return a different branch unless topology forces uniqueness globally. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.19 (Uniqueness of analytic continuation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two holomorphic continuations agree on one nonempty open subset of a connected overlap, then they agree on the entire overlap.

Solution. Their difference is holomorphic and vanishes on an open set, hence by the identity theorem vanishes everywhere on the connected overlap. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 15.20 (Continuation along a fixed chain is unique proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Given a chain of connected overlapping domains and an initial germ, at most one compatible holomorphic element can occupy each successive domain.

Solution. Apply uniqueness of analytic continuation at each overlap inductively. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 15.21 (Pathwise local uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Two continuations of the same initial germ along sufficiently close subdivisions of the same path agree wherever both are defined.

Solution. Refine the subdivisions to common overlapping neighborhoods and repeatedly apply the identity theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 15.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Analytic continuation extends local holomorphic data across overlapping domains. The identity theorem guarantees uniqueness, while singularities and monodromy determine how far continuation can proceed. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

15.13 Exercises with complete solutions

Exercise 15.23. State and apply the central principle behind Germs. Give one concrete consequence.

Hint. Continue from one disk to the next only where the local analytic representations overlap on a nonempty open set. ☐

Solution. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.24. State and apply the central principle behind Direct continuation. Give one concrete consequence.

Hint. Use the identity theorem on overlaps to show that two local continuations must agree there. ☐

Solution. Two local representatives continue one another when they agree on a connected overlap. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.25. State and apply the central principle behind Chains of continuation. Give one concrete consequence.

Hint. Record singularities encountered along the path; they determine where continuation can stop. ☐

Solution. A function can be propagated through a sequence of overlapping disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.26. State and apply the central principle behind Uniqueness. Give one concrete consequence.

Hint. To compare two paths, subdivide them into overlapping continuation charts and track the resulting germ at the endpoint. ☐

Solution. The identity theorem prevents two different continuations from branching on the same connected overlap. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.27. State and apply the central principle behind Continuation along paths. Give one concrete consequence.

Hint. In a simply connected domain without singular obstructions, path-independence of continuation is the expected monodromy conclusion. ☐

Solution. Local elements can be transported step by step along a curve when singularities are avoided. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.28. State and apply the central principle behind Natural boundaries. Give one concrete consequence.

Hint. For logarithm or roots, continue around a loop and compare the terminal germ with the initial one to detect monodromy. ☐

Solution. Some power series encounter singularities densely along their circle of convergence and cannot continue through it. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.29. State and apply the central principle behind Singular obstruction. Give one concrete consequence.

Hint. Natural boundaries are not just single singular points; test whether every boundary point obstructs continuation. ☐

Solution. Poles, branch points, and essential singularities can stop single-valued continuation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 15.30. State and apply the central principle behind Monodromy preview. Give one concrete consequence.

Hint. Separate existence of continuation along each path from single-valuedness after all possible loops. ☐

Solution. Continuation around loops may return a different branch unless topology forces uniqueness globally. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

15.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 15.31 (Rational continuation)**.** Give an analytic continuation of $\sum_{n\geq 0} z^n$ beyond $|z| < 1$.

Hint. Use its closed form. ☐

Solution. The continuation is $1/(1-z)$ on $\mathbb{C} \setminus \{1\}$. ☐

Exercise 15.32 (Continuation obstruction)**.** What point blocks continuation of $1/(1-z)$ as a holomorphic function?

Hint. Locate the pole. ☐

Solution. The point $z = 1$ is a genuine pole and cannot be crossed by a holomorphic continuation of that function. ☐

Exercise 15.33 (Overlap rule)**.** Two analytic elements agree on a nonempty open subset of a connected overlap. What follows?

Hint. Use the identity theorem. ☐

Solution. They agree on the whole connected overlap. ☐

Exercise 15.34 (Loop effect)**.** After one loop around zero, what happens to a continued square root?

Hint. Track the logarithm increment. ☐

Solution. It changes sign. ☐

Exercise 15.35 (Two loops)**.** What happens to that square root after two loops?

Hint. Apply the sign change twice. ☐

Solution. It returns to the original branch value. ☐

Proofs

Exercise 15.36 (Uniqueness of continuation)**.** Prove uniqueness of analytic continuation along a fixed chain of overlapping connected neighborhoods.

Hint. Induct across the overlaps using the identity theorem. □

Solution. Agreement on the initial germ forces agreement on the first overlap, then the next, and so on. Thus every step of the continuation chain is uniquely determined. □

Exercise 15.37 (Continuation preserves algebraic identities)**.** If two analytic functions satisfy $F^2 = g$ on the initial germ and both sides continue, show the identity persists.

Hint. Continue the difference $F^2 - g$. □

Solution. The difference is analytic and initially zero. Uniqueness and the identity theorem force it to remain zero wherever the continuation is defined. □

Exercise 15.38 (Path reversal)**.** Show that continuation along a path followed by its reverse returns the original germ when all continuations exist uniquely along the path.

Hint. Use local uniqueness step by step in reverse order. □

Solution. Each overlap transition is uniquely invertible by the same identity theorem argument, so retracing the chain recovers the starting analytic element. □

Exercise 15.39 (Simply connected monodromy principle)**.** Explain why path independent continuation is expected on a simply connected region when continuation is possible along every path.

Hint. Homotope any two paths with common endpoints while avoiding singularities. □

Solution. Local uniqueness makes the terminal germ stable under small path deformations. A homotopy between the paths then forces their terminal germs to agree. □

Counterexamples and hypothesis tests

Exercise 15.40 (Disconnected overlap)**.** Why does agreement on one component of a disconnected overlap not force agreement on another component?

Hint. The identity theorem propagates through connectedness. □

Solution. The functions may differ on another component because there is no connected analytic path of equality joining the components. □

Exercise 15.41 (Circle of convergence as barrier)**.** Is the boundary of a Taylor disk always a natural barrier to analytic continuation?

Hint. Use the geometric series as a counterexample. □

Solution. No. Its Taylor series stops at $|z| = 1$, but the function continues through most boundary points. The radius records the nearest singularity relative to the chosen center, not an automatic global barrier. □

Exercise 15.42 (Path dependence)**.** Can analytic continuation around a loop return a different germ?

Hint. Use the square root around zero. □

Solution. Yes. Multivalued analytic phenomena exhibit monodromy; the terminal germ can differ even at the same base point when the domain has nontrivial loops around branch points. □

Applications and investigations

Exercise 15.43 (Extending local formulas)**.** Why is analytic continuation useful when a power series converges only locally?

Hint. The locally defined analytic function may have compatible representations on overlapping regions. □

Solution. Continuation transports the same analytic object beyond one convergence disk while preserving identities and derivatives on overlaps. □

Exercise 15.44 (Detecting branch structure)**.** How can continuation around loops reveal a hidden Riemann surface?

Hint. Record how branch values permute after closed loops. □

Solution. Nontrivial permutations show that one planar domain cannot support a single valued version. The collection of branches and their continuation rules points toward a covering or branched surface. □

Challenge problems

Exercise 15.45 (Logarithm monodromy)**.** Describe the effect of one positive circuit around zero on an analytically continued logarithm.

Hint. Track the argument change. □

Solution. The logarithm increases by $2\pi i$. Repeated circuits add integer multiples of $2\pi i$. □

Exercise 15.46 (Algebraic branch permutation)**.** For $w^3 = z$, describe the monodromy after one positive loop around zero.

Hint. A logarithm increment of $2\pi i$ is divided by three. □

Solution. Each cube root is multiplied by $e^{2\pi i/3}$, cyclically permuting the three branches. Three loops return each branch to itself. □

Chapter 16

Möbius Transformations

Möbius transformations are the automorphisms of the Riemann sphere generated by fractional linear maps. They preserve generalized circles and provide the elementary coordinate changes of global complex analysis.

Learning goals

After this chapter, the reader should be able to:

- compute Möbius transformations from images of three points;
- determine images of circles and lines under fractional-linear transformations;
- use cross ratios to verify invariance and solve mapping constraints;
- classify fixed points and normal forms of simple Möbius maps;
- construct automorphisms of the disk and upper half-plane;
- track the point at infinity correctly in calculations on the Riemann sphere;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

16.1 Fractional linear maps

A Möbius map has form $T(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$, extended naturally to infinity.

Example 16.1 (Upper half-plane to unit disk)**.** The Möbius map

$$T(z) = \frac{z - i}{z + i}$$

sends i to 0. If $x \in \mathbb{R}$, then $|x - i| = |x + i|$, so $|T(x)| = 1$. A point such as $2i$ maps to $1/3$, showing that the upper half-plane maps to the unit disk.

16.2 Matrix representation

Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation.

16.3 Composition

Composition corresponds to matrix multiplication modulo nonzero scalars.

16.4 Three-point normalization

A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple.

Example 16.2 (Three-point normalization)**.** The cross-ratio map

$$T(z) = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

sends $z_1 \mapsto 0$, $z_2 \mapsto 1$, and $z_3 \mapsto \infty$. Thus any three distinct points on the Riemann sphere can be normalized to $0, 1, \infty$, after which many geometric questions become simpler.

16.5 Cross-ratio

The cross-ratio is invariant under Möbius transformations.

16.6 Generalized circles

Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles.

16.7 Elementary generators

Translations, dilations/rotations, and inversion generate all Möbius transformations.

Example 16.3 (A line becomes a circle)**.** For $T(z) = 1/z$, the vertical line $\operatorname{Re} z = 1$ can be written $z = 1 + iy$. If $w = 1/z = u + iv$, then solving $z = 1/w$ and imposing real part one gives

$$\left(u - \tfrac{1}{2}\right)^2 + v^2 = \tfrac{1}{4}.$$

Thus a line not through the pole maps to a circle through the origin.

16.8 Fixed points

Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation.

16.9 Disk and half-plane maps

Specific Möbius maps give conformal equivalences between the unit disk and half-plane.

16.10 Core structural results

Theorem 16.4 (Three-point theorem)**.** *Given distinct z_1, z_2, z_3 and distinct w_1, w_2, w_3 on the Riemann sphere, there is a unique Möbius transformation with $T(z_j) = w_j$.*

Proof. Construct maps sending each triple to $(0, 1, \infty)$ using cross-ratio formulas, then compose one with the inverse of the other. Uniqueness follows because a Möbius map fixing three distinct points is the identity. □

Theorem 16.5 (Cross-ratio invariance)**.** *Möbius transformations preserve the cross-ratio of four points.*

Proof. It suffices to check translations, nonzero scalings, and inversion, which generate the Möbius group; direct algebra verifies invariance for each generator. □

Theorem 16.6 (Generalized-circle preservation)**.** *Every Möbius transformation maps circles and lines to circles or lines.*

Proof. Translations and similarities preserve both classes; inversion maps circles/lines to generalized circles by an explicit equation calculation. Use the generator decomposition. □

16.11 Worked examples

Example 16.7 (Fractional linear maps diagnostic)**.** Give a rigorous worked analysis of Fractional linear maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A Möbius map has form $T(z) = (az+b)/(cz+d)$ with $ad-bc \neq 0$, extended naturally to infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.8 (Matrix representation diagnostic)**.** Give a rigorous worked analysis of Matrix representation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.9 (Composition diagnostic)**.** Give a rigorous worked analysis of Composition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Composition corresponds to matrix multiplication modulo nonzero scalars. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.10 (Three-point normalization diagnostic)**.** Give a rigorous worked analysis of Three-point normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

16.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 16.11 (Fractional linear maps diagnostic)**.** Give a rigorous worked analysis of Fractional linear maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A Möbius map has form $T(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$, extended naturally to infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.12 (Matrix representation diagnostic)**.** Give a rigorous worked analysis of Matrix representation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.13 (Composition diagnostic)**.** Give a rigorous worked analysis of Composition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Composition corresponds to matrix multiplication modulo nonzero scalars. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.14 (Three-point normalization diagnostic)**.** Give a rigorous worked analysis of Three-point normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.15 (Cross-ratio diagnostic)**.** Give a rigorous worked analysis of Cross-ratio. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The cross-ratio is invariant under Möbius transformations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.16 (Generalized circles diagnostic)**.** Give a rigorous worked analysis of Generalized circles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.17 (Elementary generators diagnostic)**.** Give a rigorous worked analysis of Elementary generators. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Translations, dilations/rotations, and inversion generate all Möbius transformations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 16.18 (Fixed points diagnostic)**.** Give a rigorous worked analysis of Fixed points. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.19 (Three-point theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Given distinct z_1, z_2, z_3 and distinct w_1, w_2, w_3 on the Riemann sphere, there is a unique Möbius transformation with $T(z_j) = w_j$.

Solution. Construct maps sending each triple to $(0, 1, \infty)$ using cross-ratio formulas, then compose one with the inverse of the other. Uniqueness follows because a Möbius map fixing three distinct points is the identity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.20 (Cross-ratio invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Möbius transformations preserve the cross-ratio of four points.

Solution. It suffices to check translations, nonzero scalings, and inversion, which generate the Möbius group; direct algebra verifies invariance for each generator. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.21 (Generalized-circle preservation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every Möbius transformation maps circles and lines to circles or lines.

Solution. Translations and similarities preserve both classes; inversion maps circles/lines to generalized circles by an explicit equation calculation. Use the generator decomposition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Möbius transformations are the automorphisms of the Riemann sphere generated by fractional linear maps. They preserve generalized circles and provide the elementary coordinate changes of global complex analysis. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

16.13 Exercises with complete solutions

Exercise 16.23. State and apply the central principle behind Fractional linear maps. Give one concrete consequence.

Hint. A Möbius map is determined by three distinct source-target pairs; solve it using cross ratios rather than four unknown coefficients when possible. □

Solution. A Möbius map has form $T(z) = (az+b)/(cz+d)$ with $ad-bc \neq 0$, extended naturally to infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.24. State and apply the central principle behind Matrix representation. Give one concrete consequence.

Hint. Include ∞ by reading the leading-coefficient ratio and denominator zeros on the Riemann sphere. $\square$

Solution. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 16.25. State and apply the central principle behind Composition. Give one concrete consequence.

Hint. Circles and lines are generalized circles; verify their images using inverse substitution or cross-ratio reality conditions. $\square$

Solution. Composition corresponds to matrix multiplication modulo nonzero scalars. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 16.26. State and apply the central principle behind Three-point normalization. Give one concrete consequence.

Hint. To map the upper half-plane to the disk, send a chosen boundary point to infinity and normalize one interior point. $\square$

Solution. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 16.27. State and apply the central principle behind Cross-ratio. Give one concrete consequence.

Hint. Automorphisms of the disk have the form phase times $(z - a)/(1 - \bar{a}z)$; enforce the normalization conditions. $\square$

Solution. The cross-ratio is invariant under Möbius transformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 16.28. State and apply the central principle behind Generalized circles. Give one concrete consequence.

Hint. Fixed points solve a quadratic equation from $T(z) = z$; include infinity if the degree drops. $\square$

Solution. Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 16.29. State and apply the central principle behind Elementary generators. Give one concrete consequence.

Hint. Cross ratios are Möbius invariant; compute one on the source and one on the target to verify a proposed map. $\square$

Solution. Translations, dilations/rotations, and inversion generate all Möbius transformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 16.30. State and apply the central principle behind Fixed points. Give one concrete consequence.

Hint. Matrix representatives are defined only up to nonzero scalar; do not treat coefficient normalization as geometric data. ☐

Solution. Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

16.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 16.31 (Evaluate a map)**.** For $T(z) = (z-1)/(z+1)$, compute $T(1)$, $T(0)$, and $T(\infty)$.

Hint. Use the leading coefficient ratio at infinity. ☐

Solution. The values are 0, -1, and 1. ☐

Exercise 16.32 (Inverse map)**.** Find the inverse of $w = (z-i)/(z+i)$.

Hint. Solve algebraically for z. ☐

Solution. One obtains $z = i(1+w)/(1-w)$. ☐

Exercise 16.33 (Pole and infinity)**.** For $T(z) = (2z+3)/(z-4)$, where does 4 map, and where does infinity map?

Hint. Use the sphere convention. ☐

Solution. The point 4 maps to infinity, and infinity maps to the leading coefficient ratio 2. ☐

Exercise 16.34 (Determinant)**.** Is $(2z+2)/(z+1)$ a genuine Möbius transformation?

Hint. Check $ad - bc$. ☐

Solution. No. The determinant is zero and the expression simplifies to the constant two away from the cancelled point. ☐

Exercise 16.35 (Fixed points)**.** Find the fixed points of $T(z) = 1/z$ on the sphere.

Hint. Solve $z = 1/z$. ☐

Solution. The finite fixed points are $z = 1$ and $z = -1$. ☐

Proofs

Exercise 16.36 (Inverse is Möbius)**.** Prove that a Möbius transformation with nonzero determinant has a Möbius inverse.

Hint. Solve $w = (az + b)/(cz + d)$ for z. □

Solution. Rearrangement gives $z = (dw - b)/(-cw + a)$, whose determinant equals the original nonzero determinant up to sign. □

Exercise 16.37 (Composition closure)**.** Show that the composition of two Möbius transformations is Möbius.

Hint. Represent each transformation by a nonsingular two by two matrix up to scalar. □

Solution. Composition corresponds to matrix multiplication. The product remains nonsingular, so the resulting fractional linear map is Möbius. □

Exercise 16.38 (Generalized circles)**.** Explain why Möbius transformations send circles and lines to circles or lines.

Hint. It suffices to check translations, nonzero scalings and rotations, and inversion. □

Solution. Affine maps preserve circles and lines. Inversion maps generalized circles to generalized circles, and every Möbius transformation decomposes into these elementary maps. □

Exercise 16.39 (Three-point uniqueness)**.** Prove that a Möbius transformation is uniquely determined by the images of three distinct points.

Hint. Compose two candidates with the inverse of one and study a map fixing three points. □

Solution. A nonidentity Möbius map has at most two fixed points because its fixed-point equation is quadratic. Hence a map fixing three distinct sphere points is the identity, proving uniqueness. □

Counterexamples and hypothesis tests

Exercise 16.40 (Zero determinant)**.** What fails when $ad - bc = 0$?

Hint. The numerator and denominator become proportional. □

Solution. The fractional expression is constant where defined and has no Möbius inverse, so it is not an automorphism of the sphere. □

Exercise 16.41 (Forgetting infinity)**.** Why must infinity be included when discussing global Möbius maps?

Hint. Denominator zeros map naturally to infinity. □

Solution. Adding infinity turns every nonsingular fractional linear expression into a bijective holomorphic map of the Riemann sphere. □

Exercise 16.42 (Interior versus boundary)**.** If a Möbius map sends the real axis to the unit circle, is checking boundary images alone enough to know which half-plane maps inside?

Hint. Test one interior point. □

Solution. No. The two complementary regions could be interchanged. Evaluating one point in the chosen half-plane determines the side. □

Applications and investigations

Exercise 16.43 (Normalize a domain)**.** Why map a half-plane to a disk before solving a boundary problem?

Hint. Standard theorems and kernels often have canonical disk forms. ☐

Solution. A Möbius normalization transfers the geometry to a standard domain where symmetry and known formulas simplify analysis, then the result can be pulled back. ☐

Exercise 16.44 (Cross-ratio invariant)**.** What geometric data can the cross-ratio preserve under Möbius transformations?

Hint. Apply the same normalization before and after the map. ☐

Solution. The cross-ratio of four ordered sphere points is invariant, providing a coordinate-free quantity for comparing configurations. ☐

Challenge problems

Exercise 16.45 (Disk automorphism)**.** Show that $T_a(z) = (z - a)/(1 - \overline{a}z)$ maps the unit disk to itself when $|a| < 1$.

Hint. Compute $1 - |T_a(z)|^2$. ☐

Solution. One obtains $(1 - |a|^2)(1 - |z|^2)/|1 - \overline{a}z|^2 > 0$ for $|z| < 1$, so the image lies in the disk; the inverse has the same form. ☐

Exercise 16.46 (Map three points)**.** Construct a Möbius transformation sending $-1, 0, 1$ to $0, 1, \infty$.

Hint. Use the cross-ratio formula or solve three equations. ☐

Solution. The map $T(z) = (z + 1)/(1 - z)$ sends -1 to zero, 0 to one, and 1 to infinity. ☐

Chapter 17

Conformal Mapping

Conformal maps preserve angles locally and reorganize domains without destroying holomorphic structure. Holomorphic maps with nonzero derivative are conformal, and the Riemann mapping theorem gives a universal model for simply connected proper planar domains.

Learning goals

After this chapter, the reader should be able to:

- verify conformality from holomorphicity and nonvanishing derivative;
- compute local angle and scale changes from the derivative;
- construct conformal equivalences by composing standard maps;
- use disk, half-plane, strip, and sector mappings in explicit domain problems;
- apply Schwarz-type principles or normalization to constrain conformal maps;
- distinguish conformal equivalence from mere homeomorphism;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

17.1 Local conformality

A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation.

Example 17.1 (A sector straightened by a power map)**.** On the sector $0 < \arg z < \pi/2$, the map $w = z^2$ doubles arguments. Hence

$$0 < \arg w < \pi,$$

so the sector maps conformally onto the upper half-plane. The derivative $2z$ is nonzero on the sector, so local angles are preserved.

17.2 Angle preservation

The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative.

17.3 Local inverse

A holomorphic function with nonzero derivative has a local holomorphic inverse.

17.4 Conformal equivalence

A biholomorphic map gives an analytic equivalence between domains.

Example 17.2 (A strip mapped to the upper half-plane)**.** For the horizontal strip $0 < \operatorname{Im} z < \pi$, the exponential map satisfies

$$\arg(e^z) = \operatorname{Im} z \in (0, \pi).$$

Thus e^z maps the strip onto the upper half-plane. Restricting the imaginary width to 2π or less is essential for injectivity.

17.5 Disk automorphisms

Automorphisms of the unit disk are Möbius transformations preserving the disk.

17.6 Half-plane equivalence

The Cayley transform identifies the upper half-plane with the unit disk.

17.7 Schwarz lemma

Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case.

Example 17.3 (Disk automorphism as a conformal normalizer)**.** For $|a| < 1$,

$$\phi_a(z) = \frac{z - a}{1 - \bar{a}z}$$

is a conformal automorphism of the unit disk and sends a to 0. Its derivative never vanishes in the disk. This normalization lets one move an arbitrary interior point to the symmetric center before applying disk estimates.

17.8 Riemann mapping theorem

Every simply connected proper plane domain is biholomorphic to the unit disk.

17.9 Normalization

Fixing one point and the argument of the derivative gives uniqueness of the Riemann map.

17.10 Boundary caution

Interior conformal equivalence does not automatically provide regular extension to arbitrary boundaries.

17.11 Core structural results

Theorem 17.4 (Holomorphic maps are conformal off critical points)**.** *If f is holomorphic and $f'(z_0) \neq 0$, then f preserves oriented angles at z_0.*

Proof. The first-order expansion is $f(z_0 + h) - f(z_0) = f'(z_0)h + o(|h|)$. Multiplication by the nonzero complex number $f'(z_0)$ rotates and scales every tangent direction by the same amount. □

Theorem 17.5 (Schwarz lemma)**.** *If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality at a nonzero point or in derivative only for rotations.*

Proof. Apply the maximum principle to $g(z) = f(z)/z$, with removable value $g(0) = f'(0)$. For every $r < 1$, maximum modulus on $|z| = r$ gives $|g(z)| \leq 1/r$ inside; let $r \to 1$. Equality invokes the maximum-modulus rigidity. □

Theorem 17.6 (Automorphisms of the disk)**.** *Every automorphism of $\mathbb{D}$ has form $e^{i\theta}(z - a)/(1 - \bar{a}z)$ for some $a \in \mathbb{D}$.*

Proof. Move the image of zero back to zero with a standard disk Möbius map. Schwarz lemma applied to the resulting automorphism and its inverse forces it to be a rotation. □

Theorem 17.7 (Normalized Riemann-map uniqueness)**.** *If two biholomorphisms $f, g : D \to \mathbb{D}$ send z_0 to zero and have positive real derivative at z_0, then $f = g$.*

Proof. The disk automorphism $g \circ f^{-1}$ fixes zero. By the disk automorphism classification it is a rotation, and the derivative normalization forces that rotation to be the identity. □

17.12 Worked examples

Example 17.8 (Local conformality diagnostic)**.** Give a rigorous worked analysis of Local conformality. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.9 (Angle preservation diagnostic)**.** Give a rigorous worked analysis of Angle preservation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.10 (Local inverse diagnostic)**.** Give a rigorous worked analysis of Local inverse. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A holomorphic function with nonzero derivative has a local holomorphic inverse. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.11 (Conformal equivalence diagnostic)**.** Give a rigorous worked analysis of Conformal equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A biholomorphic map gives an analytic equivalence between domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

17.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 17.12 (Local conformality diagnostic)**.** Give a rigorous worked analysis of Local conformality. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.13 (Angle preservation diagnostic)**.** Give a rigorous worked analysis of Angle preservation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.14 (Local inverse diagnostic)**.** Give a rigorous worked analysis of Local inverse. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic function with nonzero derivative has a local holomorphic inverse. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.15 (Conformal equivalence diagnostic)**.** Give a rigorous worked analysis of Conformal equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A biholomorphic map gives an analytic equivalence between domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.16 (Disk automorphisms diagnostic)**.** Give a rigorous worked analysis of Disk automorphisms. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Automorphisms of the unit disk are Möbius transformations preserving the disk. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.17 (Half-plane equivalence diagnostic)**.** Give a rigorous worked analysis of Half-plane equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Cayley transform identifies the upper half-plane with the unit disk. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.18 (Schwarz lemma diagnostic)**.** Give a rigorous worked analysis of Schwarz lemma. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 17.19 (Holomorphic maps are conformal off critical points proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic and $f'(z_0) \neq 0$, then f preserves oriented angles at z_0.

Solution. The first-order expansion is $f(z_0 + h) - f(z_0) = f'(z_0)h + o(|h|)$. Multiplication by the nonzero complex number $f'(z_0)$ rotates and scales every tangent direction by the same amount. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 17.20 (Schwarz lemma proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality at a nonzero point or in derivative only for rotations.

Solution. Apply the maximum principle to $g(z) = f(z)/z$, with removable value $g(0) = f'(0)$. For every $r < 1$, maximum modulus on $|z| = r$ gives $|g(z)| \leq 1/r$ inside; let $r \to 1$. Equality invokes the maximum-modulus rigidity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 17.21 (Automorphisms of the disk proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every automorphism of $\mathbb{D}$ has form $e^{i\theta}(z - a)/(1 - \bar{a}z)$ for some $a \in \mathbb{D}$.

Solution. Move the image of zero back to zero with a standard disk Möbius map. Schwarz lemma applied to the resulting automorphism and its inverse forces it to be a rotation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 17.22 (Normalized Riemann-map uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two biholomorphisms $f, g : D \to \mathbb{D}$ send z_0 to zero and have positive real derivative at z_0, then $f = g$.

Solution. The disk automorphism $g \circ f^{-1}$ fixes zero. By the disk automorphism classification it is a rotation, and the derivative normalization forces that rotation to be the identity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 17.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Conformal maps preserve angles locally and reorganize domains without destroying holomorphic structure. Holomorphic maps with nonzero derivative are conformal, and the Riemann mapping theorem gives a universal model for simply connected proper planar domains. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

17.14 Exercises with complete solutions

Exercise 17.24. State and apply the central principle behind Local conformality. Give one concrete consequence.

Hint. A holomorphic map is conformal where its derivative is nonzero; compute $f'(z_0)$ before discussing angle preservation. □

Solution. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.25. State and apply the central principle behind Angle preservation. Give one concrete consequence.

Hint. The derivative locally multiplies by $|f'|$ and rotates by $\arg f'$; read geometric distortion from that complex number. □

Solution. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.26. State and apply the central principle behind Local inverse. Give one concrete consequence.

Hint. Build complicated maps as compositions of translations, scalings, Möbius maps, exponentials, logarithms, and powers. □

Solution. A holomorphic function with nonzero derivative has a local holomorphic inverse. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.27. State and apply the central principle behind Conformal equivalence. Give one concrete consequence.

Hint. To map a strip or sector, first straighten it with an exponential or power map, then use a half-plane/disk transformation. □

Solution. A biholomorphic map gives an analytic equivalence between domains. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.28. State and apply the central principle behind Disk automorphisms. Give one concrete consequence.

Hint. Check injectivity on the chosen domain; a holomorphic local conformal map need not be globally one-to-one. □

Solution. Automorphisms of the unit disk are Möbius transformations preserving the disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.29. State and apply the central principle behind Half-plane equivalence. Give one concrete consequence.

Hint. Normalize conformal maps by fixing enough points before applying uniqueness theorems. □

Solution. The Cayley transform identifies the upper half-plane with the unit disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.30. State and apply the central principle behind Schwarz lemma. Give one concrete consequence.

Hint. Boundary correspondence may require extra hypotheses; do not infer it solely from interior conformality. ☐

Solution. Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.31. State and apply the central principle behind Riemann mapping theorem. Give one concrete consequence.

Hint. Use harmonic functions or level curves to visualize the map, but verify the analytic formula independently. ☐

Solution. Every simply connected proper plane domain is biholomorphic to the unit disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

17.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 17.32 (Quadrant to half-plane)**.** Find a simple conformal map from the first quadrant to the upper half-plane.

Hint. Double the argument. ☐

Solution. The map $w = z^2$ works. ☐

Exercise 17.33 (Strip to half-plane)**.** Map $0 < \operatorname{Im} z < \pi$ to the upper half-plane.

Hint. Use the exponential. ☐

Solution. The map $w = e^z$ works. ☐

Exercise 17.34 (Half-plane to disk)**.** Give a conformal map from the upper half-plane to the unit disk sending i to zero.

Hint. Use a Cayley transform. ☐

Solution. The map $(z - i)/(z + i)$ works. ☐

Exercise 17.35 (Angle scaling)**.** What angle does z^3 assign to a sector of opening $\pi/6$?

Hint. Multiply arguments by three. □

Solution. The image sector has opening $\pi/2$. □

Exercise 17.36 (Derivative test)**.** At which point does z^2 fail to be conformal as a local map?

Hint. Find where the derivative vanishes. □

Solution. At $z = 0$, since the derivative is $2z$. □

Proofs

Exercise 17.37 (Nonzero derivative preserves angles)**.** Prove local conformality of a holomorphic function where its derivative is nonzero.

Hint. Use the first order expansion. □

Solution. Near z_0, $f(z_0 + h) - f(z_0) = f'(z_0)h + o(|h|)$. The leading multiplication by a nonzero complex number is a rotation-dilation, so tangent angles are preserved in the limit. □

Exercise 17.38 (Inverse conformality)**.** If a holomorphic injective map has nonzero derivative, show its local inverse is holomorphic and conformal.

Hint. Use the complex inverse function theorem. □

Solution. The inverse derivative is $1/f'(z)$, which is nonzero. Hence the inverse is holomorphic and angle preserving locally. □

Exercise 17.39 (Composition)**.** Show that a composition of conformal maps is conformal where both are defined.

Hint. Use the chain rule. □

Solution. The derivative of the composition is the product of two nonzero complex derivatives, hence is nonzero and represents the composition of two rotation-dilations. □

Exercise 17.40 (Harmonic pullback)**.** Explain why composing a harmonic function with a conformal coordinate change preserves harmonicity in two dimensions.

Hint. Use the Laplacian transformation under a holomorphic map. □

Solution. The Laplacian picks up the positive factor $|f'|^2$. Thus if the original Laplacian is zero, the pulled back one is also zero. □

Counterexamples and hypothesis tests

Exercise 17.41 (Derivative zero)**.** Is z^2 conformal at zero?

Hint. Local angle preservation requires a nonzero derivative. □

Solution. No. The derivative vanishes and small angles are doubled rather than preserved. □

Exercise 17.42 (Holomorphic but not injective globally)**.** Does nonzero derivative everywhere guarantee global injectivity?

Hint. Consider the exponential. □

Solution. No. The exponential has nonzero derivative everywhere but is periodic, so it is not globally injective on the whole plane. □

Exercise 17.43 (Boundary correspondence)**.** Does a conformal map automatically extend continuously to every boundary point of arbitrary domains?

Hint. Boundary regularity requires extra hypotheses. □

Solution. No. Interior conformality alone does not guarantee a well behaved boundary extension. □

Applications and investigations

Exercise 17.44 (Dirichlet transfer)**.** How can a conformal map help solve a planar Dirichlet problem?

Hint. Move the domain to one with a known Poisson kernel. □

Solution. Solve the harmonic boundary problem in the standard domain and compose with the conformal inverse. Harmonicity is preserved by the coordinate change. □

Exercise 17.45 (Slit normalization)**.** Why are power and square-root maps useful for slit domains?

Hint. They change argument range and can unfold a cut. □

Solution. A suitable branch of a fractional power converts the slit geometry into a half-plane or sector where canonical conformal tools apply. □

Challenge problems

Exercise 17.46 (Strip to disk)**.** Construct a conformal map from $0 < \operatorname{Im} z < \pi$ to the unit disk.

Hint. First exponentiate to the upper half-plane, then apply a Cayley transform. □

Solution. One choice is $w = (e^z - i)/(e^z + i)$. □

Exercise 17.47 (Sector to disk)**.** Map the sector $0 < \arg z < \alpha$ conformally to the unit disk for $0 < \alpha < 2\pi$.

Hint. First use a power to reach the upper half-plane. □

Solution. The power $z^{\pi/\alpha}$ maps the sector to the upper half-plane on a chosen branch; composing with a Cayley transform maps it to the disk. □

Chapter 18

Schwarz–Christoffel Transformations

Schwarz–Christoffel transformations map canonical half-plane or disk domains to polygonal regions. The exponents in the derivative encode turning angles at polygon vertices.

Learning goals

After this chapter, the reader should be able to:

- derive Schwarz–Christoffel exponents from polygon interior angles;
- construct the derivative of a Schwarz–Christoffel map from prevertices;
- identify accessory constants and normalization conditions in polygon-mapping problems;
- analyze local behavior near polygon vertices from the power exponents;
- map standard half-plane or disk domains to simple polygons;
- check orientation and vertex order before interpreting a Schwarz–Christoffel formula;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

18.1 Polygonal target

A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$.

Example 18.1 (Interior angle determines the exponent)**.** Near a prevertex x_k, a Schwarz–Christoffel derivative has local form

$$f'(z) \sim C(z - x_k)^{\alpha_k/\pi - 1},$$

where α_k is the polygon interior angle. Integrating gives $f(z) - f(x_k) \sim C'(z - x_k)^{\alpha_k/\pi}$, so a half-plane angle π is transformed into the polygon angle α_k.

18.2 Derivative form

For a half-plane map with real prevertices x_k, the derivative has factors $(z - x_k)^{\alpha_k - 1}$.

18.3 Turning angles

The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex.

18.4 Integration

The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants.

Example 18.2 (Right angles give square-root factors)**.** For a rectangle, every finite vertex has interior angle $\pi/2$, hence exponent

$$\frac{\alpha}{\pi} - 1 = -\frac{1}{2}.$$

After normalizing three prevertices, the derivative therefore contains inverse square-root factors. Integrating leads to an elliptic integral, explaining why rectangle maps naturally connect Schwarz–Christoffel theory with elliptic functions.

18.5 Prevertices

Determining the prevertices is the accessory-parameter problem and is generally nonlinear.

18.6 Normalization

Three real degrees of freedom can be fixed by half-plane Möbius automorphisms.

18.7 Rectangles

Symmetry reduces rectangle maps to elliptic integrals.

Example 18.3 (Triangle exponents)**.** For a triangle with angles $\alpha_1, \alpha_2, \alpha_3$, the derivative has factors $(z - x_k)^{\alpha_k/\pi - 1}$. Since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, the three exponents sum to -2. This global balance matches the behavior required at infinity for a half-plane-to-polygon map.

18.8 Slits and degenerate polygons

Limiting polygon configurations produce maps to slit domains and strips.

18.9 Numerical construction

Practical Schwarz–Christoffel mapping often solves the parameter problem numerically.

18.10 Core structural results

Theorem 18.4 (Angle-exponent relation)**.** *Near a prevertex x_k, a Schwarz–Christoffel map with derivative behaving like $(z - x_k)^{\alpha_k - 1}$ maps a half-neighborhood to a wedge of interior angle $\alpha_k \pi$.*

Proof. Integrating gives local behavior $F(z) - F(x_k) \sim C(z - x_k)^{\alpha_k}$. The argument is therefore multiplied by α_k, sending the half-plane angle π to $\alpha_k \pi$. □

Theorem 18.5 (Sum of exponents)**.** *For an n-gon, $\sum_{k=1}^{n}(\alpha_k - 1) = -2$.*

Proof. The interior-angle sum is $(n-2)\pi$, so $\sum \alpha_k = n-2$ and subtraction of n gives -2. ☐

Theorem 18.6 (Affine freedom)**.** *Once the prevertices and exponents are fixed, two Schwarz–Christoffel maps with the same derivative factors differ by a nonzero multiplicative constant and an additive constant.*

Proof. Their derivatives are proportional by the chosen normalization, so integration yields $F_2 = AF_1 + B$ with $A \neq 0$. ☐

18.11 Worked examples

Example 18.7 (Polygonal target diagnostic)**.** Give a rigorous worked analysis of Polygonal target. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.8 (Derivative form diagnostic)**.** Give a rigorous worked analysis of Derivative form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a half-plane map with real prevertices x_k, the derivative has factors $(z - x_k)^{\alpha_k - 1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.9 (Turning angles diagnostic)**.** Give a rigorous worked analysis of Turning angles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.10 (Integration diagnostic)**.** Give a rigorous worked analysis of Integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

18.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 18.11 (Polygonal target diagnostic)**.** Give a rigorous worked analysis of Polygonal target. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 18.12 (Derivative form diagnostic)**.** Give a rigorous worked analysis of Derivative form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a half-plane map with real prevertices x_k, the derivative has factors $(z-x_k)^{\alpha_k-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.13 (Turning angles diagnostic)**.** Give a rigorous worked analysis of Turning angles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.14 (Integration diagnostic)**.** Give a rigorous worked analysis of Integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.15 (Prevertices diagnostic)**.** Give a rigorous worked analysis of Prevertices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Determining the prevertices is the accessory-parameter problem and is generally nonlinear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.16 (Normalization diagnostic)**.** Give a rigorous worked analysis of Normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Three real degrees of freedom can be fixed by half-plane Möbius automorphisms. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.17 (Rectangles diagnostic)**.** Give a rigorous worked analysis of Rectangles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Symmetry reduces rectangle maps to elliptic integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.18 (Slits and degenerate polygons diagnostic)**.** Give a rigorous worked analysis of Slits and degenerate polygons. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Limiting polygon configurations produce maps to slit domains and strips. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.19 (Angle-exponent relation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Near a prevertex x_k, a Schwarz–Christoffel map with derivative behaving like $(z - x_k)^{\alpha_k-1}$ maps a half-neighborhood to a wedge of interior angle $\alpha_k\pi$.

Solution. Integrating gives local behavior $F(z) - F(x_k) \sim C(z - x_k)^{\alpha_k}$. The argument is therefore multiplied by α_k, sending the half-plane angle π to $\alpha_k\pi$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.20 (Sum of exponents proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For an n-gon, $\sum_{k=1}^{n}(\alpha_k - 1) = -2$.

Solution. The interior-angle sum is $(n-2)\pi$, so $\sum \alpha_k = n - 2$ and subtraction of n gives -2. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.21 (Affine freedom proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Once the prevertices and exponents are fixed, two Schwarz–Christoffel maps with the same derivative factors differ by a nonzero multiplicative constant and an additive constant.

Solution. Their derivatives are proportional by the chosen normalization, so integration yields $F_2 = AF_1 + B$ with $A \neq 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Schwarz–Christoffel transformations map canonical half-plane or disk domains to polygonal regions. The exponents in the derivative encode turning angles at polygon vertices. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

18.13 Exercises with complete solutions

Exercise 18.23. State and apply the central principle behind Polygonal target. Give one concrete consequence.

Hint. For a polygon vertex with interior angle $\alpha_k\pi$, the Schwarz–Christoffel derivative contributes exponent $\alpha_k - 1$. □

Solution. A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.24. State and apply the central principle behind Derivative form. Give one concrete consequence.

Hint. Place prevertices in boundary order before integrating the product formula; wrong ordering changes the polygon. □

Solution. For a half-plane map with real prevertices x_k, the derivative has factors $(z - x_k)^{\alpha_k - 1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.25. State and apply the central principle behind Turning angles. Give one concrete consequence.

Hint. Use Möbius freedom to fix three prevertices and reduce the number of accessory parameters. □

Solution. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.26. State and apply the central principle behind Integration. Give one concrete consequence.

Hint. Near a prevertex, integrate $(z - x_k)^{\alpha_k - 1}$ to recover the local corner angle. □

Solution. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.27. State and apply the central principle behind Prevertices. Give one concrete consequence.

Hint. Determine multiplicative and additive constants from side lengths, orientation, and one base point. □

Solution. Determining the prevertices is the accessory-parameter problem and is generally nonlinear. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.28. State and apply the central principle behind Normalization. Give one concrete consequence.

Hint. For unbounded polygons, identify which prevertex corresponds to infinity and adjust the exponent bookkeeping. □

Solution. Three real degrees of freedom can be fixed by half-plane Möbius automorphisms. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.29. State and apply the central principle behind Rectangles. Give one concrete consequence.

Hint. Check branch choices of the fractional powers along the boundary so side directions are consistent. □

Solution. Symmetry reduces rectangle maps to elliptic integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.30. State and apply the central principle behind Slits and degenerate polygons. Give one concrete consequence.

Hint. Differentiate a candidate map first; verifying the Schwarz–Christoffel derivative is often simpler than integrating explicitly. □

Solution. Limiting polygon configurations produce maps to slit domains and strips. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

18.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 18.31 (Right angle exponent)**.** Find the Schwarz–Christoffel exponent for an interior angle $\pi/2$.

Hint. Compute $\alpha/\pi - 1$. ☐

Solution. The exponent is $-1/2$. ☐

Exercise 18.32 (Equilateral triangle)**.** Find the exponent at each vertex of an equilateral triangle.

Hint. Each interior angle is $\pi/3$. ☐

Solution. Each exponent is $-2/3$. ☐

Exercise 18.33 (Reflex vertex)**.** Find the exponent for interior angle $3\pi/2$.

Hint. Use the same formula. ☐

Solution. The exponent is $1/2$, reflecting a reflex corner. ☐

Exercise 18.34 (Straight angle)**.** What exponent corresponds to interior angle π?

Hint. Substitute directly. ☐

Solution. The exponent is zero, so no singular power factor is needed for a straight continuation. ☐

Exercise 18.35 (Triangle sum)**.** What is the sum of the three derivative exponents for a triangle?

Hint. Use the angle sum π. ☐

Solution. The sum is $(\pi/\pi) - 3 = -2$. ☐

Proofs

Exercise 18.36 (Local corner law)**.** Derive the local angle law from $f'(z) \sim C(z - x_k)^{\beta_k - 1}$.

Hint. Integrate the leading power. ☐

Solution. Integration gives $f(z) - f(x_k) \sim (C/\beta_k)(z - x_k)^{\beta_k}$. Arguments are multiplied by β_k, so the half-plane angle π becomes $\beta_k\pi = \alpha_k$. ☐

Exercise 18.37 (Polygon exponent sum)**.** For an n-gon, show that the finite-vertex exponents $\alpha_k/\pi - 1$ sum to -2.

Hint. Use the polygon interior angle sum. ☐

Solution. Since $\sum \alpha_k = (n-2)\pi$, the exponent sum is $(n-2) - n = -2$. ☐

Exercise 18.38 (Affine freedom)**.** Explain why multiplying a Schwarz–Christoffel map by a nonzero constant and adding a constant preserves the polygon shape up to similarity and translation.

Hint. Differentiate the transformed map. ☐

Solution. The derivative is multiplied by one nonzero complex constant, which rotates and scales all edges uniformly; the additive constant translates the image. ☐

Exercise 18.39 (Prevertex order)**.** Why must the real prevertices occur in the same cyclic order as the polygon vertices for a standard upper-half-plane map?

Hint. Follow the boundary image of the oriented real axis. ☐

Solution. The real boundary is traversed monotonically through the prevertices. Continuity maps successive real intervals to successive polygon edges, fixing the cyclic vertex order. ☐

Counterexamples and hypothesis tests

Exercise 18.40 (Wrong exponent)**.** What geometric defect results if a right-angle vertex is assigned exponent $-1/3$ instead of $-1/2$?

Hint. Convert the exponent back to an angle. ☐

Solution. The corresponding angle factor is $2/3$, giving an interior angle $2\pi/3$ rather than $\pi/2$. ☐

Exercise 18.41 (Repeated prevertices)**.** Can two distinct finite polygon vertices be assigned the same ordinary prevertex in a nondegenerate Schwarz–Christoffel map?

Hint. A single boundary point has only one local turning exponent. ☐

Solution. Not in the standard nondegenerate setup. Coincident prevertices signal a limiting or degenerate polygon configuration. ☐

Exercise 18.42 (Ignoring normalization)**.** Why are prevertex positions not all independent parameters?

Hint. The upper half-plane has a three-real-parameter Möbius automorphism group. ☐

Solution. Three real degrees of freedom can be normalized, commonly by fixing three prevertices. Failing to account for this produces redundant parameters. ☐

Applications and investigations

Exercise 18.43 (Numerical parameter problem)**.** What must be solved numerically when mapping the half-plane to a specified polygon with more than three vertices?

Hint. Angles determine exponents, but side lengths determine prevertex spacing. ☐

Solution. One solves nonlinear equations for the remaining prevertices and scale so that integrals between consecutive prevertices reproduce the prescribed side lengths. ☐

Exercise 18.44 (Flow around polygons)**.** Why are Schwarz–Christoffel maps useful in two dimensional potential flow?

Hint. Conformal maps preserve harmonic structure and angles. □

Solution. A difficult polygonal boundary can be mapped to a half-plane where analytic potentials are simple, then transported back to the physical polygonal domain. □

Challenge problems

Exercise 18.45 (Rectangle and elliptic integrals)**.** Explain why the rectangle map leads to an integral with four square-root branch points.

Hint. Each right angle contributes exponent $-1/2$. □

Solution. With four finite prevertices in a symmetric normalization, the derivative is proportional to the reciprocal square root of a quartic polynomial. Integrating this differential is an elliptic integral. □

Exercise 18.46 (Vertex at infinity)**.** How can placing one polygon vertex at infinity simplify a Schwarz–Christoffel formula?

Hint. Use Möbius freedom to send a convenient prevertex to infinity. □

Solution. Its factor is absorbed into the behavior at infinity, reducing the number of explicit finite factors and often simplifying both the integral and the parameter equations. □

Part IV

Special Functions

Chapter 19

The Gamma Function

The Gamma function extends factorials to the complex plane through an integral, functional equation, and analytic continuation. It is the prototype of a special function whose poles and growth encode arithmetic structure.

Learning goals

After this chapter, the reader should be able to:

- derive the functional equation $\Gamma(z+1) = z\Gamma(z)$ from the defining integral;
- extend the Gamma function meromorphically using recurrence;
- locate poles and compute residues at nonpositive integers;
- apply reflection or multiplication identities under the correct hypotheses;
- estimate Gamma asymptotics in elementary regimes developed in the chapter;
- translate factorial identities into Gamma-function statements and back again;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

19.1 Euler integral

For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$.

Example 19.1 (Gamma at half integers)**.** Starting from

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2}e^{-t}\,dt = \sqrt{\pi},$$

the functional equation gives

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}, \qquad \Gamma\left(\frac{5}{2}\right) = \frac{3}{4}\sqrt{\pi}.$$

More generally,

$$\Gamma\left(n+\frac{1}{2}\right) = \frac{(2n)!}{4^n n!}\sqrt{\pi}.$$

This converts many Gaussian and trigonometric integrals into factorial calculations.

19.2 Convergence

Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power.

19.3 Functional equation

Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$.

19.4 Factorials

For positive integers, $\Gamma(n+1) = n!$.

Example 19.2 (A pole and its residue)**.** Use $\Gamma(s+1) = s\Gamma(s)$. Since $\Gamma(s+1) \to 1$ as $s \to 0$,

$$\Gamma(s) = \frac{\Gamma(s+1)}{s} = \frac{1}{s} + O(1).$$

Hence $s = 0$ is a simple pole with residue 1. Repeating the functional equation yields

$$\operatorname{Res}_{s=-n} \Gamma(s) = \frac{(-1)^n}{n!}.$$

The local computation shows exactly how meromorphic continuation records the factorial recursion at negative integers.

19.5 Analyticity in the half-plane

Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$.

19.6 Meromorphic continuation

The functional equation extends Γ to the plane with simple poles at nonpositive integers.

19.7 Residues at poles

Pole residues are determined recursively from the residue at zero.

Example 19.3 (A Gamma integral with scaling)**.** For $a > 0$ and $\Re s > 0$, substitute $u = at$ in

$$I = \int_0^\infty t^{s-1} e^{-at}\, dt.$$

Then

$$I = a^{-s} \int_0^\infty u^{s-1} e^{-u}\, du = a^{-s}\Gamma(s).$$

For example, $\int_0^\infty t^{3/2} e^{-2t}\, dt = 2^{-5/2}\Gamma(5/2) = 3\sqrt{\pi}/(2^{9/2})$.

19.8 Logarithmic derivative

The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas.

19.9 Growth preview

Stirling asymptotics describe factorial-like growth in sectors.

19.10 Core structural results

Theorem 19.4 (Gamma functional equation)**.** *For* $\Re s > 0$*,* $\Gamma(s+1) = s\Gamma(s)$*.*

Proof. Integrate $\int_0^\infty t^s e^{-t} dt$ by parts. Boundary terms vanish because $t^s e^{-t}$ decays at infinity and tends to zero at the origin under the real-part condition. □

Theorem 19.5 (Meromorphic continuation)**.** *The functional equation extends* Γ *meromorphically to* $\mathbb{C}$ *with simple poles at* $0, -1, -2, \ldots$*.*

Proof. For any N, define $\Gamma(s) = \Gamma(s+N)/(s(s+1)\cdots(s+N-1))$ where the numerator integral is holomorphic for $\Re(s+N) > 0$. These local formulas agree on overlaps by the functional equation. □

Theorem 19.6 (Residue formula)**.** *At* $s = -n$*,* $n \geq 0$*,* $\mathrm{Res}(\Gamma, -n) = (-1)^n/n!$*.*

Proof. Start from $\mathrm{Res}(\Gamma, 0) = 1$ using $\Gamma(s+1) = s\Gamma(s)$ and propagate residues recursively through the functional equation. □

19.11 Worked examples

Example 19.7 (Euler integral diagnostic)**.** Give a rigorous worked analysis of Euler integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.8 (Convergence diagnostic)**.** Give a rigorous worked analysis of Convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.9 (Functional equation diagnostic)**.** Give a rigorous worked analysis of Functional equation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.10 (Factorials diagnostic)**.** Give a rigorous worked analysis of Factorials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For positive integers, $\Gamma(n+1) = n!$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

19.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 19.11 (Euler integral diagnostic)**.** Give a rigorous worked analysis of Euler integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.12 (Convergence diagnostic)**.** Give a rigorous worked analysis of Convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.13 (Functional equation diagnostic)**.** Give a rigorous worked analysis of Functional equation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.14 (Factorials diagnostic)**.** Give a rigorous worked analysis of Factorials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For positive integers, $\Gamma(n+1) = n!$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.15 (Analyticity in the half-plane diagnostic)**.** Give a rigorous worked analysis of Analyticity in the half-plane. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.16 (Meromorphic continuation diagnostic)**.** Give a rigorous worked analysis of Meromorphic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The functional equation extends Γ to the plane with simple poles at nonpositive integers. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.17 (Residues at poles diagnostic)**.** Give a rigorous worked analysis of Residues at poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Pole residues are determined recursively from the residue at zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.18 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.19 (Gamma functional equation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $\Re s > 0$, $\Gamma(s+1) = s\Gamma(s)$.

Solution. Integrate $\int_0^\infty t^s e^{-t} dt$ by parts. Boundary terms vanish because $t^s e^{-t}$ decays at infinity and tends to zero at the origin under the real-part condition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 19.20 (Meromorphic continuation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The functional equation extends Γ meromorphically to $\mathbb{C}$ with simple poles at $0, -1, -2, \dots$.

Solution. For any N, define $\Gamma(s) = \Gamma(s+N)/(s(s+1)\cdots(s+N-1))$ where the numerator integral is holomorphic for $\Re(s+N) > 0$. These local formulas agree on overlaps by the functional equation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 19.21 (Residue formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: At $s = -n$, $n \geq 0$, $\mathrm{Res}(\Gamma, -n) = (-1)^n/n!$.

Solution. Start from $\mathrm{Res}(\Gamma, 0) = 1$ using $\Gamma(s+1) = s\Gamma(s)$ and propagate residues recursively through the functional equation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 19.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Gamma function extends factorials to the complex plane through an integral, functional equation, and analytic continuation. It is the prototype of a special function whose poles and growth encode arithmetic structure. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

19.13 Exercises with complete solutions

Exercise 19.23. State and apply the central principle behind Euler integral. Give one concrete consequence.

Hint. Integrate by parts in the defining Gamma integral to obtain $\Gamma(z+1) = z\Gamma(z)$ in the initial convergence half-plane. ☐

Solution. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.24. State and apply the central principle behind Convergence. Give one concrete consequence.

Hint. Use recurrence to continue meromorphically leftward one strip at a time. ☐

Solution. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.25. State and apply the central principle behind Functional equation. Give one concrete consequence.

Hint. At $z = -n$, isolate the simple pole using the recurrence and compute its residue from the finite value at $z + n + 1$. □

Solution. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.26. State and apply the central principle behind Factorials. Give one concrete consequence.

Hint. For reflection, pair $\Gamma(z)\Gamma(1-z)$ with a contour or Beta identity and track the sine factor. □

Solution. For positive integers, $\Gamma(n+1) = n!$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.27. State and apply the central principle behind Analyticity in the half-plane. Give one concrete consequence.

Hint. Check $\Gamma(n+1) = n!$ as a normalization test for any derived identity. □

Solution. Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.28. State and apply the central principle behind Meromorphic continuation. Give one concrete consequence.

Hint. Parameter differentiation under the integral requires a uniform integrable bound; justify it before producing digamma-type formulas. □

Solution. The functional equation extends Γ to the plane with simple poles at nonpositive integers. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.29. State and apply the central principle behind Residues at poles. Give one concrete consequence.

Hint. Separate asymptotic estimates from exact identities; use the chapter's stated regime before applying Stirling-type reasoning. □

Solution. Pole residues are determined recursively from the residue at zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.30. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Locate poles before shifting contours or arguments so recurrence denominators are never divided by zero unnoticed. □

Solution. The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

19.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 19.31 (Factorial value)**.** Compute $\Gamma(7)$.

Hint. Use $\Gamma(n+1) = n!$. ☐

Solution. $\Gamma(7) = 6! = 720$. ☐

Exercise 19.32 (Half integer)**.** Compute $\Gamma(7/2)$.

Hint. Step down repeatedly to $\Gamma(1/2)$. ☐

Solution. $\Gamma(7/2) = (5/2)(3/2)(1/2)\sqrt{\pi} = 15\sqrt{\pi}/8$. ☐

Exercise 19.33 (Scaled integral)**.** Evaluate $\int_0^\infty t^2 e^{-3t}\, dt$.

Hint. Use the scaled Gamma integral with $s = 3$. ☐

Solution. The value is $3^{-3}\Gamma(3) = 2/27$. ☐

Exercise 19.34 (Residue)**.** Find $\operatorname{Res}_{s=-3} \Gamma(s)$.

Hint. Use the residue formula at negative integers. ☐

Solution. The residue is $(-1)^3/3! = -1/6$. ☐

Exercise 19.35 (Digamma recurrence)**.** From the Gamma functional equation derive a recurrence for $\psi = \Gamma'/\Gamma$.

Hint. Take a logarithmic derivative. ☐

Solution. $\psi(s+1) = \psi(s) + 1/s$. ☐

Proofs

Exercise 19.36 (Functional equation)**.** Prove $\Gamma(s+1) = s\Gamma(s)$ for $\Re s > 0$.

Hint. Integrate by parts and check both endpoint terms. ☐

Solution. Take $u = t^s$ and $dv = e^{-t}dt$. The boundary terms vanish, leaving $s\int_0^\infty t^{s-1}e^{-t}dt$. ☐

Exercise 19.37 (Holomorphy)**.** Explain why the Euler integral defines a holomorphic function on $\Re s > 0$.

Hint. Work on a compact vertical strip inside the half plane. ☐

Solution. On compact subsets, $t^{s-1}e^{-t}$ and its parameter derivatives are dominated by integrable functions near zero and infinity, so differentiation under the integral sign is valid. ☐

Exercise 19.38 (Pole locations)**.** Prove that the functional equation gives only simple poles at the nonpositive integers.

Hint. Shift the argument into the right half plane. ☐

Solution. For sufficiently large N, write $\Gamma(s) = \Gamma(s+N)/(s(s+1)\cdots(s+N-1))$. The numerator is holomorphic and nonzero at the relevant points, while each denominator factor occurs once. ☐

Exercise 19.39 (Residue recursion)**.** Prove $\operatorname{Res}_{s=-n}\Gamma(s) = (-1)^n/n!$.

Hint. Compare residues in $\Gamma(s+1) = s\Gamma(s)$. ☐

Solution. Starting with residue 1 at zero, each step to the left divides by the value of s at the new pole, producing alternating signs and factorial denominators. ☐

Counterexamples and hypothesis tests

Exercise 19.40 (Convergence at zero)**.** Does the Euler integral converge for $\Re s = 0$?

Hint. Inspect the magnitude of t^{s-1} near zero. ☐

Solution. In general no. Its magnitude behaves like t^{-1}, whose integral diverges logarithmically near zero. ☐

Exercise 19.41 (Entire or meromorphic)**.** Is Γ entire after continuation?

Hint. Recall the values forced by the functional equation at nonpositive integers. ☐

Solution. No. It is meromorphic with simple poles at $0, -1, -2, \ldots$. ☐

Exercise 19.42 (Zeros of Gamma)**.** Can the functional equation alone permit a zero of Γ in the right half plane?

Hint. Use the Euler integral only for positive real arguments, then think about the reciprocal Gamma function. ☐

Solution. The integral is positive on the positive real axis, but excluding complex zeros requires additional theory. One should not infer zero freeness on the whole half plane from positivity on one line. ☐

Applications and investigations

Exercise 19.43 (Gaussian moment)**.** Evaluate $\int_0^\infty x^4 e^{-x^2}\,dx$.

Hint. Set $t = x^2$. ☐

Solution. The integral becomes $\frac{1}{2}\Gamma(5/2) = 3\sqrt{\pi}/8$. ☐

Exercise 19.44 (Factorial asymptotics)**.** Explain what Stirling asymptotics say about $\Gamma(n+1)$ for large integers n.

Hint. Translate the asymptotic formula for Gamma to factorials. ☐

Solution. It gives $n! \sim \sqrt{2\pi n}(n/e)^n$, quantifying the factorial growth encoded by Gamma. ☐

Challenge problems

Exercise 19.45 (Reflection preparation)**.** Why is a product such as $\Gamma(s)\Gamma(1-s)$ natural when seeking identities symmetric under $s \mapsto 1-s$?

Hint. Apply the involution twice and compare pole sets. □

Solution. The product is invariant under replacing s by $1-s$. Its poles occur at integers from the two Gamma factors, matching the pole pattern of trigonometric reciprocals such as $1/\sin(\pi s)$. □

Exercise 19.46 (Meromorphic continuation by patches)**.** Explain why formulas obtained by shifting by different integers define one continuation rather than conflicting functions.

Hint. Compare them on a region where both are defined. □

Solution. Repeated use of the functional equation shows the shifted formulas agree on overlaps. The identity theorem then glues them into a unique meromorphic continuation. □

Chapter 20

Beta and Gamma Identities

The Beta function packages a two-endpoint integral and connects directly to Gamma through a change of variables. Their identities provide a compact calculus for factorials, binomial coefficients, trigonometric integrals, and parameter differentiation.

Learning goals

After this chapter, the reader should be able to:

- derive the Beta–Gamma relation by a justified change of variables in a double integral;
- compute Beta integrals and reduce them to Gamma values;
- apply reflection and duplication formulas to simplify special-function expressions;
- track convergence conditions on parameters before manipulating integrals;
- use analytic continuation to extend identities beyond their initial integral domains;
- check special cases against factorial or trigonometric values as consistency tests;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

20.1 Beta integral

For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$.

Example 20.1 (Beta at integer parameters)**.** For positive integers m, n,

$$B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} = \frac{(m-1)!(n-1)!}{(m+n-1)!}.$$

Thus $B(3,4) = 2!3!/6! = 1/60$. Directly, this says

$$\int_0^1 t^2(1-t)^3\,dt = \frac{1}{60}.$$

The Gamma quotient turns a two-endpoint integral into a factorial identity.

20.2 Symmetry

The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$.

20.3 Gamma relation

A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$.

20.4 Recurrences

Integration by parts and the Gamma relation produce rational recurrences in the parameters.

Example 20.2 (A sine power integral)**.** Using $t = \sin^2\theta$,

$$\int_0^{\pi/2} \sin^{a-1}\theta \cos^{b-1}\theta \, d\theta = \frac{1}{2} B\left(\frac{a}{2}, \frac{b}{2}\right).$$

With $a = 6$ and $b = 2$,

$$\int_0^{\pi/2} \sin^5\theta \cos\theta \, d\theta = \frac{1}{2} B(3,1) = \frac{1}{6}.$$

This agrees with the elementary substitution $u = \sin\theta$.

20.5 Trigonometric form

The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers.

20.6 Factorial identities

At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas.

20.7 Parameter derivatives

Differentiation with respect to parameters introduces logarithmic factors inside the integral.

Example 20.3 (Differentiating a parameter)**.** For $\Re x, \Re y > 0$, differentiate the Beta integral with respect to x:

$$\frac{\partial B}{\partial x}(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1} \log t \, dt.$$

From the Gamma quotient,

$$\frac{\partial B}{\partial x}(x,y) = B(x,y)\big(\psi(x) - \psi(x+y)\big).$$

At $(1,1)$, this gives $\int_0^1 \log t \, dt = -1$.

20.8 Duplication preview

Combining Beta identities yields the Gamma duplication formula.

20.9 Analytic continuation

The Gamma quotient extends Beta meromorphically beyond its defining integral region.

20.10 Core structural results

Theorem 20.4 (Beta–Gamma identity)**.** *For* $\Re x, \Re y > 0$*,* $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$*.*

Proof. Write the product of Gamma integrals over the first quadrant and substitute $u = rt$, $v = r(1-t)$ with $r > 0$, $0 < t < 1$. The Jacobian is r, and the integral factors into $\Gamma(x+y)B(x,y)$. □

Theorem 20.5 (Trigonometric Beta formula)**.** *For* $a, b > 0$*,* $\int_0^{\pi/2} \sin^{a-1}\theta \cos^{b-1}\theta \, d\theta = \frac{1}{2}B(a/2, b/2)$*.*

Proof. Set $t = \sin^2\theta$, so $dt = 2\sin\theta\cos\theta\, d\theta$ and collect the resulting powers. □

Theorem 20.6 (Beta recurrence)**.** $B(x+1, y) = \frac{x}{x+y}B(x,y)$ *in the common domain of definition.*

Proof. Use the Beta–Gamma identity and the Gamma functional equation in numerator and denominator. □

20.11 Worked examples

Example 20.7 (Beta integral diagnostic)**.** Give a rigorous worked analysis of Beta integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.8 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.9 (Gamma relation diagnostic)**.** Give a rigorous worked analysis of Gamma relation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.10 (Recurrences diagnostic)**.** Give a rigorous worked analysis of Recurrences. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

20.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 20.11 (Beta integral diagnostic)**.** Give a rigorous worked analysis of Beta integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.12 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.13 (Gamma relation diagnostic)**.** Give a rigorous worked analysis of Gamma relation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.14 (Recurrences diagnostic)**.** Give a rigorous worked analysis of Recurrences. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.15 (Trigonometric form diagnostic)**.** Give a rigorous worked analysis of Trigonometric form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.16 (Factorial identities diagnostic)**.** Give a rigorous worked analysis of Factorial identities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.17 (Parameter derivatives diagnostic)**.** Give a rigorous worked analysis of Parameter derivatives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiation with respect to parameters introduces logarithmic factors inside the integral. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.18 (Duplication preview diagnostic)**.** Give a rigorous worked analysis of Duplication preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Combining Beta identities yields the Gamma duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 20.19 (Beta–Gamma identity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $\Re x, \Re y > 0$, $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$.

Solution. Write the product of Gamma integrals over the first quadrant and substitute $u = rt$, $v = r(1-t)$ with $r > 0$, $0 < t < 1$. The Jacobian is r, and the integral factors into $\Gamma(x+y)B(x, y)$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.20 (Trigonometric Beta formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $a, b > 0$, $\int_0^{\pi/2} \sin^{a-1}\theta \cos^{b-1}\theta \, d\theta = \frac{1}{2}B(a/2, b/2)$.

Solution. Set $t = \sin^2\theta$, so $dt = 2\sin\theta\cos\theta \, d\theta$ and collect the resulting powers. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.21 (Beta recurrence proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $B(x+1, y) = \frac{x}{x+y}B(x, y)$ in the common domain of definition.

Solution. Use the Beta–Gamma identity and the Gamma functional equation in numerator and denominator. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Beta function packages a two-endpoint integral and connects directly to Gamma through a change of variables. Their identities provide a compact calculus for factorials, binomial coefficients, trigonometric integrals, and parameter differentiation. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

20.13 Exercises with complete solutions

Exercise 20.23. State and apply the central principle behind Beta integral. Give one concrete consequence.

Hint. Start from the Beta integral and use the substitution $x = u/(u+v)$, $t = u+v$ in the corresponding double integral. □

Solution. For $\Re x, \Re y > 0$, $B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.24. State and apply the central principle behind Symmetry. Give one concrete consequence.

Hint. Check $\Re a > 0$ and $\Re b > 0$ before using the integral representation of $B(a, b)$. □

Solution. The substitution $t \mapsto 1 - t$ gives $B(x, y) = B(y, x)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.25. State and apply the central principle behind Gamma relation. Give one concrete consequence.

Hint. Once $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is established, reduce special integrals algebraically. □

Solution. A two-dimensional change of variables yields $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x + y)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.26. State and apply the central principle behind Recurrences. Give one concrete consequence.

Hint. Use $B(a, b) = B(b, a)$ as a symmetry check after any substitution. □

Solution. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.27. State and apply the central principle behind Trigonometric form. Give one concrete consequence.

Hint. For duplication formulas, choose parameters that turn a Beta integral into a trigonometric integral with a half-angle substitution. □

Solution. The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.28. State and apply the central principle behind Factorial identities. Give one concrete consequence.

Hint. Analytic continuation can extend an identity after it is proved on a common open parameter domain; invoke uniqueness explicitly. □

Solution. At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.29. State and apply the central principle behind Parameter derivatives. Give one concrete consequence.

Hint. Track poles of each Gamma factor before cancelling expressions meromorphically. □

Solution. Differentiation with respect to parameters introduces logarithmic factors inside the integral. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.30. State and apply the central principle behind Duplication preview. Give one concrete consequence.

Hint. Test derived formulas at integer or half-integer parameters where Gamma values are known. □

Solution. Combining Beta identities yields the Gamma duplication formula. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

20.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 20.31 (Simple Beta value)**.** Compute $B(2,3)$.

Hint. Use the Gamma quotient. □

Solution. $B(2,3) = 1!2!/4! = 1/12$. □

Exercise 20.32 (Symmetry)**.** Compute $B(5,2)$ from $B(2,5)$.

Hint. Use $B(x,y) = B(y,x)$. □

Solution. The two values are equal, and both are $1!4!/6! = 1/30$. □

Exercise 20.33 (Trigonometric integral)**.** Evaluate $\int_0^{\pi/2} \sin^2\theta\, d\theta$ using Beta.

Hint. Use $a = 3$ and $b = 1$. □

Solution. The value is $\frac{1}{2}B(3/2,1/2) = \pi/4$. □

Exercise 20.34 (Recurrence)**.** Express $B(x+1,y)$ in terms of $B(x,y)$.

Hint. Use the Gamma functional equation. □

Solution. $B(x+1,y) = xB(x,y)/(x+y)$. □

Exercise 20.35 (Log integral)**.** Evaluate $\int_0^1 t\log t\, dt$.

Hint. Differentiate $\int_0^1 t^{x-1}dt = 1/x$ at $x = 2$. □

Solution. The value is $-1/4$. □

Proofs

Exercise 20.36 (Symmetry proof)**.** Prove $B(x, y) = B(y, x)$ directly from the integral.

Hint. Use $u = 1 - t$. □

Solution. The substitution exchanges the factors t^{x-1} and $(1 - t)^{y-1}$ and preserves the interval. □

Exercise 20.37 (Beta Gamma identity)**.** Prove the Beta Gamma identity from the product of two Gamma integrals.

Hint. Use $u = rt$ and $v = r(1 - t)$. □

Solution. The first quadrant transforms to $r > 0$, $0 < t < 1$ with Jacobian r. The product factors as $\Gamma(x + y)B(x, y)$. □

Exercise 20.38 (Trigonometric form)**.** Derive the trigonometric Beta formula.

Hint. Set $t = \sin^2\theta$. □

Solution. The differential contributes $2\sin\theta\cos\theta\, d\theta$, and the remaining powers combine to give the stated half Beta integral. □

Exercise 20.39 (Recurrence proof)**.** Prove the recurrence for $B(x + 1, y)$.

Hint. Apply the Gamma quotient to both sides. □

Solution. Use $\Gamma(x + 1) = x\Gamma(x)$ and $\Gamma(x + y + 1) = (x + y)\Gamma(x + y)$. □

Counterexamples and hypothesis tests

Exercise 20.40 (Endpoint failure)**.** What happens to the defining Beta integral if $\Re x \leq 0$?

Hint. Inspect the endpoint $t = 0$. □

Solution. The factor t^{x-1} is generally not integrable there, so the defining integral can fail even though meromorphic continuation may still exist. □

Exercise 20.41 (Symmetry domain)**.** Does the symmetry identity disappear after meromorphic continuation?

Hint. Compare the Gamma quotient under swapping variables. □

Solution. No. The quotient is symmetric wherever both sides are defined, and continuation preserves the identity meromorphically. □

Exercise 20.42 (Differentiate without control)**.** May one always differentiate the Beta integral with respect to a parameter at the boundary of its convergence region?

Hint. Check for a common integrable majorant. □

Solution. No. Differentiation under the integral requires uniform integrability control. Boundary parameters can introduce nonintegrable logarithmic singularities. □

Applications and investigations

Exercise 20.43 (Probability normalization)**.** Why does $1/B(a,b)$ normalize the density $t^{a-1}(1-t)^{b-1}$ on $(0,1)$?

Hint. Integrate the density kernel. □

Solution. Its integral is exactly $B(a,b)$, so division by this value makes the total mass equal to one. □

Exercise 20.44 (Central binomial coefficient)**.** Explain why half-integer Gamma values naturally produce central binomial coefficients.

Hint. Insert $\Gamma(n+1/2)$ into a Beta quotient. □

Solution. The half-integer formula contains $(2n)!/(4^n n!)$, the factorial ratio underlying $\binom{2n}{n}$. □

Challenge problems

Exercise 20.45 (Duplication route)**.** Sketch how Beta identities can lead to the Gamma duplication formula.

Hint. Use a trigonometric Beta integral and double-angle substitutions. □

Solution. Express the same integral in two ways, one involving $B(s,s)$ and another involving a half-integer Beta value, then translate both through Gamma quotients and simplify. □

Exercise 20.46 (Parameter second derivative)**.** What integral appears after differentiating $B(x,y)$ twice with respect to x?

Hint. Differentiate the factor t^{x-1} twice. □

Solution. The integral becomes $\int_0^1 t^{x-1}(1-t)^{y-1}(\log t)^2 dt$, linked on the Gamma side to digamma and trigamma terms. □

Chapter 21

Keyhole Contours and Branch-Cut Integrals

Keyhole contours turn branch discontinuities into computable factors. They are the standard mechanism for integrals involving fractional powers, logarithms, and branch cuts.

Learning goals

After this chapter, the reader should be able to:

- choose a branch of the logarithm appropriate for a keyhole contour;
- compute the jump of a branch-dependent integrand across the two sides of a cut;
- estimate inner- and outer-circle contributions in keyhole contours;
- derive real integrals from residue sums and branch discontinuities;
- track argument conventions and orientation signs explicitly;
- distinguish branch-cut contributions from pole residues;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

21.1 Branch choice

A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen.

Example 21.1 (Keyhole evaluation of a rational power integral)**.** Fix $0 < a < 1$ and use the branch $0 < \arg z < 2\pi$ for z^{a-1}. For

$$f(z) = \frac{z^{a-1}}{1+z},$$

the upper bank contributes $I = \int_0^\infty x^{a-1}/(1+x)\,dx$, while the lower bank contributes $-e^{2\pi i(a-1)} I$. The only enclosed pole is at -1, whose residue is $e^{i\pi(a-1)}$. Hence

$$(1 - e^{2\pi i a}) I = 2\pi i e^{i\pi(a-1)},$$

which simplifies to $I = \pi/\sin(\pi a)$.

21.2 Keyhole geometry

A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles.

21.3 Jump factor

The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut.

21.4 Vanishing circular arcs

Power estimates determine whether the small and large circular contributions disappear.

Example 21.2 (Why both circular arcs vanish)**.** For the same integrand, on the outer circle $|z| = R$,

$$|f(z)| = O(R^{a-2}),$$

so arc length $O(R)$ gives contribution $O(R^{a-1}) \to 0$ because $a < 1$. On the inner circle $|z| = \varepsilon$,

$$|f(z)| = O(\varepsilon^{a-1}),$$

so the contribution is $O(\varepsilon^a) \to 0$ because $a > 0$. These two inequalities explain the exact parameter window $0 < a < 1$.

21.5 Residue contribution

Poles away from the branch cut contribute through the residue theorem.

21.6 Logarithmic integrals

Differentiating branch-power identities with respect to exponents produces logarithms.

21.7 Orientation bookkeeping

Upper and lower sides of the cut have opposite traversal directions.

Example 21.3 (Producing a logarithm by parameter differentiation)**.** Differentiate

$$I(a) = \int_0^\infty \frac{x^{a-1}}{1+x}\,dx = \frac{\pi}{\sin(\pi a)}$$

for $0 < a < 1$. Then

$$\int_0^\infty \frac{x^{a-1}\log x}{1+x}\,dx = -\pi^2 \frac{\cos(\pi a)}{\sin^2(\pi a)}.$$

At $a = 1/2$, the right side vanishes, reflecting the symmetry $x \mapsto 1/x$.

21.8 Parameter windows

The allowed real parts of exponents are dictated by convergence at zero and infinity.

21.9 Contour variants

Hankel contours and wedge contours implement the same jump principle for other branch geometries.

21.10 Core structural results

Theorem 21.4 (Keyhole jump identity)**.** *For a branch cut along the positive real axis with argument in* $(0, 2\pi)$*, the upper and lower boundary values of* z^α *differ by the factor* $e^{2\pi i\alpha}$*.*

Proof. On the upper bank the argument tends to 0, while on the lower bank it tends to 2π. Substitution into $e^{\alpha(\log r + i \arg z)}$ gives the factor. ☐

Theorem 21.5 (Keyhole reduction principle)**.** *If the circular arcs vanish, the keyhole contour integral reduces to a scalar jump factor times the corresponding real integral along the cut.*

Proof. Write the contour as upper bank plus outer circle plus lower bank plus inner circle. Let the radii tend to their limits; the circle contributions vanish and the two bank integrals combine using their branch-value relation and opposite orientations. ☐

Theorem 21.6 (Differentiation in exponent)**.** *When a keyhole identity is locally uniformly convergent in a parameter* α*, differentiating it with respect to* α *introduces factors of* $\operatorname{Log} z$ *and yields logarithmic integral identities.*

Proof. Differentiate $z^\alpha = e^{\alpha \operatorname{Log} z}$ under the integral sign, justified by a common integrable majorant on a compact parameter subregion. ☐

21.11 Worked examples

Example 21.7 (Branch choice diagnostic)**.** Give a rigorous worked analysis of Branch choice. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.8 (Keyhole geometry diagnostic)**.** Give a rigorous worked analysis of Keyhole geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.9 (Jump factor diagnostic)**.** Give a rigorous worked analysis of Jump factor. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.10 (Vanishing circular arcs diagnostic)**.** Give a rigorous worked analysis of Vanishing circular arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Power estimates determine whether the small and large circular contributions disappear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

21.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 21.11 (Branch choice diagnostic)**.** Give a rigorous worked analysis of Branch choice. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.12 (Keyhole geometry diagnostic)**.** Give a rigorous worked analysis of Keyhole geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.13 (Jump factor diagnostic)**.** Give a rigorous worked analysis of Jump factor. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.14 (Vanishing circular arcs diagnostic)**.** Give a rigorous worked analysis of Vanishing circular arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Power estimates determine whether the small and large circular contributions disappear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.15 (Residue contribution diagnostic)**.** Give a rigorous worked analysis of Residue contribution. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles away from the branch cut contribute through the residue theorem. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.16 (Logarithmic integrals diagnostic)**.** Give a rigorous worked analysis of Logarithmic integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiating branch-power identities with respect to exponents produces logarithms. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.17 (Orientation bookkeeping diagnostic)**.** Give a rigorous worked analysis of Orientation bookkeeping. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Upper and lower sides of the cut have opposite traversal directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.18 (Parameter windows diagnostic)**.** Give a rigorous worked analysis of Parameter windows. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The allowed real parts of exponents are dictated by convergence at zero and infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.19 (Keyhole jump identity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a branch cut along the positive real axis with argument in $(0, 2\pi)$, the upper and lower boundary values of z^α differ by the factor $e^{2\pi i\alpha}$.

Solution. On the upper bank the argument tends to 0, while on the lower bank it tends to 2π. Substitution into $e^{\alpha(\log r + i \arg z)}$ gives the factor. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.20 (Keyhole reduction principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If the circular arcs vanish, the keyhole contour integral reduces to a scalar jump factor times the corresponding real integral along the cut.

Solution. Write the contour as upper bank plus outer circle plus lower bank plus inner circle. Let the radii tend to their limits; the circle contributions vanish and the two bank integrals combine using their branch-value relation and opposite orientations. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.21 (Differentiation in exponent proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: When a keyhole identity is locally uniformly convergent in a parameter α, differentiating it with respect to α introduces factors of $\operatorname{Log} z$ and yields logarithmic integral identities.

Solution. Differentiate $z^\alpha = e^{\alpha \operatorname{Log} z}$ under the integral sign, justified by a common integrable majorant on a compact parameter subregion. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Keyhole contours turn branch discontinuities into computable factors. They are the standard mechanism for integrals involving fractional powers, logarithms, and branch cuts. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

21.13 Exercises with complete solutions

Exercise 21.23. State and apply the central principle behind Branch choice. Give one concrete consequence.

Hint. Fix the branch $z^\alpha = e^{\alpha \log z}$ and write the argument on the upper and lower sides of the cut explicitly. $\square$

Solution. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.24. State and apply the central principle behind Keyhole geometry. Give one concrete consequence.

Hint. Compute the multiplicative jump across the cut before evaluating any residues; it is the source of the real integral factor. $\square$

Solution. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.25. State and apply the central principle behind Jump factor. Give one concrete consequence.

Hint. Estimate the small circle using the local exponent near zero and the large circle using decay at infinity. $\square$

Solution. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.26. State and apply the central principle behind Vanishing circular arcs. Give one concrete consequence.

Hint. Orient the two sides of the keyhole in opposite directions and keep the branch values separate. $\square$

Solution. Power estimates determine whether the small and large circular contributions disappear. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.27. State and apply the central principle behind Residue contribution. Give one concrete consequence.

Hint. Choose the cut to align with the desired real integral, usually along the positive or negative real axis. $\square$

Solution. Poles away from the branch cut contribute through the residue theorem. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.28. State and apply the central principle behind Logarithmic integrals. Give one concrete consequence.

Hint. Residues come from poles away from the cut; branch discontinuities contribute through the contour sides, not as residues. $\square$

Solution. Differentiating branch-power identities with respect to exponents produces logarithms. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. $\square$

Exercise 21.29. State and apply the central principle behind Orientation bookkeeping. Give one concrete consequence.

Hint. After the contour limit, solve the resulting algebraic equation for the real integral rather than discarding the branch factor. ☐

Solution. Upper and lower sides of the cut have opposite traversal directions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 21.30. State and apply the central principle behind Parameter windows. Give one concrete consequence.

Hint. Check parameter ranges that make both small- and large-circle estimates vanish. ☐

Solution. The allowed real parts of exponents are dictated by convergence at zero and infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

21.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 21.31 (Jump factor)**.** For the branch $0 < \arg z < 2\pi$, find the ratio of lower-bank to upper-bank values of z^α on the positive axis.

Hint. The arguments are 2π and 0. ☐

Solution. The ratio is $e^{2\pi i\alpha}$. ☐

Exercise 21.32 (Outer arc order)**.** If $f(z) = z^{a-1}/(1+z)$, what is the order of the outer-circle integral?

Hint. Multiply the magnitude estimate by arc length. ☐

Solution. It is $O(R^{a-1})$, so it vanishes when $a < 1$. ☐

Exercise 21.33 (Inner arc order)**.** For the same integrand, what is the inner-circle order?

Hint. Near zero the denominator is bounded away from zero. ☐

Solution. The contribution is $O(\varepsilon^a)$, so it vanishes when $a > 0$. ☐

Exercise 21.34 (Pole residue)**.** Find the residue of $z^{a-1}/(1+z)$ at $z = -1$ for $0 < \arg z < 2\pi$.

Hint. Use $\arg(-1) = \pi$. ☐

Solution. The residue is $(-1)^{a-1} = e^{i\pi(a-1)}$ on the chosen branch. ☐

Exercise 21.35 (Logarithmic derivative)**.** Differentiate x^{a-1} with respect to a.

Hint. Write the power as an exponential. ☐

Solution. The derivative is $x^{a-1} \log x$. ☐

Proofs

Exercise 21.36 (Keyhole reduction)**.** Prove that the two banks combine into a scalar jump factor times one real integral.

Hint. Track the opposite orientation on the lower bank. ☐

Solution. The lower bank runs from large radius to small radius and carries the branch phase. Reversing its limits produces a minus sign, so the sum is the upper integral times one minus the phase factor. ☐

Exercise 21.37 (Parameter window)**.** Prove that $0 < a < 1$ is exactly the range making both circular arcs vanish for $z^{a-1}/(1+z)$.

Hint. Estimate separately at zero and infinity. ☐

Solution. The inner contribution behaves like ε^a, requiring $a > 0$; the outer contribution behaves like R^{a-1}, requiring $a < 1$. ☐

Exercise 21.38 (Orientation sign)**.** Explain rigorously why the lower-bank integral has the opposite sign from the upper-bank integral after parametrization.

Hint. Write the lower bank as $z = xe^{2\pi i}$ with decreasing x. ☐

Solution. Its parameter runs from R down to ε, so reversing the limits contributes the negative sign. ☐

Exercise 21.39 (Differentiate under the integral)**.** Give a justification for differentiating the keyhole identity with respect to a on a compact subinterval of $(0, 1)$.

Hint. Find one integrable bound for the differentiated real integrand. ☐

Solution. On a compact parameter interval $[\delta, 1-\delta]$, the derivative is bounded by a constant times $(x^{\delta-1} + x^{-\delta})|\log x|/(1+x)$, which is integrable at zero and infinity. ☐

Counterexamples and hypothesis tests

Exercise 21.40 (Wrong branch)**.** What is wrong with using a keyhole contour before specifying a branch of z^α?

Hint. The boundary values on the two banks are branch dependent. ☐

Solution. Without a branch, the integrand is not single valued on the contour domain, so neither the jump factor nor the residue values are determined. ☐

Exercise 21.41 (Pole on the cut)**.** Can the standard residue argument be used unchanged if a pole lies on the branch cut?

Hint. The contour would pass through a singularity. ☐

Solution. No. One must indent around the pole, use a principal value prescription, or choose a different cut or contour. ☐

Exercise 21.42 (Arc estimate failure)**.** What fails when $a \geq 1$ in the standard keyhole example?

Hint. Inspect the outer-circle bound. ☐

Solution. The estimate $O(R^{a-1})$ no longer tends to zero, so the contour does not reduce to just the bank integrals and residues. ☐

Applications and investigations

Exercise 21.43 (Mellin transform)**.** Explain why keyhole contours naturally evaluate Mellin-type integrals.

Hint. Mellin kernels are complex powers. ☐

Solution. A Mellin kernel x^{s-1} acquires a controlled phase across a branch cut, so the jump converts a contour integral into the desired real Mellin integral. ☐

Exercise 21.44 (Branch-cut logarithms)**.** How can one create integrals containing $(\log x)^2$?

Hint. Differentiate twice with respect to the exponent. ☐

Solution. Two parameter derivatives of x^{a-1} produce $x^{a-1}(\log x)^2$, while the differentiated closed form provides the evaluation. ☐

Challenge problems

Exercise 21.45 (Hankel contour)**.** Why is a Hankel contour around the negative real axis useful for reciprocal Gamma representations?

Hint. Think about the jump of a complex power and exponential decay. ☐

Solution. The power has a simple phase jump across the negative axis while e^z decays along the two rays, producing a contour representation closely adapted to $1/\Gamma(s)$. ☐

Exercise 21.46 (Wedge variant)**.** How would the jump factor change if the branch cut were bounded by arguments θ_0 and $\theta_0 + 2\pi$?

Hint. Evaluate z^α on the two boundary arguments. ☐

Solution. Both values gain a common factor $e^{i\alpha\theta_0}$, while their ratio remains $e^{2\pi i\alpha}$. The essential jump is therefore unchanged. ☐

Part V

Riemann Surfaces

Chapter 22

From Analytic Continuation to Riemann Surfaces

Analytic continuation naturally leads beyond planar single-valued functions. A Riemann surface is the geometric object on which a locally holomorphic multivalued expression becomes an ordinary single-valued holomorphic function.

Learning goals

After this chapter, the reader should be able to:

- explain how multivalued analytic continuation motivates passing from the plane to a Riemann surface;
- construct local coordinate charts that make a multivalued relation single-valued;
- identify branch points and sheets in elementary algebraic examples;
- translate pathwise analytic continuation into geometric movement on a surface;
- distinguish a branched covering from an ordinary covering away from branch points;
- relate local analytic data to the global surface obtained by continuation;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

22.1 Local branches

Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches.

Example 22.1 (Two local square-root branches)**.** On a disk avoiding the origin, choose a holomorphic logarithm and set

$$\sqrt{z} = \exp\left(\frac{1}{2}\operatorname{Log} z\right).$$

The second branch is its negative. Continuing the first branch once around the origin changes $\operatorname{Log} z$ by $2\pi i$, hence multiplies the square root by $e^{\pi i} = -1$. The endpoint has the same base coordinate but the opposite germ, which is precisely why the surface of germs contains two points over a generic base point.

22.2 Branch incompatibility

Different continuations around loops can disagree after returning to the same plane point.

22.3 Surface of germs

Collecting compatible germs over plane points separates distinct analytic branches.

22.4 Projection map

A Riemann surface often comes with a holomorphic projection to the plane or sphere.

Example 22.2 (The logarithm surface as infinitely many sheets)**.** A germ of $\log z$ near 1 has values differing by $2\pi ik$. Continuing once counterclockwise around zero sends the value 0 at 1 to $2\pi i$. After m turns it becomes $2\pi im$. Thus the maximal continuation has infinitely many sheets, naturally indexed by the integer winding number.

22.5 Charts

Local coordinates make the surface look like open subsets of $\mathbb{C}$.

22.6 Holomorphic functions on surfaces

Holomorphicity is defined through compatible coordinate charts.

22.7 Branch points

Projection can fail to be locally one-to-one at ramification points.

Example 22.3 (A coordinate near a branch point)**.** For the algebraic relation $w^2 = z$, the projection to the z-plane is not a local coordinate at $(0, 0)$. Instead use w itself as the surface coordinate. Then the projection is

$$z = w^2.$$

The surface is perfectly smooth at the point even though the projection ramifies there. This separates intrinsic smoothness of the Riemann surface from singular behavior of a chosen projection.

22.8 Maximal analytic continuation

The maximal continuation of a germ can be organized as a connected Riemann surface.

22.9 Examples

The logarithm surface and the square-root surface show infinite-sheeted and two-sheeted behavior.

22.10 Core structural results

Theorem 22.4 (Local single-valuedness)**.** *Every point of a Riemann surface has a neighborhood on which a holomorphic projection that is locally biholomorphic can be used as a complex coordinate.*

Proof. This is built into the chart structure: a local coordinate is a homeomorphism to a plane domain with holomorphic transition maps. □

Theorem 22.5 (Germ-surface principle)**.** *Compatible analytic continuations of a germ can be organized into a set of germs with a natural projection to the base point and a local complex coordinate given by that projection.*

Proof. Around each germ choose a representative on a small disk. Nearby points correspond to the germs of the same representative; these neighborhoods provide charts and compatible transition maps. □

Theorem 22.6 (Single-valued lift)**.** *A multivalued analytic expression becomes single-valued when evaluated on its surface of analytic germs.*

Proof. A point of the surface records not just the base coordinate but the chosen local germ, so evaluation of that germ is unambiguous. □

22.11 Worked examples

Example 22.7 (Local branches diagnostic)**.** Give a rigorous worked analysis of Local branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.8 (Branch incompatibility diagnostic)**.** Give a rigorous worked analysis of Branch incompatibility. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Different continuations around loops can disagree after returning to the same plane point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.9 (Surface of germs diagnostic)**.** Give a rigorous worked analysis of Surface of germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Collecting compatible germs over plane points separates distinct analytic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.10 (Projection map diagnostic)**.** Give a rigorous worked analysis of Projection map. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

22.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 22.11 (Local branches diagnostic)**.** Give a rigorous worked analysis of Local branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.12 (Branch incompatibility diagnostic)**.** Give a rigorous worked analysis of Branch incompatibility. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different continuations around loops can disagree after returning to the same plane point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.13 (Surface of germs diagnostic)**.** Give a rigorous worked analysis of Surface of germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Collecting compatible germs over plane points separates distinct analytic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.14 (Projection map diagnostic)**.** Give a rigorous worked analysis of Projection map. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.15 (Charts diagnostic)**.** Give a rigorous worked analysis of Charts. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local coordinates make the surface look like open subsets of $\mathbb{C}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.16 (Holomorphic functions on surfaces diagnostic)**.** Give a rigorous worked analysis of Holomorphic functions on surfaces. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphicity is defined through compatible coordinate charts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.17 (Branch points diagnostic)**.** Give a rigorous worked analysis of Branch points. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Projection can fail to be locally one-to-one at ramification points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 22.18 (Maximal analytic continuation diagnostic)**.** Give a rigorous worked analysis of Maximal analytic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The maximal continuation of a germ can be organized as a connected Riemann surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.19 (Local single-valuedness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every point of a Riemann surface has a neighborhood on which a holomorphic projection that is locally biholomorphic can be used as a complex coordinate.

Solution. This is built into the chart structure: a local coordinate is a homeomorphism to a plane domain with holomorphic transition maps. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 22.20 (Germ-surface principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Compatible analytic continuations of a germ can be organized into a set of germs with a natural projection to the base point and a local complex coordinate given by that projection.

Solution. Around each germ choose a representative on a small disk. Nearby points correspond to the germs of the same representative; these neighborhoods provide charts and compatible transition maps. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 22.21 (Single-valued lift proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A multivalued analytic expression becomes single-valued when evaluated on its surface of analytic germs.

Solution. A point of the surface records not just the base coordinate but the chosen local germ, so evaluation of that germ is unambiguous. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 22.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Analytic continuation naturally leads beyond planar single-valued functions. A Riemann surface is the geometric object on which a locally holomorphic multivalued expression becomes an ordinary single-valued holomorphic function. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

22.13 Exercises with complete solutions

Exercise 22.23. State and apply the central principle behind Local branches. Give one concrete consequence.

Hint. Start with a multivalued function such as $\sqrt{z}$ or $\log z$ and label continuation sheets along paths. □

Solution. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.24. State and apply the central principle behind Branch incompatibility. Give one concrete consequence.

Hint. Near a nonbranch point, choose a local coordinate in which one branch is an ordinary holomorphic function. □

Solution. Different continuations around loops can disagree after returning to the same plane point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.25. State and apply the central principle behind Surface of germs. Give one concrete consequence.

Hint. At a branch point of $w^n = z$, use w as the local coordinate rather than z. □

Solution. Collecting compatible germs over plane points separates distinct analytic branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.26. State and apply the central principle behind Projection map. Give one concrete consequence.

Hint. Glue local branches along overlaps according to analytic continuation; the surface records which values belong together. □

Solution. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.27. State and apply the central principle behind Charts. Give one concrete consequence.

Hint. Project the surface to the base plane and identify where the map fails to be an ordinary covering. □

Solution. Local coordinates make the surface look like open subsets of $\mathbb{C}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.28. State and apply the central principle behind Holomorphic functions on surfaces. Give one concrete consequence.

Hint. Track a loop in the base and see which sheet the lifted path ends on; this is the geometric meaning of monodromy. □

Solution. Holomorphicity is defined through compatible coordinate charts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.29. State and apply the central principle behind Branch points. Give one concrete consequence.

Hint. Distinguish the analytic surface from a mere collection of cut planes by checking compatible holomorphic charts. □

Solution. Projection can fail to be locally one-to-one at ramification points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.30. State and apply the central principle behind Maximal analytic continuation. Give one concrete consequence.

Hint. Use the surface to make the originally multivalued expression single-valued by construction. ☐

Solution. The maximal continuation of a germ can be organized as a connected Riemann surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

22.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 22.31 (Square-root monodromy)**.** What happens to a chosen branch of $\sqrt{z}$ after one loop around zero?

Hint. Track the logarithm change by $2\pi i$. ☐

Solution. The square root changes sign. ☐

Exercise 22.32 (Logarithm monodromy)**.** What happens to $\log z$ after one positive loop around zero?

Hint. Use the argument increment. ☐

Solution. Its value increases by $2\pi i$. ☐

Exercise 22.33 (Fiber size)**.** How many germs of $\sqrt{z}$ lie over a generic nonzero point?

Hint. Solve $w^2 = z$. ☐

Solution. There are two, corresponding to the two square roots. ☐

Exercise 22.34 (Local coordinate)**.** For $w^2 = z$ near the branch point, which variable is a good surface coordinate?

Hint. Choose the variable in which the relation has no branching. ☐

Solution. The coordinate w is good; the projection is then $w \mapsto w^2$. ☐

Exercise 22.35 (Single-valued lift)**.** Why is the square root single valued on its two-sheeted surface?

Hint. A surface point remembers the branch germ. ☐

Solution. Evaluating the recorded germ gives one unambiguous value at each surface point. ☐

Proofs

Exercise 22.36 (Germ charts)**.** Explain how representatives of analytic germs define local charts on the surface of germs.

Hint. Use one representative on a small disk. ☐

Solution. Nearby points are assigned the germs of the same holomorphic representative. Projection to the disk is one to one on that neighborhood and gives the chart. ☐

Exercise 22.37 (Holomorphic transitions)**.** Prove that transition maps between overlapping germ charts are holomorphic.

Hint. Both charts use the same base coordinate where the representatives agree. ☐

Solution. On overlap the projection coordinates agree, so the transition is the identity on an open plane set and hence holomorphic. ☐

Exercise 22.38 (Maximal continuation connectedness)**.** Why is the surface generated from one germ by analytic continuation naturally connected?

Hint. Every point is reached along a continuation path from the initial germ. ☐

Solution. The lifted continuation path lies in the germ surface and connects the initial point to the target point, so all generated points lie in one connected component. ☐

Exercise 22.39 (Intrinsic holomorphy)**.** Show that the definition of a holomorphic function on a Riemann surface is independent of the chosen chart.

Hint. Insert a holomorphic transition map. ☐

Solution. If the coordinate expression is holomorphic in one chart, composing with a biholomorphic transition map gives a holomorphic expression in every compatible chart. ☐

Counterexamples and hypothesis tests

Exercise 22.40 (Plane only)**.** Can a globally single-valued square root exist on $\mathbb{C}^*$?

Hint. Continue around the unit circle. ☐

Solution. No. One loop changes the sign, contradicting single valuedness. ☐

Exercise 22.41 (Branch point as surface singularity)**.** Is the point over $z = 0$ on $w^2 = z$ singular as a Riemann-surface point?

Hint. Use w as a coordinate. ☐

Solution. No. The surface is smooth there; only the projection to the z-plane is ramified. ☐

Exercise 22.42 (One sheet for logarithm)**.** Can the maximal logarithm surface have finitely many sheets?

Hint. Count values after repeated loops. ☐

Solution. No. Each winding adds a distinct multiple of $2\pi i$, giving infinitely many sheets. ☐

Applications and investigations

Exercise 22.43 (Inverse functions)**.** Why do Riemann surfaces help with multivalued inverse functions?

Hint. Each local inverse branch is a germ. □

Solution. The surface separates different inverse germs lying over the same base point, turning the multivalued inverse into an ordinary holomorphic function upstairs. □

Exercise 22.44 (Algebraic functions)**.** How does the surface viewpoint resolve the ambiguity of an algebraic relation such as $w^3 = z(z-1)$?

Hint. Treat each compatible local solution as a sheet. □

Solution. Local roots glue into a surface on which w is single valued; branching is encoded in the projection to the z-sphere. □

Challenge problems

Exercise 22.45 (Continuation obstruction)**.** What topological feature of the base detects whether a local logarithm can be continued single valued globally?

Hint. Look at loops around the missing origin. □

Solution. Nontrivial winding around zero creates nonzero logarithmic monodromy. A simply connected domain avoiding zero removes that obstruction. □

Exercise 22.46 (Surface versus branch cut)**.** Compare using a branch cut with using the full Riemann surface for $\log z$.

Hint. A branch cut discards paths, while the surface retains them. □

Solution. A cut selects one sheet on a simply connected domain and sacrifices continuity across the cut. The Riemann surface keeps all continuations simultaneously and records their sheet changes geometrically. □

Chapter 23

Covering Maps and Monodromy

Covering maps formalize the idea of many sheets lying over one base domain. Path lifting and monodromy translate topology of loops into permutations of analytic branches.

Learning goals

After this chapter, the reader should be able to:

- verify the local lifting property for elementary covering maps;
- lift paths and homotopies once a starting point is fixed;
- compute monodromy permutations from continuation around loops;
- relate deck transformations to symmetries of a covering;
- distinguish simply connected base domains from domains with nontrivial monodromy;
- use covering-space language to organize branches of logarithms and roots;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

23.1 Covering definition

A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods.

Example 23.1 (Lifting a loop under the exponential map)**.** Consider $p(w) = e^w$ and the unit-circle loop $\gamma(t) = e^{2\pi i t}$. Starting at $w_0 = 0$, the lift

$$\widetilde{\gamma}(t) = 2\pi i t$$

satisfies $e^{\widetilde{\gamma}(t)} = \gamma(t)$. Its endpoint is $2\pi i$, not 0. Thus a closed base loop can lift to a path joining different points in the same fiber, and the endpoint records monodromy.

23.2 Sheets

Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood.

23.3 Path lifting

A path in the base has a unique lift once its starting point in the fiber is chosen.

23.4 Loop lifting

A loop may lift to a path whose endpoint is a different point of the same fiber.

Example 23.2 (Cyclic monodromy of a power map)**.** For $p(w) = w^n$ on $\mathbb{C}^*$, the fiber over 1 consists of $e^{2\pi ik/n}$. A positive loop once around zero lifts from the root $e^{2\pi ik/n}$ to $e^{2\pi i(k+1)/n}$. Hence the monodromy permutation is the n-cycle

$$(0\ 1\ \cdots\ n-1).$$

After n turns the lifted path closes.

23.5 Monodromy action

Homotopy classes of loops act by permutations on the fiber.

23.6 Deck transformations

Automorphisms of the covering commute with the projection and permute sheets globally.

23.7 Exponential covering

$z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$.

Example 23.3 (Deck transformations of the exponential covering)**.** Every translation

$$T_k(w) = w + 2\pi ik$$

satisfies $e^{T_k(w)} = e^w$, so it is a deck transformation. Conversely, a deck transformation must send 0 to another point of the fiber over 1, hence to $2\pi ik$; uniqueness of lifts then forces it to equal T_k everywhere. The deck group is therefore isomorphic to $\mathbb{Z}$.

23.8 Power maps

$z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy.

23.9 Analytic continuation

Branches of inverse functions move along lifted paths, making covering theory a natural language for continuation without branch points.

23.10 Core structural results

Theorem 23.4 (Unique path lifting)**.** *For a covering $p : X \to Y$, any path γ in Y and any $x_0 \in p^{-1}(\gamma(0))$ have a unique lift starting at x_0.*

Proof. Cover the compact parameter interval by subintervals mapping into evenly covered neighborhoods and construct the lift step by step. Local uniqueness on each sheet forces global uniqueness. ☐

Theorem 23.5 (Homotopy invariance of monodromy)**.** *Homotopic loops with fixed base point induce the same permutation of the fiber.*

Proof. Use the homotopy lifting property for coverings. The lifted endpoints vary continuously in a discrete fiber, hence remain constant through the homotopy. ☐

Theorem 23.6 (Exponential covering)**.** *The exponential map* $\exp : \mathbb{C} \to \mathbb{C}^*$ *is a covering map with deck group* $z \mapsto z + 2\pi ik$*.*

Proof. Small disks in $\mathbb{C}^*$ avoiding zero admit logarithm branches; their inverse images are disjoint translates of one logarithm branch, and exponential is biholomorphic on each translate. ☐

23.11 Worked examples

Example 23.7 (Covering definition diagnostic)**.** Give a rigorous worked analysis of Covering definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.8 (Sheets diagnostic)**.** Give a rigorous worked analysis of Sheets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.9 (Path lifting diagnostic)**.** Give a rigorous worked analysis of Path lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A path in the base has a unique lift once its starting point in the fiber is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.10 (Loop lifting diagnostic)**.** Give a rigorous worked analysis of Loop lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A loop may lift to a path whose endpoint is a different point of the same fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

23.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 23.11 (Covering definition diagnostic)**.** Give a rigorous worked analysis of Covering definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.12 (Sheets diagnostic)**.** Give a rigorous worked analysis of Sheets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.13 (Path lifting diagnostic)**.** Give a rigorous worked analysis of Path lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A path in the base has a unique lift once its starting point in the fiber is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.14 (Loop lifting diagnostic)**.** Give a rigorous worked analysis of Loop lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A loop may lift to a path whose endpoint is a different point of the same fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.15 (Monodromy action diagnostic)**.** Give a rigorous worked analysis of Monodromy action. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Homotopy classes of loops act by permutations on the fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.16 (Deck transformations diagnostic)**.** Give a rigorous worked analysis of Deck transformations. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Automorphisms of the covering commute with the projection and permute sheets globally. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.17 (Exponential covering diagnostic)**.** Give a rigorous worked analysis of Exponential covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 23.18 (Power maps diagnostic)**.** Give a rigorous worked analysis of Power maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.19 (Unique path lifting proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a covering $p : X \to Y$, any path γ in Y and any $x_0 \in p^{-1}(\gamma(0))$ have a unique lift starting at x_0.

Solution. Cover the compact parameter interval by subintervals mapping into evenly covered neighborhoods and construct the lift step by step. Local uniqueness on each sheet forces global uniqueness. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.20 (Homotopy invariance of monodromy proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Homotopic loops with fixed base point induce the same permutation of the fiber.

Solution. Use the homotopy lifting property for coverings. The lifted endpoints vary continuously in a discrete fiber, hence remain constant through the homotopy. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.21 (Exponential covering proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ is a covering map with deck group $z \mapsto z + 2\pi ik$.

Solution. Small disks in $\mathbb{C}^*$ avoiding zero admit logarithm branches; their inverse images are disjoint translates of one logarithm branch, and exponential is biholomorphic on each translate. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Covering maps formalize the idea of many sheets lying over one base domain. Path lifting and monodromy translate topology of loops into permutations of analytic branches. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

23.13 Exercises with complete solutions

Exercise 23.23. State and apply the central principle behind Covering definition. Give one concrete consequence.

Hint. To lift a path, cover it by evenly covered neighborhoods and choose the unique local lift matching the previous endpoint. □

Solution. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.24. State and apply the central principle behind Sheets. Give one concrete consequence.

Hint. Once the starting lift is fixed, uniqueness forces the entire lifted path. □

Solution. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.25. State and apply the central principle behind Path lifting. Give one concrete consequence.

Hint. For a loop, record the endpoint permutation of the fiber; composing loops composes these permutations. □

Solution. A path in the base has a unique lift once its starting point in the fiber is chosen. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.26. State and apply the central principle behind Loop lifting. Give one concrete consequence.

Hint. Deck transformations permute fibers while commuting with the covering projection; test both conditions. □

Solution. A loop may lift to a path whose endpoint is a different point of the same fiber. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.27. State and apply the central principle behind Monodromy action. Give one concrete consequence.

Hint. A simply connected base admits no nontrivial monodromy for a connected ordinary covering with fixed local branch data. □

Solution. Homotopy classes of loops act by permutations on the fiber. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.28. State and apply the central principle behind Deck transformations. Give one concrete consequence.

Hint. For $z \mapsto z^n$ on $\mathbb{C}^*$, one turn around zero cycles the n roots. □

Solution. Automorphisms of the covering commute with the projection and permute sheets globally. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.29. State and apply the central principle behind Exponential covering. Give one concrete consequence.

Hint. Homotopic loops give the same monodromy when the homotopy stays in the unbranched base. □

Solution. $z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.30. State and apply the central principle behind Power maps. Give one concrete consequence.

Hint. Separate ordinary covering behavior from branched points, where evenly covered neighborhoods fail. □

Solution. $z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

23.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 23.31 (Exponential lift)**.** Lift $\gamma(t) = e^{4\pi it}$ under e^w starting at zero.

Hint. Take a continuous logarithm along the path. ☐

Solution. The lift is $\widetilde{\gamma}(t) = 4\pi it$, ending at $4\pi i$. ☐

Exercise 23.32 (Power-map fiber)**.** List the fiber of $w \mapsto w^4$ over 1.

Hint. Find the fourth roots of unity. ☐

Solution. The fiber is $\{1, i, -1, -i\}$. ☐

Exercise 23.33 (Power monodromy)**.** Under $w \mapsto w^4$, where does the lift of one positive base loop starting at 1 end?

Hint. Advance one fourth-root step. ☐

Solution. It ends at i. ☐

Exercise 23.34 (Deck translation)**.** Which translation sends 0 to the point $-4\pi i$ in the exponential fiber?

Hint. Deck translations are by integer multiples of $2\pi i$. ☐

Solution. It is $w \mapsto w - 4\pi i$. ☐

Exercise 23.35 (Loop closure)**.** How many positive turns around zero are needed before a lift under $w \mapsto w^5$ starting at 1 closes?

Hint. The monodromy is a five cycle. ☐

Solution. Five turns are required. ☐

Proofs

Exercise 23.36 (Unique path lifting)**.** Outline a proof of unique path lifting for coverings.

Hint. Cover the compact path image by evenly covered neighborhoods. ☐

Solution. Subdivide the interval so each segment lies in one evenly covered neighborhood. The starting sheet determines a unique local lift, and the pieces concatenate uniquely. ☐

Exercise 23.37 (Homotopy invariance)**.** Why do homotopic based loops induce the same monodromy permutation?

Hint. Lift the homotopy and inspect endpoints. □

Solution. Lifted endpoints depend continuously on the homotopy parameter but lie in a discrete fiber, so they remain constant. □

Exercise 23.38 (Exponential covering)**.** Prove that $e^w : \mathbb{C} \to \mathbb{C}^*$ is a covering map.

Hint. Use local logarithm branches. □

Solution. A small disk avoiding zero admits a logarithm. Its inverse image is a disjoint union of translates of that logarithm branch by $2\pi ik$, each mapped biholomorphically onto the disk. □

Exercise 23.39 (Deck group)**.** Prove that every deck transformation of the exponential covering is a translation by $2\pi ik$.

Hint. Determine the image of one point, then use uniqueness of lifts. □

Solution. The image of zero lies in its fiber, so it is $2\pi ik$. The deck map and that translation are lifts of the same exponential map with the same starting value, hence coincide. □

Counterexamples and hypothesis tests

Exercise 23.40 (Branch point covering)**.** Is $w \mapsto w^2$ a covering map from all of $\mathbb{C}$ to all of $\mathbb{C}$?

Hint. Inspect a neighborhood of zero. □

Solution. No. The map is not locally a disjoint union of homeomorphic sheets over zero because the two sheets merge there. □

Exercise 23.41 (Endpoint depends only on loop)**.** Does the endpoint of a lifted loop depend only on the base loop and the chosen starting point?

Hint. Use uniqueness of lifts. □

Solution. Yes for the specified parametrized loop. For homotopy classes, homotopy lifting shows it depends only on the based homotopy class. □

Exercise 23.42 (Trivial monodromy)**.** Must every nontrivial covering have nontrivial monodromy for every loop?

Hint. Consider a contractible loop. □

Solution. No. Contractible loops act trivially even in a nontrivial covering. □

Applications and investigations

Exercise 23.43 (Analytic continuation)**.** How does path lifting model analytic continuation of inverse branches?

Hint. A local inverse chooses one point in a fiber. □

Solution. As the base point moves, the chosen inverse value follows the unique lifted path. Loop endpoints encode the permutation of branches. □

Exercise 23.44 (Root tracking)**.** Why is monodromy useful when numerically continuing roots of a polynomial depending on a parameter?

Hint. Follow roots along parameter loops. □

Solution. Continuation transports each root continuously and the endpoint permutation reveals how branches exchange around discriminant values. □

Challenge problems

Exercise 23.45 (Monodromy representation)**.** Explain why monodromy gives a group homomorphism from the fundamental group into a permutation group.

Hint. Compare lifting a concatenation with successive lifts. □

Solution. Lifting concatenated loops applies the first endpoint permutation and then the second, so loop multiplication corresponds to composition of permutations. □

Exercise 23.46 (Regular covering)**.** What special feature of the exponential covering makes its deck group act transitively on each fiber?

Hint. Any two fiber points differ by a period. □

Solution. Fiber points differ by $2\pi ik$, and the corresponding deck translation sends one to the other. This is the hallmark of a regular covering. □

Chapter 24

Branched Coverings

Branched coverings relax ordinary covering behavior at isolated ramification points. Locally they are modeled by power maps, and the ramification indices encode how sheets merge.

Learning goals

After this chapter, the reader should be able to:

- identify branch points and ramification indices of elementary holomorphic maps;
- write local models in the form $w = z^e$ after suitable coordinates;
- compute branching data for polynomial and rational maps;
- distinguish ramification points upstairs from branch values downstairs;
- apply degree counting with multiplicities in branched coverings;
- relate local ramification behavior to global topology in examples;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

24.1 Branched covering

Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge.

Example 24.1 (Local ramification index)**.** Consider $p(w) = w^3$ near $w = 0$. In the local coordinate w, the map already has the normal form

$$z = w^3.$$

The ramification index is $e = 3$. A small nonzero value of z has three nearby preimages, while the fiber over zero contains one point with multiplicity three. The derivative $3w^2$ vanishes at the ramification point.

24.2 Local power model

Near a ramification point, suitable coordinates often put the map in form $w = z^e$.

24.3 Ramification index

The integer e measures local multiplicity; $e > 1$ means ramification.

24.4 Branch values

Images of ramification points are branch values in the base.

Example 24.2 (Branching of a quadratic polynomial)**.** View $p(w) = w^2 - 1$ as a map of Riemann spheres. The finite critical point is $w = 0$, with branch value -1. Infinity is also ramified because in coordinates $u = 1/w$ and $v = 1/z$,

$$v = \frac{u^2}{1 - u^2} = u^2 + O(u^4).$$

Thus the degree-two map has two ramification points, each of index two.

24.5 Square-root surface

The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity.

24.6 Polynomial maps

A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points.

24.7 Monodromy around branch values

Loops around branch values permute sheets nontrivially.

Example 24.3 (Genus from Riemann–Hurwitz)**.** For a degree-two map $X \to \widehat{\mathbb{C}}$ with r simple branch points, Riemann–Hurwitz gives

$$2 - 2g(X) = 2 \cdot 2 - r.$$

Hence

$$g(X) = \frac{r - 2}{2}.$$

A double cover branched at four points has genus one, while branching at six points gives genus two.

24.8 Degree

Away from branch values, every fiber has the same number of points counted with multiplicity.

24.9 Riemann–Hurwitz preview

Total ramification contributes to the genus relation between compact surfaces.

24.10 Core structural results

Theorem 24.4 (Local ramification model)**.** *A nonconstant holomorphic map between Riemann surfaces has local coordinates in which it is $z \mapsto z^e$ for a unique integer $e \geq 1$.*

Proof. Expand the map in a local coordinate around the source point, factor out the first nonzero term of order e, and absorb the nonvanishing holomorphic factor into a new local coordinate using a holomorphic eth root. □

Theorem 24.5 (Critical-point criterion)**.** *A point is ramified iff the derivative of a local coordinate representation vanishes there.*

Proof. In the local model z^e, the derivative at zero is nonzero exactly when $e = 1$. □

Theorem 24.6 (Constant generic degree)**.** *For a proper finite branched covering, the sum of ramification multiplicities in a fiber is constant away from degenerations and equals the degree.*

Proof. Locally, zeros of the equation $p(x) = y$ are counted with multiplicity; the argument principle makes this count stable as y moves without crossing branch values. □

24.11 Worked examples

Example 24.7 (Branched covering diagnostic)**.** Give a rigorous worked analysis of Branched covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.8 (Local power model diagnostic)**.** Give a rigorous worked analysis of Local power model. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.9 (Ramification index diagnostic)**.** Give a rigorous worked analysis of Ramification index. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The integer e measures local multiplicity; $e > 1$ means ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.10 (Branch values diagnostic)**.** Give a rigorous worked analysis of Branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Images of ramification points are branch values in the base. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

24.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 24.11 (Branched covering diagnostic)**.** Give a rigorous worked analysis of Branched covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.12 (Local power model diagnostic)**.** Give a rigorous worked analysis of Local power model. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.13 (Ramification index diagnostic)**.** Give a rigorous worked analysis of Ramification index. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The integer e measures local multiplicity; $e > 1$ means ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.14 (Branch values diagnostic)**.** Give a rigorous worked analysis of Branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Images of ramification points are branch values in the base. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.15 (Square-root surface diagnostic)**.** Give a rigorous worked analysis of Square-root surface. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.16 (Polynomial maps diagnostic)**.** Give a rigorous worked analysis of Polynomial maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.17 (Monodromy around branch values diagnostic)**.** Give a rigorous worked analysis of Monodromy around branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Loops around branch values permute sheets nontrivially. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.18 (Degree diagnostic)**.** Give a rigorous worked analysis of Degree. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Away from branch values, every fiber has the same number of points counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.19 (Local ramification model proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A nonconstant holomorphic map between Riemann surfaces has local coordinates in which it is $z \mapsto z^e$ for a unique integer $e \geq 1$.

Solution. Expand the map in a local coordinate around the source point, factor out the first nonzero term of order e, and absorb the nonvanishing holomorphic factor into a new local coordinate using a holomorphic eth root. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.20 (Critical-point criterion proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A point is ramified iff the derivative of a local coordinate representation vanishes there.

Solution. In the local model z^e, the derivative at zero is nonzero exactly when $e = 1$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.21 (Constant generic degree proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a proper finite branched covering, the sum of ramification multiplicities in a fiber is constant away from degenerations and equals the degree.

Solution. Locally, zeros of the equation $p(x) = y$ are counted with multiplicity; the argument principle makes this count stable as y moves without crossing branch values. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Branched coverings relax ordinary covering behavior at isolated ramification points. Locally they are modeled by power maps, and the ramification indices encode how sheets merge. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

24.13 Exercises with complete solutions

Exercise 24.23. State and apply the central principle behind Branched covering. Give one concrete consequence.

Hint. Compute the local derivative order: if $f(z) - f(a) = c(z - a)^e + \cdots$, then e is the ramification index. □

Solution. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.24. State and apply the central principle behind Local power model. Give one concrete consequence.

Hint. A ramification point lies upstairs; its image is the branch value downstairs. ☐

Solution. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.25. State and apply the central principle behind Ramification index. Give one concrete consequence.

Hint. For a polynomial, critical points $f'(z) = 0$ are the finite ramification candidates; inspect multiplicities locally. ☐

Solution. The integer e measures local multiplicity; $e > 1$ means ramification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.26. State and apply the central principle behind Branch values. Give one concrete consequence.

Hint. Use the local model $w = z^e$ to count how many sheets merge at the branch point. ☐

Solution. Images of ramification points are branch values in the base. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.27. State and apply the central principle behind Square-root surface. Give one concrete consequence.

Hint. Degree counting uses multiplicities: a generic fiber has total multiplicity equal to the map degree. ☐

Solution. The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.28. State and apply the central principle behind Polynomial maps. Give one concrete consequence.

Hint. At infinity, switch to coordinates $u = 1/z$ and $v = 1/w$ before deciding ramification. ☐

Solution. A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.29. State and apply the central principle behind Monodromy around branch values. Give one concrete consequence.

Hint. Global branching data constrain genus through Riemann–Hurwitz when that theorem is available. ☐

Solution. Loops around branch values permute sheets nontrivially. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.30. State and apply the central principle behind Degree. Give one concrete consequence.

Hint. Do not confuse a multiple zero of the function with ramification unless it is interpreted through the map's local degree. □

Solution. Away from branch values, every fiber has the same number of points counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

24.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 24.31 (Ramification index)**.** Find the ramification index of $w \mapsto w^5$ at zero.

Hint. Read the exponent in the local power model. □

Solution. The index is 5. □

Exercise 24.32 (Derivative criterion)**.** Is $w = 0$ ramified for $p(w) = w^3 + w$?

Hint. Compute $p'(0)$. □

Solution. No. Since $p'(0) = 1$, the local index is one. □

Exercise 24.33 (Critical points)**.** Find the finite critical points of $p(w) = w^3 - 3w$.

Hint. Solve $p'(w) = 0$. □

Solution. The critical points are $w = \pm 1$. □

Exercise 24.34 (Branch values)**.** Find the corresponding finite branch values for $w^3 - 3w$.

Hint. Evaluate the polynomial at the critical points. □

Solution. The branch values are -2 at $w = 1$ and 2 at $w = -1$. □

Exercise 24.35 (Double-cover genus)**.** Find the genus of a double cover of the sphere with eight simple branch points.

Hint. Use $2 - 2g = 4 - r$. □

Solution. With $r = 8$, one gets $g = 3$. □

Proofs

Exercise 24.36 (Local power model)**.** Outline why a nonconstant holomorphic map has local form $z \mapsto z^e$.

Hint. Factor the first nonzero Taylor term and absorb the unit. □

Solution. Write the map as $z^e h(z)$ with $h(0) \neq 0$. A local holomorphic e-th root of h produces a new coordinate in which the map is exactly a power. □

Exercise 24.37 (Derivative criterion)**.** Prove that ramification is equivalent to vanishing derivative in local coordinates.

Hint. Differentiate the power normal form. □

Solution. The derivative of z^e at zero is nonzero exactly for $e = 1$. □

Exercise 24.38 (Generic degree)**.** Explain why the number of preimages counted with multiplicity is locally constant away from branch values.

Hint. Use the argument principle on the equation $p(x) - y = 0$. □

Solution. As y varies without crossing a critical value, the boundary has no zeros and the argument-principle count cannot jump. □

Exercise 24.39 (Even number of simple branch points)**.** Use Riemann–Hurwitz to show that a double cover of the sphere has an even number of simple branch points.

Hint. Solve the formula for the branch count. □

Solution. The formula gives $r = 2g + 2$, which is even. □

Counterexamples and hypothesis tests

Exercise 24.40 (Critical value confusion)**.** Is every point over a branch value necessarily ramified?

Hint. A branch value can have several preimages. □

Solution. No. At least one preimage is ramified, but other points in the same fiber can be unramified. □

Exercise 24.41 (Derivative zero means singular surface)**.** Does a vanishing derivative of the projection imply the source Riemann surface is singular?

Hint. Distinguish the map from the surface. □

Solution. No. It means the map is ramified. The source can be a perfectly smooth Riemann surface. □

Exercise 24.42 (Degree from distinct points only)**.** Can degree be computed by counting only distinct points in a branch fiber?

Hint. Remember multiplicities. □

Solution. No. At branch fibers several sheets merge, so multiplicities must be counted to recover the degree. □

Applications and investigations

Exercise 24.43 (Hyperelliptic curve)**.** For $y^2 = \prod_{j=1}^{2g+2}(x - a_j)$ with distinct roots, explain the natural map to the sphere.

Hint. Project to the x-coordinate. ☐

Solution. The projection is a degree-two branched cover, with simple branching over the roots and genus g by Riemann–Hurwitz. ☐

Exercise 24.44 (Polynomial dynamics)**.** Why are critical values central when studying a polynomial as a branched cover?

Hint. They are exactly where local inverse branches fail to remain separate. ☐

Solution. Away from critical values the map is a genuine covering, while crossing or looping around critical values changes the organization and monodromy of inverse branches. ☐

Challenge problems

Exercise 24.45 (Infinity ramification)**.** Determine the ramification index of a degree d polynomial at infinity as a map of spheres.

Hint. Use coordinates $u = 1/w$ and $v = 1/p(w)$. ☐

Solution. The leading term gives $v \sim c^{-1}u^d$, so infinity has ramification index d. ☐

Exercise 24.46 (Cubic sphere map)**.** Check Riemann–Hurwitz for a generic cubic polynomial map of the sphere.

Hint. There are two finite simple critical points and ramification at infinity. ☐

Solution. The two finite points contribute one each and infinity contributes two, for total ramification four. Then $2 = 3 \cdot 2 - 4$, as required for a sphere source. ☐

Chapter 25

Construction by Gluing

Gluing constructs Riemann surfaces by identifying open pieces through holomorphic transition maps. This is the global counterpart of defining a manifold from compatible charts.

Learning goals

After this chapter, the reader should be able to:

- construct Riemann surfaces by gluing charts along biholomorphic overlap maps;
- verify cocycle compatibility of transition functions on multiple overlaps;
- check Hausdorff and manifold conditions in explicit gluing constructions;
- build standard surfaces from polygons, annuli, or slit planes;
- track how holomorphic functions and differentials transform across glued charts;
- distinguish equivalent gluings from genuinely different complex structures;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

25.1 Gluing data

Start with complex domains and specify identifications between open subsets.

Example 25.1 (Gluing two charts into the sphere)**.** Take two copies $U_0, U_\infty \cong \mathbb{C}$ with coordinates z and w. Identify nonzero points by

$$w = \frac{1}{z}.$$

The transition map is biholomorphic on $\mathbb{C}^*$ and is its own inverse. The quotient is the Riemann sphere: U_0 covers all finite points, while $w = 0$ in the second chart represents infinity.

25.2 Transition maps

Identifications must be biholomorphic and satisfy inverse and cocycle compatibility.

25.3 Quotient space

The disjoint union modulo the identifications becomes the candidate surface.

25.4 Hausdorff condition

Gluing must be controlled so the quotient remains Hausdorff.

Example 25.2 (A torus from a parallelogram)**.** Let P be a parallelogram generated by noncollinear periods ω_1, ω_2. Identify opposite sides by translations $z \mapsto z+\omega_1$ and $z \mapsto z+\omega_2$. The transition maps are holomorphic translations and the corner identifications are compatible. The quotient is topologically a torus and inherits a complex structure from the planar coordinate.

25.5 Charts from pieces

Each original domain descends locally to a coordinate chart on the quotient.

25.6 Sphere from two charts

Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere.

25.7 Algebraic curves

Local branches of an algebraic relation can be glued into a smooth Riemann surface.

Example 25.3 (Two slit planes for a square root)**.** Take two copies of the plane slit along the nonnegative real axis. Glue the upper bank of the first copy to the lower bank of the second and vice versa. Moving around the origin switches sheets, while two turns return to the starting sheet. The resulting surface carries a single-valued coordinate w with projection $z = w^2$.

25.8 Cut-and-paste constructions

Copies of slit planes or polygons can be glued to produce root surfaces and tori.

25.9 Transition-cocycle viewpoint

The global complex structure is completely encoded by holomorphic chart transition functions.

25.10 Core structural results

Theorem 25.4 (Holomorphic gluing theorem)**.** *If Hausdorff quotient gluing data have biholomorphic transition maps satisfying the cocycle conditions, the quotient inherits a unique Riemann-surface structure for which the piece maps are local charts.*

Proof. Use the images of the original domains as charts. On overlaps their transition maps are exactly the prescribed biholomorphisms, so the atlas is compatible and defines the complex structure. □

Theorem 25.5 (Sphere gluing)**.** *Gluing two copies of* $\mathbb{C}$ *by* $w = 1/z$ *on* $\mathbb{C}^*$ *produces a compact Riemann surface biholomorphic to* $\widehat{\mathbb{C}}$*.*

Proof. Map one chart to finite complex numbers and the origin of the second chart to infinity. The transition relation is precisely the standard coordinate change at infinity. ☐

Theorem 25.6 (Cocycle consistency)**.** *For triple overlaps, transition maps must satisfy* $g_{ij} \circ g_{jk} = g_{ik}$*.*

Proof. A point represented through three charts must have the same coordinate identification independent of the path through the overlap graph; the cocycle condition is exactly that independence. ☐

25.11 Worked examples

Example 25.7 (Gluing data diagnostic)**.** Give a rigorous worked analysis of Gluing data. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Start with complex domains and specify identifications between open subsets. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.8 (Transition maps diagnostic)**.** Give a rigorous worked analysis of Transition maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.9 (Quotient space diagnostic)**.** Give a rigorous worked analysis of Quotient space. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The disjoint union modulo the identifications becomes the candidate surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.10 (Hausdorff condition diagnostic)**.** Give a rigorous worked analysis of Hausdorff condition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Gluing must be controlled so the quotient remains Hausdorff. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

25.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 25.11 (Gluing data diagnostic)**.** Give a rigorous worked analysis of Gluing data. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Start with complex domains and specify identifications between open subsets. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 25.12 (Transition maps diagnostic). Give a rigorous worked analysis of Transition maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.13 (Quotient space diagnostic). Give a rigorous worked analysis of Quotient space. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The disjoint union modulo the identifications becomes the candidate surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.14 (Hausdorff condition diagnostic). Give a rigorous worked analysis of Hausdorff condition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Gluing must be controlled so the quotient remains Hausdorff. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.15 (Charts from pieces diagnostic). Give a rigorous worked analysis of Charts from pieces. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Each original domain descends locally to a coordinate chart on the quotient. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.16 (Sphere from two charts diagnostic). Give a rigorous worked analysis of Sphere from two charts. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.17 (Algebraic curves diagnostic). Give a rigorous worked analysis of Algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local branches of an algebraic relation can be glued into a smooth Riemann surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.18 (Cut-and-paste constructions diagnostic). Give a rigorous worked analysis of Cut-and-paste constructions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Copies of slit planes or polygons can be glued to produce root surfaces and tori. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.19 (Holomorphic gluing theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If Hausdorff quotient gluing data have biholomorphic transition maps satisfying the cocycle conditions, the quotient inherits a unique Riemann-surface structure for which the piece maps are local charts.

Solution. Use the images of the original domains as charts. On overlaps their transition maps are exactly the prescribed biholomorphisms, so the atlas is compatible and defines the complex structure. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 25.20 (Sphere gluing proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Gluing two copies of $\mathbb{C}$ by $w = 1/z$ on $\mathbb{C}^*$ produces a compact Riemann surface biholomorphic to $\widehat{\mathbb{C}}$.

Solution. Map one chart to finite complex numbers and the origin of the second chart to infinity. The transition relation is precisely the standard coordinate change at infinity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 25.21 (Cocycle consistency proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For triple overlaps, transition maps must satisfy $g_{ij} \circ g_{jk} = g_{ik}$.

Solution. A point represented through three charts must have the same coordinate identification independent of the path through the overlap graph; the cocycle condition is exactly that independence. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 25.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Gluing constructs Riemann surfaces by identifying open pieces through holomorphic transition maps. This is the global counterpart of defining a manifold from compatible charts. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

25.13 Exercises with complete solutions

Exercise 25.23. State and apply the central principle behind Gluing data. Give one concrete consequence.

Hint. Write every overlap map explicitly and check it is biholomorphic on the overlap region. □

Solution. Start with complex domains and specify identifications between open subsets. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.24. State and apply the central principle behind Transition maps. Give one concrete consequence.

Hint. On triple overlaps, verify $g_{ij}g_{jk} = g_{ik}$; this cocycle identity makes chart identifications consistent. □

Solution. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.25. State and apply the central principle behind Quotient space. Give one concrete consequence.

Hint. To check Hausdorffness, separate inequivalent points using neighborhoods that survive the quotient. □

Solution. The disjoint union modulo the identifications becomes the candidate surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.26. State and apply the central principle behind Hausdorff condition. Give one concrete consequence.

Hint. Polygon edge identifications should be orientation-compatible with the intended complex structure; track the local coordinates near vertices. □

Solution. Gluing must be controlled so the quotient remains Hausdorff. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.27. State and apply the central principle behind Charts from pieces. Give one concrete consequence.

Hint. Functions glue only when their local expressions agree after composing with transition maps. □

Solution. Each original domain descends locally to a coordinate chart on the quotient. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.28. State and apply the central principle behind Sphere from two charts. Give one concrete consequence.

Hint. Differentials glue with the derivative of the transition function; check the transformation law on overlaps. □

Solution. Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.29. State and apply the central principle behind Algebraic curves. Give one concrete consequence.

Hint. Equivalent refinements or chart changes do not alter the underlying Riemann surface; compare transition atlases. □

Solution. Local branches of an algebraic relation can be glued into a smooth Riemann surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.30. State and apply the central principle behind Cut-and-paste constructions. Give one concrete consequence.

Hint. Watch for quotient identifications that create nonmanifold points; local neighborhoods must still look like disks. □

Solution. Copies of slit planes or polygons can be glued to produce root surfaces and tori. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

25.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 25.31 (Sphere transition)**.** What is the transition map between the finite and infinite charts of the sphere?

Hint. Use the reciprocal coordinate. □

Solution. It is $w = 1/z$ on $\mathbb{C}^*$. □

Exercise 25.32 (Inverse transition)**.** What is the inverse of $w = 1/z$?

Hint. The reciprocal map is an involution. □

Solution. It is $z = 1/w$. □

Exercise 25.33 (Torus edge map)**.** What holomorphic map identifies opposite vertical edges of a lattice parallelogram?

Hint. Use translation by one period. □

Solution. It is a translation $z \mapsto z + \omega_1$ or its inverse, depending on the chosen edge orientation. □

Exercise 25.34 (Cocycle check)**.** If $g_{12}(z) = z + 1$ and $g_{23}(z) = z + i$, what must g_{13} be on a triple overlap?

Hint. Compose the first two transitions. □

Solution. The cocycle condition gives $g_{13}(z) = z + 1 + i$. □

Exercise 25.35 (Square-root sheet switch)**.** How many crossings of the slit are needed to return to the original sheet in the two-sheeted square-root gluing?

Hint. Each crossing switches sheets. □

Solution. Two crossings return to the original sheet. □

Proofs

Exercise 25.36 (Gluing theorem)**.** Explain why compatible biholomorphic transition maps define a complex structure on a Hausdorff quotient.

Hint. Use the images of the original pieces as charts. □

Solution. Each piece descends locally homeomorphically to the quotient, and on overlaps the chart transitions are exactly the prescribed biholomorphisms. Thus the atlas is holomorphically compatible. □

Exercise 25.37 (Sphere compactness)**.** Why is the two-chart sphere gluing compact?

Hint. Relate it to the one-point compactification of the plane. □

Solution. The second chart provides a neighborhood of the added infinity point, and the resulting quotient is homeomorphic to the compact Riemann sphere. □

Exercise 25.38 (Cocycle necessity)**.** Why is the cocycle condition necessary on triple overlaps?

Hint. A point can be transferred between charts by two different routes. □

Solution. Without $g_{ij} \circ g_{jk} = g_{ik}$, the two routes assign different coordinates to the same equivalence class, so the quotient atlas is inconsistent. □

Exercise 25.39 (Translation transitions)**.** Prove that a lattice quotient inherits a Riemann-surface structure.

Hint. Use disks smaller than the shortest nonzero lattice displacement. □

Solution. Such a disk has disjoint lattice translates. Projection is one to one on it, and any overlap transition between lifted disks is a translation, hence holomorphic. □

Counterexamples and hypothesis tests

Exercise 25.40 (Noninvertible transition)**.** Can $w = z^2$ be used as a chart transition across a neighborhood containing zero?

Hint. A transition must be biholomorphic. □

Solution. No. Near zero the map is not locally one to one and its derivative vanishes. □

Exercise 25.41 (Non-Hausdorff quotient)**.** Is holomorphic compatibility alone enough if the quotient topology is non-Hausdorff?

Hint. Recall the manifold separation axiom. □

Solution. No. A Riemann surface must be Hausdorff, so pathological gluing can fail before complex analysis is considered. □

Exercise 25.42 (Arbitrary boundary identification)**.** May opposite polygon edges be glued by any continuous map and still produce a Riemann surface?

Hint. The transition must respect complex structure. □

Solution. No. The local identifications used as chart transitions must be holomorphic or antiholomorphic only if one is deliberately changing orientation; for a Riemann surface atlas they must be biholomorphic. □

Applications and investigations

Exercise 25.43 (Polygon models)**.** Why are polygon gluings useful for constructing compact surfaces?

Hint. They encode global topology through finitely many edge identifications. □

Solution. A single polygon with paired edges can represent the whole surface, while affine or translational edge maps often make the induced complex charts explicit. □

Exercise 25.44 (Algebraic curve normalization)**.** How can gluing local branches help construct a smooth surface from an algebraic relation?

Hint. Use separate local parameters near branch behavior. □

Solution. Smooth local branches are treated as charts and glued where their coordinate descriptions agree, separating sheets that appear multivalued in the plane projection. □

Challenge problems

Exercise 25.45 (Sphere from disks)**.** Construct the sphere by gluing two unit disks along annular collars rather than two full planes.

Hint. Use reciprocal coordinates on the overlap annuli. □

Solution. Choose annuli where $w = 1/z$ maps one collar biholomorphically to the other. The resulting compact quotient has the same two-chart complex structure as the sphere. □

Exercise 25.46 (Corner consistency on a torus)**.** Why do the four vertices of a fundamental parallelogram become one point without creating a singularity?

Hint. Use translations around the corner. □

Solution. Neighborhood sectors from the four corners glue by translations into one ordinary disk neighborhood. The total angle and holomorphic coordinate match smoothly, so no cone singularity appears. □

Chapter 26

Compactification and Genus

Compactification adds missing points in a way compatible with the complex structure, while genus records the basic topological type of a compact connected Riemann surface.

Learning goals

After this chapter, the reader should be able to:

- compactify basic noncompact Riemann surfaces by adding points with valid local coordinates;
- compute or identify genus from polygonal, covering, or Euler-characteristic data;
- distinguish genus zero, one, and higher-genus compact surfaces in standard examples;
- relate punctures, compactification points, and meromorphic extension;
- use Riemann–Hurwitz-type data where available to constrain genus;
- check that the added points produce a compact Riemann surface rather than only a topological compactification;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

26.1 Adding infinity

The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$.

Example 26.1 (Compactifying the complex plane)**.** Add one point ∞ to $\mathbb{C}$. Near infinity use the coordinate

$$w = \frac{1}{z}.$$

A neighborhood of ∞ corresponds to $|z| > R$ together with infinity, which maps to $|w| < 1/R$. The reciprocal transition is holomorphic away from zero, giving the one-point compactification its standard Riemann-surface structure.

26.2 Puncture filling

A puncture can sometimes be filled when the local complex structure extends across it.

26.3 Compact algebraic curves

Affine algebraic curves often acquire points at infinity in their projective compactification.

26.4 Genus

For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$.

Example 26.2 (Genus from Euler characteristic)**.** For a compact orientable surface,

$$\chi = 2 - 2g.$$

The sphere has $\chi = 2$, hence $g = 0$. A torus has a cell decomposition with one vertex, two edges and one face, so $\chi = 1 - 2 + 1 = 0$, hence $g = 1$. This topological invariant matches the analytic distinction between the sphere and complex tori.

26.5 Sphere and torus

The sphere has genus zero; complex tori have genus one.

26.6 Meromorphic functions

On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere.

26.7 Branched-cover viewpoint

Maps to the sphere reveal genus through their degree and ramification.

Example 26.3 (Riemann–Hurwitz for a torus double cover)**.** Let $X \to \widehat{\mathbb{C}}$ be a double cover branched simply at four points. Then

$$2 - 2g(X) = 2 \cdot 2 - 4 = 0,$$

so $g(X) = 1$. Thus any such compact double cover has torus topology. The calculation is the bridge from branch data to global genus.

26.8 Riemann–Hurwitz formula

For finite branched covers, Euler characteristics differ by a total ramification correction.

26.9 Compactness consequences

Holomorphic functions, divisors, and differential forms become globally constrained on compact surfaces.

26.10 Core structural results

Theorem 26.4 (Holomorphic functions on compact surfaces). *Every holomorphic function from a compact connected Riemann surface to $\mathbb{C}$ is constant.*

Proof. The absolute value attains a maximum by compactness. In a local coordinate the maximum-modulus principle forces local constancy, and connectedness propagates it globally. □

Theorem 26.5 (Riemann–Hurwitz formula). *For a degree-d holomorphic map $f : X \to Y$ of compact Riemann surfaces, $2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X}(e_p - 1)$.*

Proof. Triangulate the target with branch values among vertices and lift the triangulation. Faces and edges lift d-fold, while vertices lose exactly the total ramification multiplicity. Comparing Euler characteristics yields the formula. □

Theorem 26.6 (Genus of a double cover). *A degree-two cover of the sphere branched simply at r points has genus $g = (r - 2)/2$.*

Proof. Apply Riemann–Hurwitz with $d = 2$, target genus zero, and total ramification r: $2g - 2 = -4 + r$. □

26.11 Worked examples

Example 26.7 (Adding infinity diagnostic). Give a rigorous worked analysis of Adding infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.8 (Puncture filling diagnostic). Give a rigorous worked analysis of Puncture filling. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A puncture can sometimes be filled when the local complex structure extends across it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.9 (Compact algebraic curves diagnostic). Give a rigorous worked analysis of Compact algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Affine algebraic curves often acquire points at infinity in their projective compactification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.10 (Genus diagnostic). Give a rigorous worked analysis of Genus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

26.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 26.11 (Adding infinity diagnostic)**.** Give a rigorous worked analysis of Adding infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.12 (Puncture filling diagnostic)**.** Give a rigorous worked analysis of Puncture filling. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A puncture can sometimes be filled when the local complex structure extends across it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.13 (Compact algebraic curves diagnostic)**.** Give a rigorous worked analysis of Compact algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Affine algebraic curves often acquire points at infinity in their projective compactification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.14 (Genus diagnostic)**.** Give a rigorous worked analysis of Genus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.15 (Sphere and torus diagnostic)**.** Give a rigorous worked analysis of Sphere and torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sphere has genus zero; complex tori have genus one. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.16 (Meromorphic functions diagnostic)**.** Give a rigorous worked analysis of Meromorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.17 (Branched-cover viewpoint diagnostic)**.** Give a rigorous worked analysis of Branched-cover viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Maps to the sphere reveal genus through their degree and ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.18 (Riemann–Hurwitz formula diagnostic)**.** Give a rigorous worked analysis of Riemann–Hurwitz formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For finite branched covers, Euler characteristics differ by a total ramification correction. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.19 (Holomorphic functions on compact surfaces proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every holomorphic function from a compact connected Riemann surface to $\mathbb{C}$ is constant.

Solution. The absolute value attains a maximum by compactness. In a local coordinate the maximum-modulus principle forces local constancy, and connectedness propagates it globally. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.20 (Riemann–Hurwitz formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a degree-d holomorphic map $f : X \to Y$ of compact Riemann surfaces, $2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X}(e_p - 1)$.

Solution. Triangulate the target with branch values among vertices and lift the triangulation. Faces and edges lift d-fold, while vertices lose exactly the total ramification multiplicity. Comparing Euler characteristics yields the formula. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.21 (Genus of a double cover proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A degree-two cover of the sphere branched simply at r points has genus $g = (r - 2)/2$.

Solution. Apply Riemann–Hurwitz with $d = 2$, target genus zero, and total ramification r: $2g - 2 = -4 + r$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Compactification adds missing points in a way compatible with the complex structure, while genus records the basic topological type of a compact connected Riemann surface. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

26.13 Exercises with complete solutions

Exercise 26.23. State and apply the central principle behind Adding infinity. Give one concrete consequence.

Hint. To add a point at infinity, find a reciprocal or other coordinate in which the punctured neighborhood becomes a disk around zero. □

Solution. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.24. State and apply the central principle behind Puncture filling. Give one concrete consequence.

Hint. A meromorphic function extends across a compactification point when its local expression has at worst a pole in the new coordinate. ☐

Solution. A puncture can sometimes be filled when the local complex structure extends across it. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 26.25. State and apply the central principle behind Compact algebraic curves. Give one concrete consequence.

Hint. Compute Euler characteristic from a cell or polygon decomposition, then use $\chi = 2-2g$ for a compact orientable surface. ☐

Solution. Affine algebraic curves often acquire points at infinity in their projective compactification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 26.26. State and apply the central principle behind Genus. Give one concrete consequence.

Hint. For a branched covering, combine degree and ramification data to constrain genus via Riemann–Hurwitz. ☐

Solution. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 26.27. State and apply the central principle behind Sphere and torus. Give one concrete consequence.

Hint. Adding one point to $\mathbb{C}$ gives the sphere because $w = 1/z$ supplies the second chart near infinity. ☐

Solution. The sphere has genus zero; complex tori have genus one. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 26.28. State and apply the central principle behind Meromorphic functions. Give one concrete consequence.

Hint. Do not infer compactness from bounded coordinates; prove the added surface is topologically compact. ☐

Solution. On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 26.29. State and apply the central principle behind Branched-cover viewpoint. Give one concrete consequence.

Hint. Track punctures separately from handles: filling punctures changes compactness without necessarily changing genus. ☐

Solution. Maps to the sphere reveal genus through their degree and ramification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.30. State and apply the central principle behind Riemann–Hurwitz formula. Give one concrete consequence.

Hint. Use known models—sphere, torus, higher-genus polygon—to sanity-check the computed genus. □

Solution. For finite branched covers, Euler characteristics differ by a total ramification correction. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

26.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 26.31 (Sphere genus)**.** Find the genus of a compact surface with Euler characteristic 2.

Hint. Use $\chi = 2 - 2g$. □

Solution. The genus is zero. □

Exercise 26.32 (Two-handle Euler characteristic)**.** Find the Euler characteristic of a genus-two surface.

Hint. Use $\chi = 2 - 2g$. □

Solution. It is -2. □

Exercise 26.33 (Torus genus)**.** A compact orientable surface has Euler characteristic zero. What is its genus?

Hint. Solve the genus formula. □

Solution. Its genus is one. □

Exercise 26.34 (Double-cover branch count)**.** How many simple branch points does a degree-two sphere cover of genus two have?

Hint. Use Riemann–Hurwitz. □

Solution. The equation $2 - 4 = 4 - r$ gives $r = 6$. □

Exercise 26.35 (Compact holomorphic function)**.** What can be said about a holomorphic map from a compact connected Riemann surface to $\mathbb{C}$?

Hint. Apply the maximum modulus principle. □

Solution. It must be constant. □

Proofs

Exercise 26.36 (Compact maximum principle)**.** Prove that every holomorphic function on a compact connected Riemann surface is constant.

Hint. The modulus attains a maximum. □

Solution. Compactness gives a maximum point. In a local chart the maximum modulus principle forces local constancy, and connectedness propagates it globally. □

Exercise 26.37 (Sphere compactification)**.** Show that the reciprocal coordinate produces a compatible chart at infinity.

Hint. Check the overlap transition. □

Solution. On the overlap with the finite chart, $w = 1/z$ is biholomorphic. Therefore adjoining the point $w = 0$ defines a valid complex chart around infinity. □

Exercise 26.38 (Torus Euler characteristic)**.** Prove from a square cell decomposition that a torus has Euler characteristic zero.

Hint. Count vertices, edges and faces after identifications. □

Solution. All four vertices identify to one, opposite edge pairs give two edges, and there is one face, so $\chi = 1 - 2 + 1 = 0$. □

Exercise 26.39 (Riemann–Hurwitz genus)**.** Derive $g = (r - 2)/2$ for a double cover of the sphere with r simple branch points.

Hint. Insert degree two and total ramification r. □

Solution. The formula gives $2 - 2g = 4 - r$, hence $g = (r - 2)/2$. □

Counterexamples and hypothesis tests

Exercise 26.40 (Remove compactness)**.** Does the theorem that every holomorphic function is constant remain true on a noncompact Riemann surface?

Hint. Use the complex plane. □

Solution. No. The identity function on $\mathbb{C}$ is a nonconstant holomorphic example. □

Exercise 26.41 (Every puncture fills)**.** Can every puncture of a Riemann surface be filled holomorphically?

Hint. Consider essential singular behavior. □

Solution. No. Filling requires the complex structure and relevant functions or maps to extend appropriately; essential or topological behavior can obstruct a chosen extension. □

Exercise 26.42 (Genus from local charts)**.** Can genus be read from one local coordinate chart?

Hint. Genus is global topology. □

Solution. No. Every surface is locally disk-like; genus measures global handle structure and requires global information. □

Applications and investigations

Exercise 26.43 (Projective completion)**.** Why does adding points at infinity help when studying affine algebraic curves?

Hint. Compactness gives global constraints. □

Solution. The projective completion packages behavior at infinity into finitely many surface points and makes tools such as divisors, meromorphic functions and Riemann–Hurwitz global. □

Exercise 26.44 (Meromorphic maps)**.** Why are meromorphic functions on a compact Riemann surface naturally viewed as holomorphic maps to the sphere?

Hint. Treat poles as values equal to infinity. □

Solution. Near a pole, the reciprocal is holomorphic and vanishes, so adjoining the sphere point at infinity turns the meromorphic function into a holomorphic map. □

Challenge problems

Exercise 26.45 (Degree and ramification)**.** A degree-three map from a genus-two surface to the sphere has total ramification R. Find R.

Hint. Apply Riemann–Hurwitz. □

Solution. $2 - 2 \cdot 2 = 3 \cdot 2 - R$, so $-2 = 6 - R$ and $R = 8$. □

Exercise 26.46 (Why genus is invariant)**.** Explain why biholomorphic compact Riemann surfaces must have the same genus.

Hint. A biholomorphism is in particular a homeomorphism. □

Solution. Genus is a topological invariant of compact orientable surfaces, and a biholomorphism preserves the underlying topology. □

Part VI

Elliptic Functions

Chapter 27

Lattices and Complex Tori

A lattice in the complex plane is a discrete rank-two subgroup, and quotienting by it produces a compact complex torus. This is the analytic model of every genus-one Riemann surface after choosing a base point.

Learning goals

After this chapter, the reader should be able to:

- construct a complex torus as $\mathbb{C}/\Lambda$ from a lattice;
- choose fundamental parallelograms and identify equivalent boundary points;
- verify that holomorphic structure descends through the lattice quotient;
- classify lattice changes induced by basis transformations and scaling;
- compute basic topological and complex-analytic invariants of the torus;
- relate periodic functions on $\mathbb{C}$ to meromorphic functions on the torus;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

27.1 Lattices

A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$.

Example 27.1 (Reducing a point to a fundamental parallelogram)**.** For the square lattice $\Lambda = \mathbb{Z} + i\mathbb{Z}$, take $z = 3.4 - 2.7i$. Subtract the lattice element $3 - 3i$:

$$z - (3 - 3i) = 0.4 + 0.3i.$$

Thus z and $0.4 + 0.3i$ represent the same torus point. Reduction modulo the two periods is the concrete quotient operation behind $\mathbb{C}/\Lambda$.

27.2 Fundamental parallelogram

Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications.

27.3 Complex torus

The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations.

27.4 Quotient projection

The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map.

Example 27.2 (Changing lattice basis)**.** Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and define

$$\omega_1' = \omega_1 + \omega_2, \qquad \omega_2' = \omega_2.$$

The change matrix has determinant one, so the new pair generates the same lattice. Conversely,

$$\omega_1 = \omega_1' - \omega_2', \qquad \omega_2 = \omega_2'.$$

The torus depends on the lattice subgroup, not on a particular basis.

27.5 Change of lattice basis

Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice.

27.6 Normalized modulus

After scaling, a lattice can be represented by $\mathbb{Z} + \mathbb{Z}\tau$ with $\Im\tau > 0$.

27.7 Modular equivalence

Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori.

Example 27.3 (Normalizing a lattice)**.** Given noncollinear periods ω_1, ω_2, scale by $1/\omega_1$. The lattice becomes

$$\mathbb{Z} + \mathbb{Z}\tau, \qquad \tau = \frac{\omega_2}{\omega_1}.$$

After reversing a basis vector if necessary, choose $\Im\tau > 0$. This isolates the complex shape parameter while removing the irrelevant overall scale.

27.8 Holomorphic differentials

The form dz descends to a nowhere-vanishing holomorphic differential on the torus.

27.9 Genus one

The torus has Euler characteristic zero and genus one.

27.10 Core structural results

Theorem 27.4 (Complex structure on the quotient)**.** *For a lattice Λ, the quotient $\mathbb{C}/\Lambda$ has a unique Riemann-surface structure making the projection locally biholomorphic.*

Proof. Choose disks small enough that distinct lattice translates are disjoint. Projection restricts to a homeomorphism on each such disk, and transition maps between lifted charts are translations, hence holomorphic. □

Theorem 27.5 (Compactness of the torus)**.** *$\mathbb{C}/\Lambda$ is compact.*

Proof. The closure of a fundamental parallelogram is compact and its projection surjects onto the quotient. A continuous image of a compact set is compact. □

Theorem 27.6 (Basis-change invariance)**.** *If $(\omega_1', \omega_2') = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ with an integral matrix of determinant ± 1, then the generated lattice is unchanged.*

Proof. The new generators are integral combinations of the old, and invertibility over $\mathbb{Z}$ expresses the old generators as integral combinations of the new. □

27.11 Worked examples

Example 27.7 (Lattices diagnostic)**.** Give a rigorous worked analysis of Lattices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.8 (Fundamental parallelogram diagnostic)**.** Give a rigorous worked analysis of Fundamental parallelogram. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.9 (Complex torus diagnostic)**.** Give a rigorous worked analysis of Complex torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.10 (Quotient projection diagnostic)**.** Give a rigorous worked analysis of Quotient projection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

27.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 27.11 (Lattices diagnostic)**.** Give a rigorous worked analysis of Lattices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.12 (Fundamental parallelogram diagnostic)**.** Give a rigorous worked analysis of Fundamental parallelogram. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.13 (Complex torus diagnostic)**.** Give a rigorous worked analysis of Complex torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.14 (Quotient projection diagnostic)**.** Give a rigorous worked analysis of Quotient projection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.15 (Change of lattice basis diagnostic)**.** Give a rigorous worked analysis of Change of lattice basis. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.16 (Normalized modulus diagnostic)**.** Give a rigorous worked analysis of Normalized modulus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. After scaling, a lattice can be represented by $\mathbb{Z} + \mathbb{Z}\tau$ with $\Im\tau > 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.17 (Modular equivalence diagnostic)**.** Give a rigorous worked analysis of Modular equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.18 (Holomorphic differentials diagnostic)**.** Give a rigorous worked analysis of Holomorphic differentials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The form dz descends to a nowhere-vanishing holomorphic differential on the torus. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 27.19 (Complex structure on the quotient proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a lattice Λ, the quotient $\mathbb{C}/\Lambda$ has a unique Riemann-surface structure making the projection locally biholomorphic.

Solution. Choose disks small enough that distinct lattice translates are disjoint. Projection restricts to a homeomorphism on each such disk, and transition maps between lifted charts are translations, hence holomorphic. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.20 (Compactness of the torus proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $\mathbb{C}/\Lambda$ is compact.

Solution. The closure of a fundamental parallelogram is compact and its projection surjects onto the quotient. A continuous image of a compact set is compact. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.21 (Basis-change invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $(\omega_1', \omega_2') = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ with an integral matrix of determinant ± 1, then the generated lattice is unchanged.

Solution. The new generators are integral combinations of the old, and invertibility over $\mathbb{Z}$ expresses the old generators as integral combinations of the new. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. A lattice in the complex plane is a discrete rank-two subgroup, and quotienting by it produces a compact complex torus. This is the analytic model of every genus-one Riemann surface after choosing a base point. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

27.13 Exercises with complete solutions

Exercise 27.23. State and apply the central principle behind Lattices. Give one concrete consequence.

Hint. Choose two $\mathbb{R}$-linearly independent periods ω_1, ω_2; their integer span is the lattice. ☐

Solution. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.24. State and apply the central principle behind Fundamental parallelogram. Give one concrete consequence.

Hint. Represent every torus point by a point in one fundamental parallelogram, with opposite edges identified. ☐

Solution. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.25. State and apply the central principle behind Complex torus. Give one concrete consequence.

Hint. Translations by lattice vectors are holomorphic, so the quotient inherits a well-defined complex atlas. ☐

Solution. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.26. State and apply the central principle behind Quotient projection. Give one concrete consequence.

Hint. Changing the lattice basis by an element of $SL_2(\mathbb{Z})$ does not change the underlying lattice. ☐

Solution. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.27. State and apply the central principle behind Change of lattice basis. Give one concrete consequence.

Hint. Scaling the lattice by $\lambda \neq 0$ gives a biholomorphic torus after scaling the coordinate by λ. ☐

Solution. Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.28. State and apply the central principle behind Normalized modulus. Give one concrete consequence.

Hint. A doubly periodic meromorphic function on $\mathbb{C}$ descends to a meromorphic function on the compact torus. ☐

Solution. After scaling, a lattice can be represented by $\mathbb{Z}+\mathbb{Z}\tau$ with $\Im\tau > 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.29. State and apply the central principle behind Modular equivalence. Give one concrete consequence.

Hint. Use compactness to rule out nonconstant holomorphic functions on the torus without poles. ☐

Solution. Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.30. State and apply the central principle behind Holomorphic differentials. Give one concrete consequence.

Hint. Distinguish the lattice basis from the lattice itself; basis-dependent formulas should be checked for modular invariance. ☐

Solution. The form dz descends to a nowhere-vanishing holomorphic differential on the torus. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

27.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 27.31 (Square lattice reduction)**.** Reduce $2.3 + 4.8i$ modulo $\mathbb{Z} + i\mathbb{Z}$ to the unit square.

Hint. Subtract integer real and imaginary parts. ☐

Solution. One representative is $0.3 + 0.8i$. ☐

Exercise 27.32 (Basis determinant)**.** Does the pair $\omega_1' = 2\omega_1 + \omega_2$, $\omega_2' = \omega_1 + \omega_2$ generate the same lattice?

Hint. Compute the determinant of the integer matrix. ☐

Solution. Yes. The determinant is $2 \cdot 1 - 1 \cdot 1 = 1$. ☐

Exercise 27.33 (Normalized modulus)**.** Normalize the lattice generated by 2 and $1 + 3i$.

Hint. Divide both periods by 2. ☐

Solution. It becomes $\mathbb{Z} + \mathbb{Z}((1 + 3i)/2)$. ☐

Exercise 27.34 (Torus covering)**.** What is the deck group of $\mathbb{C} \to \mathbb{C}/\Lambda$?

Hint. Deck maps are period translations. ☐

Solution. It is the translation group $z \mapsto z + \lambda$ for $\lambda \in \Lambda$. ☐

Exercise 27.35 (Euler characteristic)**.** What is the genus of $\mathbb{C}/\Lambda$?

Hint. A fundamental parallelogram has torus edge identifications. ☐

Solution. It has genus one. ☐

Proofs

Exercise 27.36 (Quotient complex structure)**.** Prove that $\mathbb{C}/\Lambda$ has a Riemann-surface structure.

Hint. Use sufficiently small disks with disjoint lattice translates. ☐

Solution. Projection is injective on each small disk and transitions between different lifts are translations, hence holomorphic. ☐

Exercise 27.37 (Compactness)**.** Prove that the complex torus is compact.

Hint. Project a closed fundamental parallelogram. ☐

Solution. The closed parallelogram is compact and its projection is continuous and surjective, so the quotient is compact. ☐

Exercise 27.38 (Basis invariance)**.** Prove that an integral basis change with determinant ± 1 leaves the lattice unchanged.

Hint. Use the integer inverse matrix. ☐

Solution. New generators are integer combinations of old ones, and the inverse matrix with integer entries expresses the old generators as combinations of the new ones. ☐

Exercise 27.39 (Descending differential)**.** Explain why dz descends to the torus.

Hint. Check invariance under deck translations. ☐

Solution. Translations have differential one, so $d(z + \lambda) = dz$. Thus the form is invariant and defines a holomorphic one-form downstairs. ☐

Counterexamples and hypothesis tests

Exercise 27.40 (Collinear generators)**.** Do 1 and 2 generate a complex lattice of rank two?

Hint. Check real linear independence. ☐

Solution. No. Their ratio is real, so the subgroup is one-dimensional rather than a discrete rank-two lattice spanning the plane. ☐

Exercise 27.41 (Determinant two)**.** Does the basis change matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ preserve the lattice?

Hint. Its determinant is not ± 1. ☐

Solution. Not in general. It generates a sublattice of index two. ☐

Exercise 27.42 (Noncompact quotient)**.** Would quotienting by only one nonzero period give a compact Riemann surface?

Hint. The remaining transverse direction is unbounded. ☐

Solution. No. The quotient is a cylinder and remains noncompact. ☐

Applications and investigations

Exercise 27.43 (Periodic functions)**.** Why does a doubly periodic meromorphic function descend to $\mathbb{C}/\Lambda$?

Hint. It is constant on lattice equivalence classes. □

Solution. Periodicity makes the function well defined on quotient points, and local quotient charts preserve meromorphicity. □

Exercise 27.44 (Genus-one moduli)**.** What information does the normalized parameter τ retain?

Hint. Overall scaling was removed. □

Solution. It records the complex shape of the lattice, modulo changes of lattice basis by modular transformations. □

Challenge problems

Exercise 27.45 (Modular basis move)**.** Show that swapping a normalized basis sends τ to $-1/\tau$ after rescaling.

Hint. Use the new basis $(\tau, -1)$ and normalize the first generator. □

Solution. Dividing by τ gives generators 1 and $-1/\tau$, so the normalized parameter transforms accordingly. □

Exercise 27.46 (No holomorphic coordinate globally)**.** Why can the torus not admit one global chart into an open subset of $\mathbb{C}$?

Hint. A chart is a homeomorphism onto an open plane set. □

Solution. The torus is compact, while a nonempty open subset of $\mathbb{C}$ is not compact. Hence no single global planar chart exists. □

Chapter 28

Elliptic Functions

Elliptic functions are meromorphic functions periodic with respect to a rank-two lattice. Compactness of the associated torus forces strong balance laws for zeros, poles, and residues.

Learning goals

After this chapter, the reader should be able to:

- verify double periodicity of candidate elliptic functions with respect to a lattice;
- use compactness of the torus to prove global constraints on elliptic functions;
- count zeros and poles of elliptic functions with multiplicity in a fundamental domain;
- apply residue balance to elliptic functions on the torus;
- construct elliptic functions from lattice-periodic meromorphic data;
- distinguish singly periodic functions from genuinely doubly periodic elliptic functions;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

28.1 Double periodicity

An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$.

Example 28.1 (Residues in one period cell)**.** Let f be elliptic for Λ, with no poles on the boundary of a fundamental parallelogram P. Opposite edge integrals cancel by periodicity, so

$$\int_{\partial P} f(z)\,dz = 0.$$

The residue theorem gives

$$\sum_{a\in P} \operatorname{Res}(f,a) = 0.$$

Therefore an elliptic function cannot have exactly one simple pole modulo the lattice with nonzero residue.

28.2 Descent to the torus

Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane.

28.3 Poles in a period cell

A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant.

28.4 Residue balance

The sum of residues in a fundamental parallelogram is zero.

Example 28.2 (Zero and pole count from the logarithmic derivative)**.** For a nonzero elliptic function f, the logarithmic derivative f'/f is also elliptic. Opposite edges cancel, hence

$$\frac{1}{2\pi i}\int_{\partial P}\frac{f'}{f}\,dz = 0.$$

By the argument principle, the left side equals the number of zeros minus the number of poles in P, counted with multiplicity. Thus every period cell contains equally many zeros and poles.

28.5 Zero-pole balance

The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity.

28.6 Abel-type position constraint

The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity.

28.7 Constructing elliptic functions

Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist.

Example 28.3 (Why an elliptic function must have a pole)**.** Suppose an elliptic function f had no poles. It would descend to a holomorphic function on the compact torus $\mathbb{C}/\Lambda$. By the compact maximum principle it would be constant. Hence every nonconstant elliptic function has at least one pole in every fundamental cell.

28.8 Even and odd examples

The Weierstrass $\wp$-function is even; its derivative is odd.

28.9 Function-field preview

For a fixed lattice, elliptic functions are generated algebraically by $\wp$ and $\wp'$.

28.10 Core structural results

Theorem 28.4 (Residue sum theorem)**.** *The sum of residues of an elliptic function in a fundamental parallelogram containing no poles on its boundary is zero.*

Proof. Integrate around the boundary. Opposite edges cancel because the function is periodic and the orientations are opposite. The residue theorem then forces the interior residue sum to vanish. □

Theorem 28.5 (Equal zero and pole count)**.** *A nonzero elliptic function has equally many zeros and poles in a fundamental parallelogram, counted with multiplicity.*

Proof. Apply the argument principle to f'/f. This quotient is elliptic, so opposite-edge contour integrals cancel, leaving zero; hence zeros minus poles equals zero. □

Theorem 28.6 (No single simple pole)**.** *There is no elliptic function with exactly one simple pole modulo the lattice.*

Proof. Such a function would have one nonzero residue in a fundamental parallelogram, contradicting the residue-sum theorem. □

28.11 Worked examples

Example 28.7 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.8 (Descent to the torus diagnostic)**.** Give a rigorous worked analysis of Descent to the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.9 (Poles in a period cell diagnostic)**.** Give a rigorous worked analysis of Poles in a period cell. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.10 (Residue balance diagnostic)**.** Give a rigorous worked analysis of Residue balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The sum of residues in a fundamental parallelogram is zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

28.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 28.11 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.12 (Descent to the torus diagnostic)**.** Give a rigorous worked analysis of Descent to the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.13 (Poles in a period cell diagnostic)**.** Give a rigorous worked analysis of Poles in a period cell. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.14 (Residue balance diagnostic)**.** Give a rigorous worked analysis of Residue balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sum of residues in a fundamental parallelogram is zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.15 (Zero-pole balance diagnostic)**.** Give a rigorous worked analysis of Zero-pole balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.16 (Abel-type position constraint diagnostic)**.** Give a rigorous worked analysis of Abel-type position constraint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.17 (Constructing elliptic functions diagnostic)**.** Give a rigorous worked analysis of Constructing elliptic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.18 (Even and odd examples diagnostic)**.** Give a rigorous worked analysis of Even and odd examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Weierstrass $\wp$-function is even; its derivative is odd. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 28.19 (Residue sum theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The sum of residues of an elliptic function in a fundamental parallelogram containing no poles on its boundary is zero.

Solution. Integrate around the boundary. Opposite edges cancel because the function is periodic and the orientations are opposite. The residue theorem then forces the interior residue sum to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 28.20 (Equal zero and pole count proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A nonzero elliptic function has equally many zeros and poles in a fundamental parallelogram, counted with multiplicity.

Solution. Apply the argument principle to f'/f. This quotient is elliptic, so opposite-edge contour integrals cancel, leaving zero; hence zeros minus poles equals zero. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 28.21 (No single simple pole proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: There is no elliptic function with exactly one simple pole modulo the lattice.

Solution. Such a function would have one nonzero residue in a fundamental parallelogram, contradicting the residue-sum theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 28.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Elliptic functions are meromorphic functions periodic with respect to a rank-two lattice. Compactness of the associated torus forces strong balance laws for zeros, poles, and residues. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

28.13 Exercises with complete solutions

Exercise 28.23. State and apply the central principle behind Double periodicity. Give one concrete consequence.

Hint. Verify both independent periods, not just one, before calling a meromorphic function elliptic. ☐

Solution. An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.24. State and apply the central principle behind Descent to the torus. Give one concrete consequence.

Hint. Integrate over the boundary of a fundamental parallelogram; opposite-edge contributions cancel by periodicity. ☐

Solution. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.25. State and apply the central principle behind Poles in a period cell. Give one concrete consequence.

Hint. The residue theorem on the parallelogram shows the sum of residues in a fundamental domain is zero. ☐

Solution. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.26. State and apply the central principle behind Residue balance. Give one concrete consequence.

Hint. Apply the argument principle to count zeros and poles; periodic boundary cancellation forces equal total multiplicities. ☐

Solution. The sum of residues in a fundamental parallelogram is zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.27. State and apply the central principle behind Zero-pole balance. Give one concrete consequence.

Hint. A nonconstant elliptic function must have poles because it descends to a nonconstant meromorphic function on a compact surface. ☐

Solution. The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.28. State and apply the central principle behind Abel-type position constraint. Give one concrete consequence.

Hint. Construct new elliptic functions from sums, products, derivatives, or rational functions of known elliptic functions. ☐

Solution. The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.29. State and apply the central principle behind Constructing elliptic functions. Give one concrete consequence.

Hint. Shift zeros and poles back into one fundamental domain before counting them. ☐

Solution. Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 28.30. State and apply the central principle behind Even and odd examples. Give one concrete consequence.

Hint. Check whether apparent periods are primitive; a smaller lattice may reveal additional symmetry but does not destroy ellipticity. ☐

Solution. The Weierstrass $\wp$-function is even; its derivative is odd. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

28.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 28.31 (Residue balance)**.** An elliptic function has two simple poles in a cell with residues 3 and r. Find r.

Hint. The residue sum is zero. ☐

Solution. $r = -3$. ☐

Exercise 28.32 (Zero count)**.** An elliptic function has poles of total multiplicity five in a fundamental cell. How many zeros does it have there, counted with multiplicity?

Hint. Use zero-pole balance. ☐

Solution. It has five zeros counted with multiplicity. ☐

Exercise 28.33 (Single pole)**.** Can an elliptic function have one simple pole modulo the lattice?

Hint. Use residue balance. ☐

Solution. No, because its nonzero residue could not be canceled. ☐

Exercise 28.34 (Derivative periodicity)**.** If f is elliptic, is f' elliptic?

Hint. Differentiate the period identity. ☐

Solution. Yes. Differentiating $f(z+\omega) = f(z)$ gives $f'(z+\omega) = f'(z)$. ☐

Exercise 28.35 (Log derivative)**.** Why is f'/f elliptic where defined?

Hint. Both numerator and denominator have the same periods. ☐

Solution. The quotient inherits every period of f, with meromorphic singularities at zeros and poles. ☐

Proofs

Exercise 28.36 (Residue theorem)**.** Prove residue balance for an elliptic function.

Hint. Integrate around a fundamental parallelogram. □

Solution. Each edge integral cancels with the opposite translated edge. The boundary integral is zero, so the residue theorem forces the interior residue sum to vanish. □

Exercise 28.37 (Zero-pole theorem)**.** Prove that zeros and poles balance in a period cell.

Hint. Apply the argument principle to f'/f. □

Solution. The boundary integral cancels by periodicity and equals $2\pi i$ times zeros minus poles, so the difference is zero. □

Exercise 28.38 (Nonconstant implies pole)**.** Prove that a nonconstant elliptic function must have a pole.

Hint. Descend to the compact torus. □

Solution. Without poles it would be holomorphic on the compact torus and therefore constant, contradiction. □

Exercise 28.39 (Position constraint idea)**.** Explain why integrating $zf'(z)/f(z)$ around a period cell leads to a lattice-valued constraint on sums of zeros and poles.

Hint. Opposite-edge cancellation now leaves period multiples. □

Solution. Residues give the weighted sum of zero locations minus pole locations. Boundary terms differ by lattice periods, so the resulting difference is determined modulo the lattice. □

Counterexamples and hypothesis tests

Exercise 28.40 (One double pole)**.** Can an elliptic function have exactly one double pole modulo the lattice?

Hint. A double pole can have zero residue. □

Solution. Yes. Residue balance does not forbid it; the Weierstrass $\wp$-function is the standard example. □

Exercise 28.41 (Entire doubly periodic)**.** Can a nonconstant entire doubly periodic function exist?

Hint. Descend to the compact torus. □

Solution. No. It would define a holomorphic function on a compact connected surface and hence be constant. □

Exercise 28.42 (Boundary pole)**.** May one apply the standard period-cell residue count with a pole exactly on the chosen boundary?

Hint. The contour passes through a singularity. □

Solution. Not directly. Shift the parallelogram slightly or use a boundary convention that avoids poles. □

Applications and investigations

Exercise 28.43 (Constructing principal parts)**.** What compatibility condition must proposed simple-pole residues satisfy before an elliptic function can realize them?

Hint. Use the residue sum theorem. □

Solution. Their sum over one period cell must be zero. □

Exercise 28.44 (Compactification viewpoint)**.** Why are elliptic-function balance laws stronger than analogous local meromorphic facts in the plane?

Hint. The quotient torus is compact. □

Solution. Compactness removes boundary escape and forces global residue and divisor-degree constraints within one fundamental cell. □

Challenge problems

Exercise 28.45 (Order of an elliptic function)**.** Why is the total pole multiplicity in a cell often called the order of a nonconstant elliptic function?

Hint. It equals the total zero multiplicity. □

Solution. The zero-pole theorem shows the same integer counts generic inverse images of values, so it is the degree of the corresponding meromorphic map from the torus to the sphere. □

Exercise 28.46 (No order one)**.** Show that a nonconstant elliptic function cannot have order one.

Hint. Order one would mean one simple pole. □

Solution. A single simple pole has a nonzero residue, contradicting residue balance. Thus the smallest possible order is two. □

Chapter 29

The Weierstrass $\wp$-Function

The Weierstrass $\wp$-function is the canonical elliptic function attached to a lattice. Its carefully regularized lattice sum gives a doubly periodic even function with one double pole per period cell and an algebraic differential equation.

Learning goals

After this chapter, the reader should be able to:

- construct the Weierstrass $\wp$-function from its lattice series and justify convergence on compacta away from the lattice;
- derive evenness, double periodicity, and pole structure of $\wp$;
- compute the differential equation satisfied by $\wp$ and identify the invariants g_2, g_3;
- use $\wp'$ and local expansions to analyze zeros and ramification;
- relate lattice symmetries to special values of the Weierstrass function;
- distinguish the analytic definition of $\wp$ from the algebraic cubic relation it later parametrizes;

Conceptual roadmap

$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$

29.1 Regularized lattice sum

Define $\wp(z) = z^{-2} + \sum_{\omega \in \Lambda \setminus \{0\}} [(z-\omega)^{-2} - \omega^{-2}]$.

Example 29.1 (Why the Weierstrass sum converges)**.** For bounded z and large lattice points ω,

$$\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} = \frac{2z\omega - z^2}{\omega^2(z-\omega)^2} = O(|\omega|^{-3}).$$

The lattice sum $\sum_{\omega \neq 0} |\omega|^{-3}$ converges. Therefore the regularized series for $\wp(z)$ converges normally on compact sets avoiding Λ. The subtraction term is exactly what improves the decay from quadratic to cubic order.

29.2 Normal convergence

The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice.

29.3 Evenness

Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$.

29.4 Double periodicity

The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic.

Example 29.2 (A half-period critical point)**.** Let a be a nonzero half-period, so $2a \in \Lambda$. Periodicity and oddness of $\wp'$ give

$$\wp'(a) = \wp'(a - 2a) = \wp'(-a) = -\wp'(a).$$

Hence $\wp'(a) = 0$. The three nonzero half-period classes therefore map to the three finite branch values of the degree-two map $\wp : \mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$.

29.5 Pole structure

Modulo the lattice, $\wp$ has one double pole at the origin with zero residue.

29.6 Derivative

The derivative $\wp'$ is odd and has a triple pole at the lattice points.

29.7 Half-period values

At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation.

Example 29.3 (Recovering the cubic differential equation)**.** Near zero,

$$\wp(z) = z^{-2} + O(z^2), \qquad \wp'(z) = -2z^{-3} + O(z).$$

Choose lattice invariants g_2, g_3 so that the principal part of

$$F = (\wp')^2 - 4\wp^3 + g_2\wp + g_3$$

cancels. Then F is elliptic and has no poles, hence is constant on the compact torus. Matching the constant term gives $F = 0$, so

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

29.8 Differential equation

There are lattice invariants g_2, g_3 with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

29.9 Second derivative relation

Differentiating the cubic gives $\wp'' = 6\wp^2 - g_2/2$.

29.10 Uniformization preview

The map $z \mapsto (\wp(z), \wp'(z))$ parametrizes a nonsingular cubic curve when the discriminant is nonzero.

29.11 Core structural results

Theorem 29.4 (Normal convergence of the Weierstrass series)**.** *The defining series for $\wp$ converges normally on compact subsets of $\mathbb{C} \setminus \Lambda$.*

Proof. For bounded z and large $|\omega|$, expand $(z-\omega)^{-2} - \omega^{-2} = O(|z|/|\omega|^3)$. The lattice sum $\sum_{\omega \neq 0} |\omega|^{-3}$ converges, so the Weierstrass M-test applies away from lattice points. □

Theorem 29.5 (Elliptic differential equation)**.** *There exist constants $g_2 = 60 \sum_{\omega \neq 0} \omega^{-4}$ and $g_3 = 140 \sum_{\omega \neq 0} \omega^{-6}$ such that $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.*

Proof. Compare Laurent expansions at zero. The difference $(\wp')^2 - 4\wp^3 + g_2\wp + g_3$ is elliptic and has its principal part cancelled, hence is entire on the torus and therefore constant. The chosen coefficients make the constant zero. □

Theorem 29.6 (Half-period critical points)**.** *If $2a \in \Lambda$ but $a \notin \Lambda$, then $\wp'(a) = 0$.*

Proof. Because $\wp'$ is odd and periodic, $\wp'(a) = \wp'(a - 2a) = \wp'(-a) = -\wp'(a)$. □

Theorem 29.7 (Degree-two map to the sphere)**.** *The function $\wp$ induces a degree-two meromorphic map $\mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$, branched at the three nonzero half-periods and the origin.*

Proof. Generically $\wp(z) = \wp(-z)$ gives two points in a fiber. Fixed points of the involution $z \mapsto -z$ modulo the lattice are exactly the four half-period classes, where the local multiplicity rises to two. □

29.12 Worked examples

Example 29.8 (Regularized lattice sum diagnostic)**.** Give a rigorous worked analysis of Regularized lattice sum. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Define $\wp(z) = z^{-2} + \sum_{\omega \in \Lambda \setminus \{0\}} [(z-\omega)^{-2} - \omega^{-2}]$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.9 (Normal convergence diagnostic)**.** Give a rigorous worked analysis of Normal convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.10 (Evenness diagnostic)**.** Give a rigorous worked analysis of Evenness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.11 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

29.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 29.12 (Regularized lattice sum diagnostic)**.** Give a rigorous worked analysis of Regularized lattice sum. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Define $\wp(z) = z^{-2} + \sum_{\omega\in\Lambda\setminus\{0\}}[(z-\omega)^{-2} - \omega^{-2}]$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 29.13 (Normal convergence diagnostic)**.** Give a rigorous worked analysis of Normal convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 29.14 (Evenness diagnostic)**.** Give a rigorous worked analysis of Evenness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 29.15 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 29.16 (Pole structure diagnostic)**.** Give a rigorous worked analysis of Pole structure. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Modulo the lattice, $\wp$ has one double pole at the origin with zero residue. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 29.17 (Derivative diagnostic)**.** Give a rigorous worked analysis of Derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The derivative $\wp'$ is odd and has a triple pole at the lattice points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.18 (Half-period values diagnostic)**.** Give a rigorous worked analysis of Half-period values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.19 (Normal convergence of the Weierstrass series proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The defining series for $\wp$ converges normally on compact subsets of $\mathbb{C} \setminus \Lambda$.

Solution. For bounded z and large $|\omega|$, expand $(z-\omega)^{-2} - \omega^{-2} = O(|z|/|\omega|^3)$. The lattice sum $\sum_{\omega \neq 0} |\omega|^{-3}$ converges, so the Weierstrass M-test applies away from lattice points. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.20 (Elliptic differential equation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: There exist constants $g_2 = 60 \sum_{\omega \neq 0} \omega^{-4}$ and $g_3 = 140 \sum_{\omega \neq 0} \omega^{-6}$ such that $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

Solution. Compare Laurent expansions at zero. The difference $(\wp')^2 - 4\wp^3 + g_2\wp + g_3$ is elliptic and has its principal part cancelled, hence is entire on the torus and therefore constant. The chosen coefficients make the constant zero. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.21 (Half-period critical points proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $2a \in \Lambda$ but $a \notin \Lambda$, then $\wp'(a) = 0$.

Solution. Because $\wp'$ is odd and periodic, $\wp'(a) = \wp'(a-2a) = \wp'(-a) = -\wp'(a)$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.22 (Degree-two map to the sphere proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The function $\wp$ induces a degree-two meromorphic map $\mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$, branched at the three nonzero half-periods and the origin.

Solution. Generically $\wp(z) = \wp(-z)$ gives two points in a fiber. Fixed points of the involution $z \mapsto -z$ modulo the lattice are exactly the four half-period classes, where the local multiplicity rises to two. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Weierstrass $\wp$-function is the canonical elliptic function attached to a lattice. Its carefully regularized lattice sum gives a doubly periodic even function with one double pole per period cell and an algebraic differential equation. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

29.14 Exercises with complete solutions

Exercise 29.24. State and apply the central principle behind Regularized lattice sum. Give one concrete consequence.

Hint. Pair lattice points ω and $-\omega$ in the defining series to expose evenness and improve convergence control. ☐

Solution. Define $\wp(z) = z^{-2} + \sum_{\omega\in\Lambda\setminus\{0\}}[(z-\omega)^{-2} - \omega^{-2}]$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.25. State and apply the central principle behind Normal convergence. Give one concrete consequence.

Hint. Differentiate the locally uniformly convergent lattice series termwise away from lattice points to obtain $\wp'$. ☐

Solution. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.26. State and apply the central principle behind Evenness. Give one concrete consequence.

Hint. Periodicity follows by reindexing the lattice sum; perform the index shift rather than appealing only to the name 'elliptic'. ☐

Solution. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.27. State and apply the central principle behind Double periodicity. Give one concrete consequence.

Hint. Near zero, isolate the z^{-2} term and expand the remaining lattice sum holomorphically. ☐

Solution. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.28. State and apply the central principle behind Pole structure. Give one concrete consequence.

Hint. Compare elliptic functions with the same principal parts to derive the differential equation for $\wp$. ☐

Solution. Modulo the lattice, $\wp$ has one double pole at the origin with zero residue. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.29. State and apply the central principle behind Derivative. Give one concrete consequence.

Hint. Define g_2 and g_3 from lattice sums and track their scaling weights under $\Lambda \mapsto \lambda\Lambda$. ☐

Solution. The derivative $\wp'$ is odd and has a triple pole at the lattice points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.30. State and apply the central principle behind Half-period values. Give one concrete consequence.

Hint. Zeros of $\wp'$ occur at nontrivial half-period classes; use oddness and periodicity to locate them. ☐

Solution. At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.31. State and apply the central principle behind Differential equation. Give one concrete consequence.

Hint. Keep the analytic lattice-series definition separate from the cubic equation until the identity has actually been proved. ☐

Solution. There are lattice invariants g_2, g_3 with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

29.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 29.32 (Principal part)**.** What is the principal part of $\wp$ at a lattice point?

Hint. Translate the expansion at zero. ☐

Solution. It is $1/(z-\omega)^2$. ☐

Exercise 29.33 (Residue)**.** What is the residue of $\wp$ at a lattice point?

Hint. A pure double pole has no $(z-\omega)^{-1}$ term. ☐

Solution. The residue is zero. ☐

Exercise 29.34 (Parity)**.** State the parity of $\wp$ and $\wp'$.

Hint. Differentiate an even function. ☐

Solution. $\wp$ is even and $\wp'$ is odd. ☐

Exercise 29.35 (Half-period derivative)**.** What is $\wp'(a)$ at a nonzero half-period class?

Hint. Use periodicity and oddness. ☐

Solution. It is zero. ☐

Exercise 29.36 (Second derivative)**.** Differentiate the cubic equation to find $\wp''$ away from zeros of $\wp'$, then extend by analyticity.

Hint. Differentiate both sides and cancel $2\wp'$. ☐

Solution. $\wp'' = 6\wp^2 - g_2/2$. ☐

Proofs

Exercise 29.37 (Normal convergence)**.** Prove normal convergence of the regularized lattice series away from the lattice.

Hint. Use the $O(|\omega|^{-3})$ estimate. ☐

Solution. On a compact set, the estimate is uniform for large ω, and the lattice sum of inverse cubes converges, so the Weierstrass M-test applies. ☐

Exercise 29.38 (Evenness)**.** Prove $\wp(-z) = \wp(z)$.

Hint. Pair lattice points ω and $-\omega$. ☐

Solution. Replacing z by $-z$ and reindexing the lattice by $\omega \mapsto -\omega$ leaves the regularized sum unchanged. ☐

Exercise 29.39 (Half-period zeros)**.** Prove that $\wp'$ vanishes at all three nonzero half-period classes.

Hint. Use $a \equiv -a \pmod{\Lambda}$. ☐

Solution. Periodicity gives $\wp'(a) = \wp'(-a)$, while oddness gives $\wp'(-a) = -\wp'(a)$, hence the value is zero. ☐

Exercise 29.40 (Differential equation)**.** Explain why canceling the principal part in the cubic expression forces a global identity.

Hint. An elliptic function with no poles is constant. ☐

Solution. After cancellation the difference descends to a holomorphic function on the compact torus, so it is constant. Matching one Laurent coefficient determines that constant as zero. ☐

Counterexamples and hypothesis tests

Exercise 29.41 (Unregularized sum)**.** Why is $\sum_{\omega\in\Lambda}(z-\omega)^{-2}$ not used naively as the definition of $\wp$?

Hint. Absolute convergence in two lattice dimensions is borderline. ☐

Solution. The inverse-square lattice sum is not absolutely convergent in the required manner. Regularization removes the problematic leading tail and yields normal convergence. ☐

Exercise 29.42 (Simple pole)**.** Could $\wp$ have a simple pole at zero?

Hint. Use evenness. ☐

Solution. No. An even Laurent series has no odd-power term such as z^{-1}; the leading pole is double. ☐

Exercise 29.43 (Discriminant zero)**.** Does the cubic $y^2 = 4x^3 - g_2x - g_3$ remain a smooth elliptic curve when its discriminant vanishes?

Hint. A repeated root creates a singular point. □

Solution. No. Vanishing discriminant makes the cubic singular, so the smooth torus uniformization requires nonzero discriminant. □

Applications and investigations

Exercise 29.44 (Degree-two map)**.** Why does $\wp$ generically identify exactly the pair z and $-z$?

Hint. Use evenness and the pole order. □

Solution. Evenness gives the pair, and the function has degree two as a meromorphic map on the torus, so a generic value has exactly those two preimages counted with multiplicity. □

Exercise 29.45 (Cubic uniformization)**.** How does the differential equation turn analytic data into an algebraic curve?

Hint. Set $x = \wp(z)$ and $y = \wp'(z)$. □

Solution. The pair satisfies the cubic equation, producing a holomorphic map from the torus into the corresponding projective cubic. □

Challenge problems

Exercise 29.46 (Branch points of wp)**.** Identify the four branch points of the degree-two map $\wp$ on the torus.

Hint. They are the fixed points of $z \mapsto -z$ modulo the lattice. □

Solution. They are the origin class and the three nonzero half-period classes. The origin maps to infinity and the other three map to the finite values e_1, e_2, e_3. □

Exercise 29.47 (Invariant weights)**.** Predict how g_2 and g_3 scale when the lattice is multiplied by a nonzero complex number c.

Hint. Inspect their lattice sums of fourth and sixth inverse powers. □

Solution. They scale as $g_2(c\Lambda) = c^{-4}g_2(\Lambda)$ and $g_3(c\Lambda) = c^{-6}g_3(\Lambda)$. □

Chapter 30

Addition Formulas

Addition formulas express the group structure of the complex torus directly in the values of $\wp$ and $\wp'$. They are the analytic form of the chord-and-tangent law on the associated cubic curve.

Learning goals

After this chapter, the reader should be able to:

- derive addition formulas for $\wp$ and related elliptic functions from analytic identities;
- specialize the addition formula to obtain duplication formulas;
- track poles and symmetry to verify candidate elliptic identities;
- use the identity theorem to prove equality of elliptic functions once principal parts and enough values agree;
- translate addition on the torus into algebraic formulas for $\wp$ and $\wp'$;
- apply addition identities to compute special values or group-law coordinates;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

30.1 Translation on the torus

Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$.

Example 30.1 (Using the addition formula)**.** For points u, v with $\wp(u) \neq \wp(v)$, set

$$m = \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}.$$

Then

$$\wp(u+v) = -\wp(u) - \wp(v) + \frac{m^2}{4}.$$

Thus the sum coordinate is determined by the two points and the chord slope. The same expression is exactly the x-coordinate computation in the chord law on $y^2 = 4x^3 - g_2x - g_3$.

30.2 Need for an addition law

Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data.

30.3 Weierstrass addition formula

A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$.

30.4 Duplication formula

Taking $v \to u$ yields the tangent or duplication formula.

Example 30.2 (Duplication as a tangent limit)**.** Let $v \to u$. The chord quotient satisfies

$$\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \longrightarrow \frac{\wp''(u)}{\wp'(u)}$$

when $\wp'(u) \neq 0$. Therefore

$$\wp(2u) = -2\wp(u) + \frac{1}{4}\left(\frac{\wp''(u)}{\wp'(u)}\right)^2.$$

The chord has become the tangent, matching the geometric duplication law on the cubic.

30.5 Subtraction formula

Replacing v by $-v$ gives a companion formula for $\wp(u-v)$.

30.6 Pole comparison proof

Differences of candidate expressions can be shown constant by comparing elliptic principal parts.

30.7 Cubic geometry

The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic.

Example 30.3 (Adding a half-period)**.** Let a be a nonzero half-period, so $\wp'(a) = 0$. In the addition formula with $v = a$, the slope becomes

$$\frac{\wp'(u)}{\wp(u) - \wp(a)}.$$

Hence translation by a two-torsion point is represented by a rational expression in $\wp(u)$ and $\wp'(u)$. The simplification reflects the fact that half-periods are points of order two on the torus.

30.8 Special values

Half-periods and torsion points simplify the formulas because $\wp'$ may vanish.

30.9 Division phenomena

Iterating addition generates rational functions describing multiplication-by-n on the elliptic curve.

30.10 Core structural results

Theorem 30.4 (Weierstrass addition formula)**.** *For $u, v, u \pm v \notin \Lambda$ and $\wp(u) \neq \wp(v)$, $\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2$.*

Proof. As a function of one variable, compare both sides after shifting by the other argument. Their poles and principal parts agree; the difference is an elliptic function without poles, hence constant. Evaluate at a convenient limiting point to determine the constant. □

Theorem 30.5 (Duplication formula)**.** *The limit $v \to u$ in the addition formula gives $\wp(2u) = -2\wp(u) + \frac{1}{4}(\wp''(u)/\wp'(u))^2$ whenever the displayed quotient is defined.*

Proof. Apply l'Hôpital's rule to the chord-slope quotient as $v \to u$. □

Theorem 30.6 (Chord interpretation)**.** *Under $z \mapsto (\wp(z), \wp'(z))$, the analytic addition formula agrees with the chord-and-tangent law on $y^2 = 4x^3 - g_2x - g_3$.*

Proof. The squared chord slope determines the third intersection's x-coordinate by Vieta's formula; reflection in the x-axis changes the sign of y, matching the torus inverse $z \mapsto -z$. □

30.11 Worked examples

Example 30.7 (Translation on the torus diagnostic)**.** Give a rigorous worked analysis of Translation on the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.8 (Need for an addition law diagnostic)**.** Give a rigorous worked analysis of Need for an addition law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.9 (Weierstrass addition formula diagnostic)**.** Give a rigorous worked analysis of Weierstrass addition formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.10 (Duplication formula diagnostic)**.** Give a rigorous worked analysis of Duplication formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Taking $v \to u$ yields the tangent or duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

30.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 30.11 (Translation on the torus diagnostic)**.** Give a rigorous worked analysis of Translation on the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.12 (Need for an addition law diagnostic)**.** Give a rigorous worked analysis of Need for an addition law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.13 (Weierstrass addition formula diagnostic)**.** Give a rigorous worked analysis of Weierstrass addition formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.14 (Duplication formula diagnostic)**.** Give a rigorous worked analysis of Duplication formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Taking $v \to u$ yields the tangent or duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.15 (Subtraction formula diagnostic)**.** Give a rigorous worked analysis of Subtraction formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Replacing v by $-v$ gives a companion formula for $\wp(u-v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.16 (Pole comparison proof diagnostic)**.** Give a rigorous worked analysis of Pole comparison proof. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differences of candidate expressions can be shown constant by comparing elliptic principal parts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.17 (Cubic geometry diagnostic)**.** Give a rigorous worked analysis of Cubic geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 30.18 (Special values diagnostic)**.** Give a rigorous worked analysis of Special values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Half-periods and torsion points simplify the formulas because $\wp'$ may vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 30.19 (Weierstrass addition formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $u, v, u \pm v \notin \Lambda$ and $\wp(u) \neq \wp(v)$, $\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2$.

Solution. As a function of one variable, compare both sides after shifting by the other argument. Their poles and principal parts agree; the difference is an elliptic function without poles, hence constant. Evaluate at a convenient limiting point to determine the constant. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 30.20 (Duplication formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The limit $v \to u$ in the addition formula gives $\wp(2u) = -2\wp(u) + \frac{1}{4}(\wp''(u)/\wp'(u))^2$ whenever the displayed quotient is defined.

Solution. Apply l'Hôpital's rule to the chord-slope quotient as $v \to u$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 30.21 (Chord interpretation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Under $z \mapsto (\wp(z), \wp'(z))$, the analytic addition formula agrees with the chord-and-tangent law on $y^2 = 4x^3 - g_2x - g_3$.

Solution. The squared chord slope determines the third intersection's x-coordinate by Vieta's formula; reflection in the x-axis changes the sign of y, matching the torus inverse $z \mapsto -z$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 30.22 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Addition formulas express the group structure of the complex torus directly in the values of $\wp$ and $\wp'$. They are the analytic form of the chord-and-tangent law on the associated cubic curve. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

30.13 Exercises with complete solutions

Exercise 30.23. State and apply the central principle behind Translation on the torus. Give one concrete consequence.

Hint. Form the elliptic function whose poles cancel between the two sides of the proposed addition identity. ☐

Solution. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.24. State and apply the central principle behind Need for an addition law. Give one concrete consequence.

Hint. After principal parts cancel, the difference is entire on the compact torus and hence constant; evaluate one convenient point to find the constant. ☐

Solution. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.25. State and apply the central principle behind Weierstrass addition formula. Give one concrete consequence.

Hint. For the standard $\wp$ addition formula, use the slope $(\wp'(u) - \wp'(v))/(\wp(u) - \wp(v))$ and track symmetry in u, v. ☐

Solution. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.26. State and apply the central principle behind Duplication formula. Give one concrete consequence.

Hint. Obtain the duplication formula by taking the limit $v \to u$ rather than substituting into a singular expression directly. ☐

Solution. Taking $v \to u$ yields the tangent or duplication formula. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.27. State and apply the central principle behind Subtraction formula. Give one concrete consequence.

Hint. Check parity: $\wp$ is even and $\wp'$ is odd, which simplifies many addition identities. ☐

Solution. Replacing v by $-v$ gives a companion formula for $\wp(u-v)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.28. State and apply the central principle behind Pole comparison proof. Give one concrete consequence.

Hint. Translate $u+v$ on the torus into the third intersection point of a line with the cubic only after the cubic parametrization is established. ☐

Solution. Differences of candidate expressions can be shown constant by comparing elliptic principal parts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.29. State and apply the central principle behind Cubic geometry. Give one concrete consequence.

Hint. Use pole locations to decide where a candidate formula can be analytically continued across apparent singular denominators. ☐

Solution. The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.30. State and apply the central principle behind Special values. Give one concrete consequence.

Hint. Verify special cases such as $v = -u$ or half-periods as consistency checks on signs and constants. ☐

Solution. Half-periods and torsion points simplify the formulas because $\wp'$ may vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

30.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 30.31 (Chord slope)**.** Write the slope quantity appearing in the Weierstrass addition formula.

Hint. Use the difference of derivative values divided by the difference of function values. ☐

Solution. It is $(\wp'(u) - \wp'(v))/(\wp(u) - \wp(v))$. ☐

Exercise 30.32 (Inverse point)**.** What happens to $(\wp(z), \wp'(z))$ when z is replaced by $-z$?

Hint. Use parity. ☐

Solution. The first coordinate is unchanged and the second changes sign. ☐

Exercise 30.33 (Two-torsion)**.** Why is a half-period a point of order two on the torus?

Hint. Double it modulo the lattice. ☐

Solution. Because $2a \in \Lambda$, so $2[a] = [0]$. ☐

Exercise 30.34 (Duplication slope)**.** What replaces the chord slope in the limit $v \to u$?

Hint. Apply l'Hopital to numerator and denominator. ☐

Solution. It becomes $\wp''(u)/\wp'(u)$. ☐

Exercise 30.35 (Subtraction)**.** How is a formula for $\wp(u - v)$ obtained from the addition formula?

Hint. Replace v by $-v$. □

Solution. Use $\wp(-v) = \wp(v)$ and $\wp'(-v) = -\wp'(v)$ in the same addition identity. □

Proofs

Exercise 30.36 (Pole comparison)**.** Explain the standard elliptic-function proof strategy for the addition formula.

Hint. Treat one side minus the other as a function of one variable. □

Solution. Show the difference is elliptic and that all poles and principal parts cancel. It is then holomorphic on the compact torus and hence constant; evaluate a convenient limit to determine the constant. □

Exercise 30.37 (Duplication formula)**.** Derive the duplication formula from the addition formula.

Hint. Let $v \to u$ and use l'Hopital. □

Solution. The chord quotient tends to $\wp''(u)/\wp'(u)$, yielding the tangent formula for $\wp(2u)$. □

Exercise 30.38 (Chord geometry)**.** Why does the squared slope determine the third intersection x-coordinate on the Weierstrass cubic?

Hint. Substitute the line equation into the cubic and use Vieta. □

Solution. The resulting cubic in x has the two known intersection abscissas and a third one. Comparing the quadratic coefficient expresses the third abscissa in terms of the slope squared and the first two abscissas. □

Exercise 30.39 (Compatibility with inverse)**.** Prove that torus inversion corresponds to reflection across the x-axis on the cubic.

Hint. Use parity of $\wp$ and $\wp'$. □

Solution. Under $z \mapsto -z$, $x = \wp(z)$ stays fixed while $y = \wp'(z)$ changes sign, exactly the cubic group inverse. □

Counterexamples and hypothesis tests

Exercise 30.40 (Equal x-coordinates)**.** Can the chord version of the addition formula be used directly when $\wp(u) = \wp(v)$?

Hint. The displayed denominator vanishes. □

Solution. Not directly. One must use a limiting tangent case or recognize that the points may be inverses or otherwise special. □

Exercise 30.41 (Half-period derivative)**.** Does the duplication formula containing $\wp''/\wp'$ apply directly at a half-period?

Hint. The denominator vanishes there. □

Solution. No. A separate limiting analysis is required at two-torsion points. □

Exercise 30.42 (Only wp values)**.** Can generic addition be recovered from $\wp(u)$ and $\wp(v)$ alone?

Hint. $\wp$ forgets the signs of the points. ☐

Solution. No. Derivative data distinguish z from $-z$ and are needed to choose the correct group sum generically. ☐

Applications and investigations

Exercise 30.43 (Fast multiplication)**.** How can duplication formulas help compute nP on an elliptic curve?

Hint. Use repeated doubling and addition. ☐

Solution. They provide rational recurrences for the coordinates, enabling the same double-and-add strategy used in algebraic elliptic-curve arithmetic. ☐

Exercise 30.44 (Torsion detection)**.** Why do special addition formulas simplify at torsion points?

Hint. Some multiples become the identity and derivative values can vanish. ☐

Solution. Algebraic relations among repeated sums impose polynomial conditions on $\wp$-values, leading to division polynomials and torsion equations. ☐

Challenge problems

Exercise 30.45 (Third intersection)**.** Explain the sign change needed after finding the third intersection of a chord with the cubic.

Hint. The group sum is the inverse of that third point. ☐

Solution. Three collinear points sum to the identity, so $P+Q$ is obtained by reflecting the third intersection across the x-axis. ☐

Exercise 30.46 (Multiplication map degree)**.** Predict the degree of the multiplication-by-n map on a complex torus.

Hint. Count preimages of the origin modulo the lattice. ☐

Solution. The kernel consists of n^2 classes $(a\omega_1 + b\omega_2)/n$, so the map has degree n^2. ☐

Chapter 31

Elliptic Curves as Riemann Surfaces

The Weierstrass parametrization identifies a complex torus with a smooth projective cubic curve. Analytic addition on the torus becomes the algebraic group law of an elliptic curve.

Learning goals

After this chapter, the reader should be able to:

- verify that $(\wp(z), \wp'(z))$ lies on the Weierstrass cubic;
- extend the Weierstrass parametrization across the lattice class to the point at infinity;
- prove that the torus-to-cubic map is bijective and hence biholomorphic under the nonsingularity hypothesis;
- identify the point at infinity as the group identity and inversion as $z \mapsto -z$;
- show that torus addition agrees with the chord-and-tangent group law on the cubic;
- relate lattice scaling, the invariants g_2, g_3, and the genus-one moduli viewpoint;

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

31.1 Weierstrass cubic

The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$.

Example 31.1 (Checking the cubic equation)**.** Set

$$x = \wp(z), \qquad y = \wp'(z).$$

The differential equation gives immediately

$$y^2 = 4x^3 - g_2x - g_3.$$

Thus every torus point away from the lattice maps to the affine Weierstrass cubic. Near the lattice origin, the pole expansions send the point to the unique point at infinity in projective closure.

31.2 Parametrization

The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity.

31.3 Point at infinity

The unique point at infinity is the identity of the cubic group law.

31.4 Injectivity modulo sign and derivative

The pair $(\wp, \wp')$ separates torus points modulo the lattice.

Example 31.2 (The invariant differential)**.** On the cubic $y^2 = 4x^3 - g_2x - g_3$, consider dx/y. Under the Weierstrass parametrization,

$$dx = \wp'(z)\, dz = y\, dz.$$

Therefore

$$\frac{dx}{y} = dz.$$

The nowhere-vanishing holomorphic differential on the torus is exactly the pullback of the algebraic invariant differential.

31.5 Surjectivity

Every point of the smooth cubic arises from a torus point.

31.6 Biholomorphism

The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism.

31.7 Group law

Addition of arguments corresponds to the chord-and-tangent group law on the cubic.

Example 31.3 (A nonsingularity check)**.** For

$$F(x, y) = y^2 - 4x^3 + g_2x + g_3,$$

a singular affine point would satisfy $F = F_x = F_y = 0$. Thus $y = 0$ and $12x^2 = g_2$, while x is also a repeated root of $4x^3 - g_2x - g_3$. This occurs exactly when the discriminant $g_2^3 - 27g_3^2$ vanishes. Nonzero discriminant therefore gives a smooth cubic.

31.8 Invariant differential

The form dx/y pulls back to dz up to the normalization determined by the chosen cubic equation.

31.9 Moduli viewpoint

Changing the lattice by scaling changes g_2, g_3 with weights and preserves the modular invariant j.

31.10 Genus-one synthesis

Complex tori, smooth cubic curves, elliptic functions, and genus-one Riemann surfaces with a base point describe the same analytic geometry.

31.11 Core structural results

Theorem 31.4 (Weierstrass map lies on the cubic)**.** *For every $z \notin \Lambda$, the pair $(\wp(z), \wp'(z))$ satisfies $y^2 = 4x^3 - g_2x - g_3$.*

Proof. This is exactly the differential equation for $\wp$. ☐

Theorem 31.5 (Extension at the origin)**.** *The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends holomorphically at the lattice class of zero to the projective point at infinity.*

Proof. Using $\wp(z) = z^{-2} + O(z^2)$ and $\wp'(z) = -2z^{-3} + O(z)$, rescale homogeneous coordinates by a suitable power of z to obtain a finite nonzero projective limit equal to the unique point at infinity. ☐

Theorem 31.6 (Torus-cubic biholomorphism)**.** *If the discriminant is nonzero, the extended Weierstrass map $\mathbb{C}/\Lambda \to E$ is a biholomorphism onto the smooth cubic E.*

Proof. The map is nonconstant and holomorphic between compact Riemann surfaces. The pair $\wp, \wp'$ separates points modulo the lattice, giving injectivity; compactness makes the image closed, while nonconstant holomorphic maps are open, so the image is all of the connected cubic. A holomorphic bijection between compact Riemann surfaces has holomorphic inverse. ☐

Theorem 31.7 (Compatibility of group laws)**.** *Under the Weierstrass biholomorphism, addition on $\mathbb{C}/\Lambda$ corresponds to the cubic chord-and-tangent group law.*

Proof. The addition formula computes the coordinates of the image of $u + v$ and matches the algebraic chord construction; the identity is the lattice origin mapped to infinity and inversion is $z \mapsto -z$, which changes the sign of $\wp'$. ☐

31.12 Worked examples

Example 31.8 (Weierstrass cubic diagnostic)**.** Give a rigorous worked analysis of Weierstrass cubic. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.9 (Parametrization diagnostic)**.** Give a rigorous worked analysis of Parametrization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.10 (Point at infinity diagnostic)**.** Give a rigorous worked analysis of Point at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The unique point at infinity is the identity of the cubic group law. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.11 (Injectivity modulo sign and derivative diagnostic)**.** Give a rigorous worked analysis of Injectivity modulo sign and derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

31.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 31.12 (Weierstrass cubic diagnostic)**.** Give a rigorous worked analysis of Weierstrass cubic. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.13 (Parametrization diagnostic)**.** Give a rigorous worked analysis of Parametrization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.14 (Point at infinity diagnostic)**.** Give a rigorous worked analysis of Point at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The unique point at infinity is the identity of the cubic group law. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.15 (Injectivity modulo sign and derivative diagnostic)**.** Give a rigorous worked analysis of Injectivity modulo sign and derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.16 (Surjectivity diagnostic)**.** Give a rigorous worked analysis of Surjectivity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every point of the smooth cubic arises from a torus point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.17 (Biholomorphism diagnostic)**.** Give a rigorous worked analysis of Biholomorphism. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.18 (Group law diagnostic)**.** Give a rigorous worked analysis of Group law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Addition of arguments corresponds to the chord-and-tangent group law on the cubic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 31.19 (Weierstrass map lies on the cubic proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For every $z \notin \Lambda$, the pair $(\wp(z), \wp'(z))$ satisfies $y^2 = 4x^3 - g_2x - g_3$.

Solution. This is exactly the differential equation for $\wp$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.20 (Extension at the origin proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends holomorphically at the lattice class of zero to the projective point at infinity.

Solution. Using $\wp(z) = z^{-2} + O(z^2)$ and $\wp'(z) = -2z^{-3} + O(z)$, rescale homogeneous coordinates by a suitable power of z to obtain a finite nonzero projective limit equal to the unique point at infinity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.21 (Torus-cubic biholomorphism proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If the discriminant is nonzero, the extended Weierstrass map $\mathbb{C}/\Lambda \to E$ is a biholomorphism onto the smooth cubic E.

Solution. The map is nonconstant and holomorphic between compact Riemann surfaces. The pair $\wp, \wp'$ separates points modulo the lattice, giving injectivity; compactness makes the image closed, while nonconstant holomorphic maps are open, so the image is all of the connected cubic. A holomorphic bijection between compact Riemann surfaces has holomorphic inverse. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.22 (Compatibility of group laws proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Under the Weierstrass biholomorphism, addition on $\mathbb{C}/\Lambda$ corresponds to the cubic chord-and-tangent group law.

Solution. The addition formula computes the coordinates of the image of $u + v$ and matches the algebraic chord construction; the identity is the lattice origin mapped to infinity and inversion is $z \mapsto -z$, which changes the sign of $\wp'$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.23 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Weierstrass parametrization identifies a complex torus with a smooth projective cubic curve. Analytic addition on the torus becomes the algebraic group law of an elliptic curve. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

31.14 Exercises with complete solutions

Exercise 31.24. State and apply the central principle behind Weierstrass cubic. Give one concrete consequence.

Hint. Substitute $x = \wp(z)$, $y = \wp'(z)$ into the differential equation before doing any projective geometry. □

Solution. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.25. State and apply the central principle behind Parametrization. Give one concrete consequence.

Hint. Near the lattice origin, use $\wp(z) = z^{-2} + O(z^2)$ and $\wp'(z) = -2z^{-3} + O(z)$ to identify the limiting projective point. □

Solution. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.26. State and apply the central principle behind Point at infinity. Give one concrete consequence.

Hint. The discriminant condition $g_2^3 - 27g_3^2 \neq 0$ is what guarantees the cubic is smooth; check it before invoking compact Riemann-surface arguments. □

Solution. The unique point at infinity is the identity of the cubic group law. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.27. State and apply the central principle behind Injectivity modulo sign and derivative. Give one concrete consequence.

Hint. To prove injectivity, use that the pair $(\wp, \wp')$ separates points modulo the lattice, not $\wp$ alone. □

Solution. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.28. State and apply the central principle behind Surjectivity. Give one concrete consequence.

Hint. A nonconstant holomorphic map from a compact surface has closed image; combine this with openness to obtain surjectivity onto the connected cubic. □

Solution. Every point of the smooth cubic arises from a torus point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.29. State and apply the central principle behind Biholomorphism. Give one concrete consequence.

Hint. Once the map is a holomorphic bijection between compact Riemann surfaces, use the standard inverse-holomorphicity theorem for biholomorphism. ☐

Solution. The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.30. State and apply the central principle behind Group law. Give one concrete consequence.

Hint. Under inversion $z \mapsto -z$, $\wp$ is unchanged and $\wp'$ changes sign; this matches the cubic group inverse. ☐

Solution. Addition of arguments corresponds to the chord-and-tangent group law on the cubic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.31. State and apply the central principle behind Invariant differential. Give one concrete consequence.

Hint. Use the addition formula to compare torus addition with the chord-and-tangent coordinates, including the point at infinity as identity. ☐

Solution. The form dx/y pulls back to dz up to the normalization determined by the chosen cubic equation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

31.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 31.32 (Curve point)**.** If $x = \wp(z)$, what is the corresponding y-coordinate in the Weierstrass parametrization?

Hint. Use the derivative coordinate. ☐

Solution. It is $y = \wp'(z)$. ☐

Exercise 31.33 (Identity point)**.** Which point of the projective cubic is the group identity?

Hint. Look at the image of the lattice origin. ☐

Solution. The unique point at infinity is the identity. □

Exercise 31.34 (Inverse)**.** What is the inverse of (x, y) in the cubic group law?

Hint. Match torus inversion. □

Solution. It is $(x, -y)$. □

Exercise 31.35 (Differential pullback)**.** Compute the pullback of dx/y.

Hint. Use $x = \wp(z)$ and $dx = \wp'(z)dz$. □

Solution. The pullback is dz. □

Exercise 31.36 (Discriminant condition)**.** What condition on g_2, g_3 ensures the cubic is nonsingular?

Hint. Use the Weierstrass discriminant. □

Solution. One requires $g_2^3 - 27g_3^2 \neq 0$. □

Proofs

Exercise 31.37 (Map lies on cubic)**.** Prove that the Weierstrass parametrization lands on the cubic.

Hint. Use the differential equation for $\wp$. □

Solution. Substituting $x = \wp(z)$ and $y = \wp'(z)$ into the cubic equation gives an identity. □

Exercise 31.38 (Extension at infinity)**.** Explain why the parametrization extends over the lattice origin to the projective point at infinity.

Hint. Use the Laurent orders 2 and 3 of $\wp$ and $\wp'$. □

Solution. The affine coordinates blow up with controlled orders. In projective coordinates, rescaling by an appropriate power of z yields a finite limit equal to the unique point at infinity. □

Exercise 31.39 (Biholomorphism)**.** Why does an injective nonconstant holomorphic map from the compact torus to a smooth connected cubic become a biholomorphism?

Hint. Use compactness and the open mapping theorem. □

Solution. The image is compact and therefore closed, and nonconstant holomorphic maps are open. A nonempty subset that is both open and closed in the connected cubic is all of it. A holomorphic bijection between compact Riemann surfaces has holomorphic inverse. □

Exercise 31.40 (Group-law compatibility)**.** Prove conceptually that torus addition matches the chord-and-tangent law.

Hint. Use the Weierstrass addition formula. □

Solution. The analytic formula gives the coordinates of the image of $u + v$ and agrees with the coordinate formula obtained from the cubic chord slope. Identity and inversion also match, so the group laws coincide. □

Counterexamples and hypothesis tests

Exercise 31.41 (Zero discriminant)**.** What fails when $g_2^3 - 27g_3^2 = 0$?

Hint. The cubic has a repeated root. ☐

Solution. The projective cubic becomes singular, so it is no longer a smooth genus-one Riemann surface. ☐

Exercise 31.42 (wp alone injective)**.** Is $z \mapsto \wp(z)$ injective on the torus?

Hint. Use evenness. ☐

Solution. No. Generically z and $-z$ have the same $\wp$-value. ☐

Exercise 31.43 (Affine cubic compact)**.** Is the affine equation alone a compact Riemann surface?

Hint. Inspect behavior at infinity. ☐

Solution. No. One must add the projective point at infinity to obtain the compact elliptic curve. ☐

Applications and investigations

Exercise 31.44 (Analytic algebraic dictionary)**.** What does the torus-cubic equivalence translate between?

Hint. Match periods and elliptic functions with algebraic coordinates. ☐

Solution. It identifies lattice-periodic analytic functions with rational functions on a cubic, torus addition with the algebraic group law, and dz with the invariant differential dx/y. ☐

Exercise 31.45 (Moduli invariant)**.** Why is a scale-invariant quantity such as j useful for classifying complex elliptic curves?

Hint. Scaling changes g_2 and g_3 with different weights. ☐

Solution. A suitable ratio cancels those weights, so it depends on the complex isomorphism class rather than the chosen scaling of the lattice or cubic equation. ☐

Challenge problems

Exercise 31.46 (Separation by wp pair)**.** Why does the pair $(\wp, \wp')$ separate torus points even though $\wp$ alone does not?

Hint. Equal $\wp$-values generically give opposite points. ☐

Solution. If two points have the same $\wp$-value, they are generically negatives. Their $\wp'$-values then have opposite signs, so equality of both coordinates forces the same torus class, including the branch cases. ☐

Exercise 31.47 (Genus-one synthesis)**.** State the four equivalent viewpoints assembled in this chapter.

Hint. Connect topology, quotient geometry, function theory and algebraic geometry. ☐

Solution. A pointed genus-one compact Riemann surface can be viewed as a complex torus, as a smooth projective cubic, through its elliptic-function field generated by $\wp$ and $\wp'$, and as an analytic group uniformized by the complex plane modulo a lattice. ☐